0.1 - Linear Algebra

Numerical Optimization

0.1.1 Supremum and Infimum

- A minimizer is a point where the lowest value of a function is reached. Some functions get as close as we like to a value without ever reaching it, and to talk about such a value we need two words that go beyond the minimum and the maximum.

Definition: A set $A \subseteq \mathbb{R}$ is **bounded from above** if there is a $y \in \mathbb{R}$ with $x \leq y$ for all $x \in A$, and **bounded from below** if there is a $y \in \mathbb{R}$ with $x \geq y$ for all $x \in A$. Such a y is called an upper bound or a lower bound of A.

Definition: The smallest upper bound of A is called its **supremum** and the largest lower bound is called its **infimum**, written as;

$$\sup A \qquad \text{and} \qquad \inf A$$

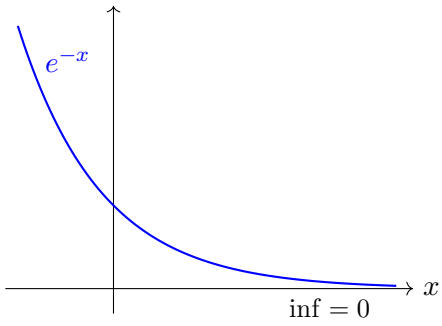
- Meaning $\sup A$ is the tightest ceiling we can put on the set and $\inf A$ is the tightest floor under it.

Example: Let $A = \{x : 1 \leq x < 3\}$. Find $\sup A$ and $\inf A$. Does A have a maximum and a minimum?

- Every point of A is below 3, so 3 is an upper bound. No smaller number works, since for any $y < 3$ there are points of A between y and 3. Hence; $\sup A = 3$. However; $3 \notin A$ and whichever point of A we pick there is a larger one in it, so A has no maximum.
- From below 1 is a lower bound and no larger number is one, so $\inf A = 1$. This time $1 \in A$, so it is also the minimum of A.
- So the supremum and the infimum do not need to be in the set. If the supremum is in the set it is the maximum, and if the infimum is in the set it is the minimum.
- In optimization the set that we look at is the set of the values that f takes on the feasible set. For example $f(x) = e^{-x}$ takes every positive value on $\mathbb{R}$ and only those, so;

$$\inf_{x \in \mathbb{R}} e^{-x} = 0$$

but no x gives $e^{-x} = 0$. This is the lower bound that we never reach of the Optimization Problems pages.



- So; a minimizer exists exactly when the infimum is reached, and then the infimum is the minimum.
- A set of points in $\mathbb{R}^n$ has no ceiling and no floor, because vectors have no order. There we call a set bounded when all of its points stay inside a ball around the origin. To measure a distance in $\mathbb{R}^n$ we first need some linear algebra, and this is where the rest of the topic goes.

Worked Examples

Example: Let $f(x) = \dfrac{x}{1+x}$ on the feasible set $x \geq 0$.

a) Find the infimum and the supremum of the values of f.

- We first write f in a form that is easier to read;

$$f(x) = \frac{x}{1+x} = \frac{(1+x)-1}{1+x} = 1 - \frac{1}{1+x}$$

- For $x \geq 0$ the fraction $\frac{1}{1+x}$ is at most 1 and it is always positive. So; $0 \leq f(x) < 1$.
- At $x = 0$ we have the value 0, so the infimum is 0. As x grows the fraction dies and f gets as close to 1 as we like, so no number below 1 is an upper bound and the supremum is 1.

b) Does f have a minimizer and a maximizer on this set?

- With the infimum 0 and the supremum 1 from part a; we reach the infimum at $x = 0$, so $x^* = 0$ is the minimizer with the minimum value 0.
- We never reach the supremum since $f(x) < 1$ for every x. So; f has no maximizer on this set, although it is bounded from above.

Your Turn

a) Find the supremum and the infimum of $A = \{2 - 1/n : n = 1, 2, 3, \dots\}$. Which of them is in A?

b) Find the supremum and the infimum of $B = \{x : 0 < x \leq 5\}$. Does B have a maximum and a minimum?

c) Find the supremum and the infimum of the values of $f(x) = \dfrac{1}{1 + x^2}$ on $\mathbb{R}$.

d) Does the function f of part c have a maximizer and a minimizer?

0.1.2 Vector Spaces and Subspaces

- The design variables of a problem are collected in a vector, and all we ever do with vectors is to add them and to scale them, as in the step $\vec{x}_k + \alpha_k \vec{p}_k$. A vector space is a set where these two operations never take us outside.
- From here on I will put an arrow on every vector so they do not mix up with the numbers.

Definition: A non empty set V with an addition and a multiplication with numbers is called a **vector space** if for all $\vec{x}, \vec{y} \in V$ and all $\alpha \in \mathbb{R}$;

$$\vec{x} + \vec{y} \in V \qquad \text{and} \qquad \alpha\,\vec{x} \in V$$

and the usual rules of calculation hold;

1) $\vec{x} + \vec{y} = \vec{y} + \vec{x}$ and $\vec{x} + (\vec{y} + \vec{z}) = (\vec{x} + \vec{y}) + \vec{z}$

2) There is a zero vector $\vec{0} \in V$ with $\vec{x} + \vec{0} = \vec{x}$, and every $\vec{x}$ has a $-\vec{x} \in V$ with $\vec{x} + (-\vec{x}) = \vec{0}$

3) $1\,\vec{x} = \vec{x}$ and $\alpha(\beta\,\vec{x}) = (\alpha\beta)\,\vec{x}$

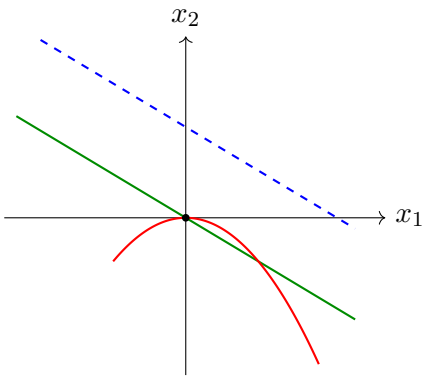
4) $\alpha(\vec{x} + \vec{y}) = \alpha\,\vec{x} + \alpha\,\vec{y}$ and $(\alpha + \beta)\,\vec{x} = \alpha\,\vec{x} + \beta\,\vec{x}$

- The first line is the important one. We say that V is closed under addition and closed under scalar multiplication.
- The standard examples are $\mathbb{R}$ and $\mathbb{R}^n$, where we add and scale component by component. Another one is $C[a, b]$, the set of the continuous functions on the interval $[a, b]$, because the sum of two continuous functions and a multiple of one are continuous again. The vectors with n complex components form a vector space as well, which we write as $\mathbb{C}^n$.
- So a vector does not have to be an arrow. A function is one too.

Definition: If V and W are vector spaces with the same operations and $V \subseteq W$, then V is called a **subspace** of W.

- The rules of calculation already hold in W, so for a subset V we only need to check the two closures. Taking $\alpha = 0$ in the second one we have $\vec{0} \in V$, so a set that misses the origin is never a subspace.
- The definition allows $V = W$, so every vector space is a subspace of itself. The smallest subspace we can have is the set $\{\vec{0}\}$.

- Let's look at three sets in $\mathbb{R}^2$. The green line passes through the origin. Adding two of its points or scaling one keeps us on the line, so it is a subspace.
- The blue dashed line does not pass through the origin, so it is not a subspace.
- The red curve passes through the origin but the sum of two of its points does not lie on the curve, so it is not a subspace either.



Definition: For a subspace V and a vector $\vec{x}_0$ the set;

$$\vec{x}_0 + V = \{\vec{x}_0 + \vec{v} : \vec{v} \in V\}$$

is called an **affine space**.

- Meaning we take a subspace and shift it away from the origin by $\vec{x}_0$. The blue dashed line of the sketch is the green one shifted up, so it is an affine space.

Example: Let $\vec{a} \in \mathbb{R}^n$ be a nonzero vector and $b \neq 0$ a number, and let;

$$V = \{\vec{x} \in \mathbb{R}^n : \vec{a}^T\vec{x} = 0\} \qquad \text{and} \qquad A = \{\vec{x} \in \mathbb{R}^n : \vec{a}^T\vec{x} = b\}$$

where $\vec{a}^T\vec{x} = a_1x_1 + \cdots + a_nx_n$.

a) Show that V is a subspace of $\mathbb{R}^n$.

- Let $\vec{x}, \vec{y} \in V$, so $\vec{a}^T\vec{x} = 0$ and $\vec{a}^T\vec{y} = 0$. For the sum and for a multiple we have;

$$\vec{a}^T(\vec{x} + \vec{y}) = \vec{a}^T\vec{x} + \vec{a}^T\vec{y} = 0 + 0 = 0$$
$$\vec{a}^T(\alpha\,\vec{x}) = \alpha\,\vec{a}^T\vec{x} = \alpha \times 0 = 0$$

  so both of them are in V again. Hence; V is a subspace.

b) Show that A is not a subspace of $\mathbb{R}^n$.

- The origin has $\vec{a}^T\vec{0} = 0 \neq b$, so A misses the origin. The closures fail as well. For $\vec{x}, \vec{y} \in A$ we have;

$$\vec{a}^T(\vec{x} + \vec{y}) = \vec{a}^T\vec{x} + \vec{a}^T\vec{y} = b + b = 2b \neq b$$

  since $b \neq 0$. So; A is not closed under addition.

c) Write A as an affine space.

- First we need one point of A. Since $\vec{a} \neq \vec{0}$ it has a component $a_j \neq 0$. Let's take $\vec{x}_0$ with b/a_j in its j-th slot and zeros in the others, so $\vec{a}^T\vec{x}_0 = a_j \times b/a_j = b$ and $\vec{x}_0 \in A$.
- For any $\vec{v} \in V$ we have $\vec{a}^T(\vec{x}_0 + \vec{v}) = b + 0 = b$, so $\vec{x}_0 + \vec{v}$ is in A. The other way around, for any $\vec{x} \in A$ we have $\vec{a}^T(\vec{x} - \vec{x}_0) = b - b = 0$, so $\vec{x} - \vec{x}_0$ is in V. Hence;

$$A = \vec{x}_0 + V$$

- For $\vec{a} = (1, 1)^T$ and $b = 4$ the subspace V is the line $x_1 + x_2 = 0$, and with $j = 1$ we have $A = (4, 0)^T + V$, the line $x_1 + x_2 = 4$. Notice that this A is the boundary line that we walked along in the Walking in a Feasible Set pages. The solutions of a linear equation always form an affine space like this.

Worked Examples

Example: Decide which of the following sets are subspaces of $\mathbb{R}^3$;

$$V_1 = \{\vec{x} : x_1 + x_2 + x_3 = 0\} \qquad V_2 = \{\vec{x} : x_1 + x_2 + x_3 = 1\}$$
$$V_3 = \{\vec{x} : x_1 \geq 0\} \qquad V_4 = \{\vec{x} : x_1 x_2 = 0\}$$

- For each set we check the two closures, and one counter example is enough to say no.
- V_1 is a subspace. If the components of $\vec{x}$ and of $\vec{y}$ add up to zero then the components of $\vec{x} + \vec{y}$ add up to $0 + 0 = 0$ and the components of $\alpha\,\vec{x}$ add up to $\alpha \times 0 = 0$.
- V_2 is not a subspace since it misses the origin, $0 + 0 + 0 \neq 1$. If we shift V_1 by $(1, 0, 0)^T$ we have V_2, so it is an affine space.
- V_3 is not a subspace. The vector $(1, 0, 0)^T$ is in it but its multiple with $\alpha = -1$ is $(-1, 0, 0)^T$, which is not. So; it is closed under addition but not under scalar multiplication.
- V_4 is not a subspace. The vectors $(1, 0, 0)^T$ and $(0, 1, 0)^T$ are in it but their sum $(1, 1, 0)^T$ has $x_1 x_2 = 1 \neq 0$. This one is closed under scalar multiplication but not under addition.
- So one should check both of the closures, a set can pass one of them and fail the other.

Your Turn

Decide for each one of the following sets if it is a subspace, an affine space that is not a subspace, or neither of them. Give a reason, and write the affine space in the form $\vec{x}_0 + V$.

a) $\{\vec{x} \in \mathbb{R}^2 : x_2 = 3x_1\}$

b) $\{\vec{x} \in \mathbb{R}^2 : x_2 = 3x_1 + 1\}$

c) $\{\vec{x} \in \mathbb{R}^2 : x_2 = x_1^2\}$

d) $\{\vec{x} \in \mathbb{R}^3 : x_1 = x_2 \text{ and } x_3 = 0\}$

0.1.3 Span and Linear Independence

- A vector space has infinitely many vectors but we can describe all of them with a few. For a handful of vectors we want to know two things, which vectors we can build from them and whether any one of them is wasted.

Definition: A **linear combination** of the vectors $\vec{x}_1, \ldots, \vec{x}_k$ is a vector of the form;

$$\vec{x} = \sum_{i=1}^{k} \alpha_i \, \vec{x}_i = \alpha_1 \vec{x}_1 + \cdots + \alpha_k \vec{x}_k$$

with numbers α_i. The set of all of their linear combinations is called their **span**. If every vector of a space S can be written this way, we say that the vectors span S and write $\text{span}\{\vec{x}_1, \ldots, \vec{x}_k\} = S$.

Example: Show that $\vec{v}_1 = (1,0)^T$ and $\vec{v}_2 = (1,1)^T$ span $\mathbb{R}^2$, and write $(5,2)^T$ as their linear combination.

- We take any vector $(x_1, x_2)^T$ and look for numbers α and β with;

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \alpha \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha + \beta \\ \beta \end{bmatrix}$$

- The second row gives $\beta = x_2$ and then the first row gives $\alpha = x_1 - x_2$. These numbers exist for every x_1 and x_2, so the two vectors span $\mathbb{R}^2$.

- For $(5,2)^T$ we have $\beta = 2$ and $\alpha = 5 - 2 = 3$ so; $(5,2)^T = 3\,\vec{v}_1 + 2\,\vec{v}_2$.

- The span of a single vector is the line through the origin along it. The span of two vectors that are not on the same line is a plane through the origin, which in $\mathbb{R}^2$ is everything. A span is always a subspace, since sums and multiples of linear combinations are linear combinations again.

Definition: The vectors $\vec{x}_1, \ldots, \vec{x}_k$ are **linearly independent** if;

$$\sum_{i=1}^{k} \alpha_i \, \vec{x}_i = \vec{0} \quad \Rightarrow \quad \alpha_i = 0 \quad \text{for all} \;\; i$$

Otherwise they are called **linearly dependent**.

- Meaning the zero vector can always be built by taking every $\alpha_i = 0$, and independence says that this is the only way to build it.

- To see what this has to do with a wasted vector let's take dependent vectors. Then there is a combination that gives $\vec{0}$ with some $\alpha_j \neq 0$, and we can solve it for $\vec{x}_j$;

$$\alpha_j \, \vec{x}_j = -\sum_{i \neq j} \alpha_i \, \vec{x}_i \quad \Rightarrow \quad \vec{x}_j = \sum_{i \neq j} \beta_i \, \vec{x}_i \qquad \text{with} \qquad \beta_i = -\frac{\alpha_i}{\alpha_j}$$

- So; one of the vectors is a linear combination of the others. It adds nothing to the span, we can throw it away and still build the same vectors. In an independent set there is no such vector.

Example: Let $\vec{v}_1 = (1,0)^T$, $\vec{v}_2 = (1,1)^T$ and $\vec{v}_3 = (2,1)^T$.

a) Show that $\vec{v}_1$ and $\vec{v}_2$ are linearly independent.

- We set a combination of the two to zero;

$$\alpha\,\vec{v}_1 + \beta\,\vec{v}_2 = \begin{bmatrix} \alpha+\beta \\ \beta \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

The second row gives $\beta = 0$ and then the first row gives $\alpha = 0$. There is no other way, so they are linearly independent.

b) Show that the three vectors together are linearly dependent, and write one of them with the others.

- Now we set $\alpha_1\vec{v}_1 + \alpha_2\vec{v}_2 + \alpha_3\vec{v}_3 = \vec{0}$, which is;

$$\begin{aligned} \alpha_1 + \alpha_2 + 2\alpha_3 &= 0 \\ \alpha_2 + \alpha_3 &= 0 \end{aligned}$$

- These are two equations for three unknowns, so we are free to choose one. With $\alpha_3 = 1$ the second row gives $\alpha_2 = -1$ and the first row gives $\alpha_1 = 1 - 2 = -1$. So;

$$-\vec{v}_1 - \vec{v}_2 + \vec{v}_3 = \vec{0} \quad \Rightarrow \quad \vec{v}_3 = \vec{v}_1 + \vec{v}_2$$

which is a combination with nonzero numbers. Hence; they are linearly dependent.

- Notice that this had to happen. The first two vectors already span $\mathbb{R}^2$, so any third vector is a combination of them.

Worked Examples

Example: Let $\vec{v}_1 = (1,1,0)^T$, $\vec{v}_2 = (0,1,1)^T$ and $\vec{v}_3 = (1,0,1)^T$.

a) Show that the vectors are linearly independent.

- The rows of $\alpha_1\vec{v}_1 + \alpha_2\vec{v}_2 + \alpha_3\vec{v}_3 = \vec{0}$ are;

$$\begin{aligned} \alpha_1 + \alpha_3 &= 0 \\ \alpha_1 + \alpha_2 &= 0 \\ \alpha_2 + \alpha_3 &= 0 \end{aligned}$$

- The first two rows give $\alpha_3 = -\alpha_1$ and $\alpha_2 = -\alpha_1$. Then the third row is $-2\alpha_1 = 0$, so $\alpha_1 = 0$ and with it $\alpha_2 = \alpha_3 = 0$. Hence; the vectors are linearly independent.

b) Write $(2,3,1)^T$ as their linear combination.

- The same rows with $(2,3,1)^T$ on the right are;

$$\begin{aligned} \alpha_1 + \alpha_3 &= 2 \\ \alpha_1 + \alpha_2 &= 3 \\ \alpha_2 + \alpha_3 &= 1 \end{aligned}$$

- Adding the three rows we have $2(\alpha_1 + \alpha_2 + \alpha_3) = 6$, so the sum of the three numbers is 3. Subtracting each row from this sum gives;

$$\alpha_2 = 3 - 2 = 1 \qquad \alpha_3 = 3 - 3 = 0 \qquad \alpha_1 = 3 - 1 = 2$$

- So; $(2,3,1)^T = 2\,\vec{v}_1 + \vec{v}_2$. Let's check, $(2,2,0) + (0,1,1) = (2,3,1)$.

Your Turn

Let $\vec{v}_1 = (1, 0, 1)^T$, $\vec{v}_2 = (0, 1, 1)^T$ and $\vec{v}_3 = (1, 1, 2)^T$.

a) Show that $\vec{v}_1$ and $\vec{v}_2$ are linearly independent.

b) Are the three vectors together linearly independent? If not, write one of them with the others.

c) Is $(2, 3, 5)^T$ in the span of $\vec{v}_1$ and $\vec{v}_2$? If yes, write it as their linear combination.

d) Is $(1, 1, 1)^T$ in the span of $\vec{v}_1$ and $\vec{v}_2$?

0.1.4 Basis and Dimension

- A set that spans the space with no vector wasted is the best description of the space we can have. With such a set we can give every vector an address, and the address is unique.

Definition: A set $\mathcal{B} = \{\vec{x}_1, \ldots, \vec{x}_k\}$ of vectors is called a **basis** of V if it is linearly independent and it spans V.

- The vectors $\vec{v}_1 = (1,0)^T$ and $\vec{v}_2 = (1,1)^T$ of the previous subtopic span $\mathbb{R}^2$ and they are linearly independent, so $\{\vec{v}_1, \vec{v}_2\}$ is a basis of $\mathbb{R}^2$.

- The basis of $\mathbb{R}^n$ that we use the most is the **standard basis** $\{\vec{e}_1, \ldots, \vec{e}_n\}$, where $\vec{e}_i$ has a 1 in the ith place and zeros everywhere else. Every vector is $\vec{x} = x_1\vec{e}_1 + \cdots + x_n\vec{e}_n$, so they span $\mathbb{R}^n$, and this sum is $\vec{0}$ only if every x_i is zero, so they are linearly independent.

Remark: A vector space does not have a unique basis. We have just seen two different ones for $\mathbb{R}^2$.

Proposition: If $\mathcal{B} = \{\vec{x}_1, \ldots, \vec{x}_k\}$ is a basis of V, then every $\vec{x} \in V$ can be written as $\vec{x} = \sum_i \alpha_i \vec{x}_i$ in exactly one way.

Proof: Since $\mathcal{B}$ spans V there is at least one way. Suppose that there are two, with the numbers α_i and α_i'.

- Subtracting the two from each other we have;

$$\vec{0} = \vec{x} - \vec{x} = \sum_i \left(\alpha_i - \alpha_i'\right) \vec{x}_i$$

- Since the vectors of $\mathcal{B}$ are linearly independent, all the numbers of this combination are zero;

$$\therefore \quad \alpha_i = \alpha_i' \quad \text{for all} \quad i$$

Definition: The numbers α_i are called the **coordinates** of $\vec{x}$ in the basis $\mathcal{B}$, collected in the vector;

$$[\vec{x}]_{\mathcal{B}} = (\alpha_1, \ldots, \alpha_k)^T$$

- So a vector and its coordinates are two different things. The vector is the point itself, and the coordinates are its address in a chosen basis. The same vector has different coordinates in different bases, and what we usually write as $(x_1, x_2)^T$ is its coordinates in the standard basis.

- To find the coordinates we solve a linear system. Let A be the matrix whose columns are the vectors of the basis. Then a linear combination of them is A times the vector of its numbers, so;

$$A\,[\vec{x}]_{\mathcal{B}} = \vec{x} \quad \Rightarrow \quad [\vec{x}]_{\mathcal{B}} = A^{-1}\,\vec{x}$$

Example: Let $\mathcal{B} = \{\vec{v}_1, \vec{v}_2\}$ with $\vec{v}_1 = (1,0)^T$ and $\vec{v}_2 = (1,1)^T$.

a) Find the coordinates of a vector $\vec{x} = (x_1, x_2)^T$ in $\mathcal{B}$.

- We put the two vectors into the columns of A. A combination with the numbers α_1 and α_2 is then;

$$\alpha_1\vec{v}_1 + \alpha_2\vec{v}_2 = \begin{bmatrix}\alpha_1+\alpha_2\\ \alpha_2\end{bmatrix} = \begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}\begin{bmatrix}\alpha_1\\ \alpha_2\end{bmatrix} = A\,\vec{\alpha}$$

- The determinant of A is 1 and its inverse is;

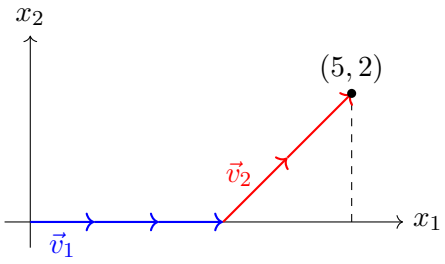
$$A^{-1} = \begin{bmatrix}1 & -1\\ 0 & 1\end{bmatrix} \quad\Rightarrow\quad [\vec{x}]_{\mathcal{B}} = A^{-1}\,\vec{x} = \begin{bmatrix}x_1 - x_2\\ x_2\end{bmatrix}$$

b) Find the coordinates of $(5,2)^T$ in $\mathcal{B}$ and check them.

- For $(5,2)^T$ the coordinates $[\vec{x}]_{\mathcal{B}} = (x_1 - x_2,\ x_2)^T$ from part a give;

$$[\vec{x}]_{\mathcal{B}} = (5-2,\ 2)^T = (3,2)^T$$

- Let's check, $3\,\vec{v}_1 + 2\,\vec{v}_2 = (3,0) + (2,2) = (5,2)$.
- On the sketch we reach the point with three steps of $\vec{v}_1$ and then two steps of $\vec{v}_2$. In the standard basis the same point is five steps of $\vec{e}_1$ and two steps of $\vec{e}_2$, the dashed way.



Theorem: Any two bases of a vector space have the same number of vectors.

Definition: The number of the vectors in a basis of V is called the **dimension** of V, written as $\dim V$.

- I will not prove the theorem here, but without it the dimension would depend on the basis we happen to pick. The standard basis has n vectors so; $\dim \mathbb{R}^n = n$.
- A subspace has a dimension of its own. The line $x_1 + x_2 = 0$ of the Vector Spaces and Subspaces pages is spanned by the single vector $(1,-1)^T$, so its dimension is 1.

Remark: A basis is a *maximal* independent set. If we add any other vector of V to it, the new vector is already a combination of the basis and the set becomes dependent. A basis is also a *minimal* spanning set. If we remove a vector from it, the removed one can not be built from the rest and the set stops spanning V.

- So; in a space of dimension n more than n vectors are always linearly dependent and fewer than n vectors never span the space.

Worked Examples

Example: Let $V = \{\vec{x} \in \mathbb{R}^3 : x_1 + x_2 + x_3 = 0\}$.

a) Find a basis of V and its dimension.

- The equation lets us choose x_1 and x_2 freely and then fixes $x_3 = -x_1 - x_2$. So every vector of V is;

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ -x_1 - x_2 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$$

- So the vectors $(1, 0, -1)^T$ and $(0, 1, -1)^T$ span V. They are linearly independent, because the first two rows of the combination above are x_1 and x_2 themselves, so it is $\vec{0}$ only for $x_1 = x_2 = 0$.

- Hence; these two vectors are a basis of V and $\dim V = 2$. The set V is a plane through the origin in $\mathbb{R}^3$.

b) Find the coordinates of $(2, 3, -5)^T$ in this basis.

- From the form $\vec{x} = x_1(1, 0, -1)^T + x_2(0, 1, -1)^T$ of part a we read the coordinates directly, they are the first two components, $(2, 3)^T$. Let's check the third one, $-2 - 3 = -5$.

Your Turn

Let $V = \{\vec{x} \in \mathbb{R}^3 : x_1 = 2x_2\}$.

a) Find a basis of V and its dimension.

b) Find the coordinates of $(4, 2, 7)^T$ in your basis.

c) Is $\{(2, 1, 0)^T,\ (0, 0, 1)^T,\ (2, 1, 1)^T\}$ a basis of V?

0.1.5 Metric and Norm

- The points $\vec{x}_0, \vec{x}_1, \vec{x}_2, \dots$ of a walk should get close to the minimizer, and a step should have a length. Both of these need a way to measure. There are two tools for it that look alike, the distance between two points and the length of one vector.

Definition: A function d that takes two points x, y of a set A and returns a number $d(x, y) \in \mathbb{R}$ is called a **metric** or a **distance** on A if for all $x, y, z \in A$;

1) $d(x, y) \geq 0$ and $d(x, y) = 0 \iff x = y$

2) $d(x, y) = d(y, x)$

3) $d(x, y) \leq d(x, z) + d(z, y)$

- Meaning a distance is never negative and we are at the distance zero only from the point itself, the way there is as long as the way back, and we never find a shortcut by a detour over a third point z. We call the third rule the **triangle inequality**.

Definition: A **norm** on a vector space V is a function $\|\cdot\| : V \to \mathbb{R}$, $\vec{x} \mapsto \|\vec{x}\|$ which satisfies for all $\vec{x}, \vec{y} \in V$ and $\alpha \in \mathbb{R}$;

1) $\|\vec{x}\| \geq 0$ and $\|\vec{x}\| = 0 \iff \vec{x} = \vec{0}$

2) $\|\alpha\, \vec{x}\| = |\alpha|\, \|\vec{x}\|$

3) $\|\vec{x} + \vec{y}\| \leq \|\vec{x}\| + \|\vec{y}\|$

The number $\|\vec{x}\|$ is also called the **length** of the vector.

- So what is the difference between the two? With a norm we measure one vector, its length, which is how far it is from the origin. For this we need a vector space, because the second rule scales a vector and the third one adds two of them.

- With a metric we measure the gap between two points and we never add or scale them. So; it makes sense on any set, even on one that is not a vector space.

- Coming from physics the two look like the same thing, because there the distance between two points is always the length of the vector from one to the other. This is exactly how a norm produces a metric.

Remark: Every norm gives a metric on its vector space;

$$d(\vec{x}, \vec{y}) = \|\vec{x} - \vec{y}\|$$

- Let's check the three rules of a metric. The first one is the first rule of the norm for the vector $\vec{x} - \vec{y}$. For the second one we use the scaling with $\alpha = -1$, and for the third one we add and subtract $\vec{z}$;

$$\|\vec{y} - \vec{x}\| = \|(-1)(\vec{x} - \vec{y})\| = |-1|\, \|\vec{x} - \vec{y}\| = \|\vec{x} - \vec{y}\|$$
$$\|\vec{x} - \vec{y}\| = \|(\vec{x} - \vec{z}) + (\vec{z} - \vec{y})\| \leq \|\vec{x} - \vec{z}\| + \|\vec{z} - \vec{y}\|$$

  where the last two lengths are $d(\vec{x}, \vec{z})$ and $d(\vec{z}, \vec{y})$.

- The other way around does not hold. There are metrics that come from no norm, and the following one is the simplest.

Example: For the points of $\mathbb{R}^2$ let $d(\vec{x},\vec{y}) = 1$ if $\vec{x} \neq \vec{y}$ and $d(\vec{x},\vec{y}) = 0$ if $\vec{x} = \vec{y}$.

a) Show that d is a metric.

- The values of d are 0 and 1, it is 0 only for equal points and swapping the points changes nothing. So the first two rules hold.
- For the triangle inequality let $\vec{x} \neq \vec{y}$, otherwise the left hand side is 0 and there is nothing to show. A third point $\vec{z}$ can not be equal to both of them, so at least one of $d(\vec{x},\vec{z})$ and $d(\vec{z},\vec{y})$ is 1 and;

$$d(\vec{x},\vec{y}) = 1 \leq d(\vec{x},\vec{z}) + d(\vec{z},\vec{y})$$

b) Show that d does not come from a norm.

- Suppose that $d(\vec{x},\vec{y}) = \|\vec{x} - \vec{y}\|$ for some norm and take two points with $\vec{x} \neq \vec{y}$. By the scaling rule of the norm we would have;

$$d(2\vec{x},2\vec{y}) = \|2\vec{x} - 2\vec{y}\| = 2\,\|\vec{x} - \vec{y}\| = 2\,d(\vec{x},\vec{y}) = 2$$

- However; $2\vec{x} \neq 2\vec{y}$ so by its definition $d(2\vec{x},2\vec{y}) = 1$. So; there is no such norm.

- This metric only tells whether two points are the same or not, and stretching the plane does not change its distances. A norm always feels the stretch. In this course we work in $\mathbb{R}^n$ and we measure with a norm, so the distance between two points is the norm of their difference.

Worked Examples

Example: Show that $\|\vec{x}\| = |x_1| + |x_2|$ is a norm on $\mathbb{R}^2$.

- We check the three rules one by one. For numbers we already know $|a+b| \leq |a| + |b|$ and $|\alpha a| = |\alpha|\,|a|$.
- An absolute value is never negative, so $\|\vec{x}\| \geq 0$. A sum of two such numbers is zero only if both of them are zero, which is $x_1 = x_2 = 0$. So; $\|\vec{x}\| = 0 \iff \vec{x} = \vec{0}$.
- For the scaling we have;

$$\|\alpha\,\vec{x}\| = |\alpha x_1| + |\alpha x_2| = |\alpha|\,|x_1| + |\alpha|\,|x_2| = |\alpha|\,\|\vec{x}\|$$

- For the triangle inequality we use the rule for numbers in each component;

$$\begin{aligned} \|\vec{x} + \vec{y}\| &= |x_1 + y_1| + |x_2 + y_2| \\ &\leq |x_1| + |y_1| + |x_2| + |y_2| \\ &= \|\vec{x}\| + \|\vec{y}\| \end{aligned}$$

- All three rules hold. Hence; it is a norm.

Your Turn

a) Is $g(\vec{x}) = |x_1 - x_2|$ a norm on $\mathbb{R}^2$?

b) Show that $h(\vec{x}) = w_1|x_1| + w_2|x_2|$ is a norm on $\mathbb{R}^2$ for any weights $w_1, w_2 > 0$. In which rules do you use that the weights are positive, and which rule breaks if one of them is zero?

c) Take the weights $w_1 = 2$ and $w_2 = 3$. Find the distance between $\vec{x} = (1, 1)^T$ and $\vec{y} = (3, 0)^T$ with the metric $d(\vec{x}, \vec{y}) = h(\vec{x} - \vec{y})$.

d) Check the triangle inequality of this metric for the two points of part c over the point $\vec{z} = (3, 1)^T$.

0.1.6 L_p Norms and Unit Balls

- There is more than one way to measure the length of a vector, and the choice changes the answer of an optimization problem. We have used three of them in the Optimization Problems pages. Here is the family they belong to.

Definition: For a number $p \geq 1$ the L_p **norm** of $\vec{x} \in \mathbb{R}^n$ is;

$$\|\vec{x}\|_p = \left(\sum_{i=1}^{n} |x_i|^p \right)^{1/p}$$

- The members that we use are;

$$\|\vec{x}\|_1 = \sum_{i=1}^{n} |x_i| \qquad \|\vec{x}\|_2 = \left(\sum_{i=1}^{n} x_i^2 \right)^{1/2} \qquad \|\vec{x}\|_\infty = \max_{1 \leq i \leq n} |x_i|$$

  where the L_2 norm is the usual length, also called the **Euclidean norm**. The L_∞ norm is what the L_p norm becomes as $p \to \infty$, because for a large p the largest component takes over the whole sum.

Example: Let $\vec{x} = (3, -4)^T$. Find $\|\vec{x}\|_1$, $\|\vec{x}\|_2$, $\|\vec{x}\|_3$ and $\|\vec{x}\|_\infty$.

- Putting the components into the definitions we have;

$$\begin{aligned} \|\vec{x}\|_1 &= 3 + 4 = 7 \\ \|\vec{x}\|_2 &= (9 + 16)^{1/2} = 5 \\ \|\vec{x}\|_3 &= (27 + 64)^{1/3} = 91^{1/3} \approx 4.50 \\ \|\vec{x}\|_\infty &= \max\{3, 4\} = 4 \end{aligned}$$

- Notice that the norm gets smaller as p grows and it comes down towards the largest component.

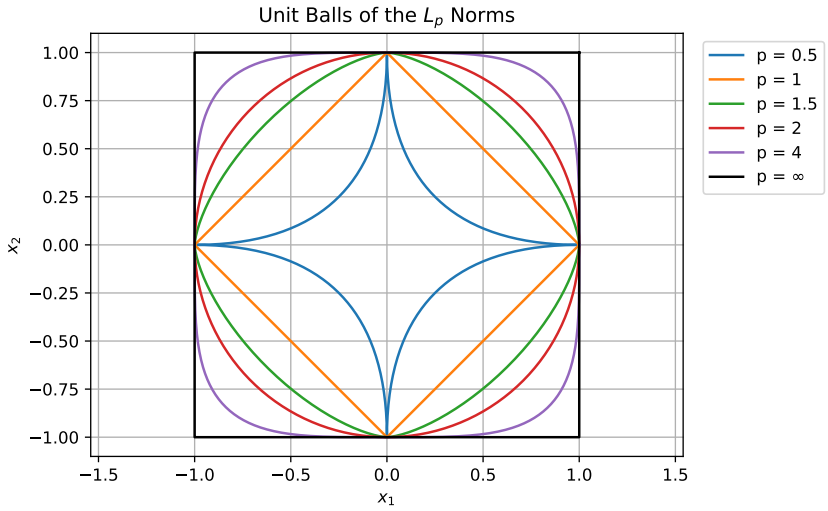
- The number p does not have to be a natural number, with every real $p \geq 1$ we have a norm. Below 1 we lose the triangle inequality. For $p = 1/2$ with $\vec{x} = (1, 0)^T$ and $\vec{y} = (0, 1)^T$ both vectors have the length 1, but their sum has $(1 + 1)^2 = 4 > 1 + 1$.

- With n components there is always a largest one, so we write max. A function $f \in C[a, b]$ is like a vector with infinitely many components, its values, and there the largest one may not be reached. So for functions we write the same norm with a supremum, $\|f\|_\infty = \sup |f(x)|$. In $\mathbb{R}^n$ the two mean the same thing.

Definition: The **unit ball** of a norm is the set of the vectors whose length is at most 1;

$$B = \{\vec{x} \in \mathbb{R}^n : \|\vec{x}\| \leq 1\}$$

- Every norm has its own unit ball. The edges $\|\vec{x}\|_p = 1$ of the unit balls in $\mathbb{R}^2$ are plotted for several p in the following graph as;



- For $p = 2$ the ball is $x_1^2 + x_2^2 \leq 1$, the usual disc. For $p = 1$ it is $|x_1| + |x_2| \leq 1$, a square standing on its corner with the corners at ± 1 on the axes.

- For $p = \infty$ it is $\max\{|x_1|, |x_2|\} \leq 1$. The larger one of two numbers is at most 1 exactly when both of them are, so this is $|x_1| \leq 1$ and $|x_2| \leq 1$ together, the full square with the corners $(\pm 1, \pm 1)$.

- As p grows the ball grows from the standing square through the disc to the full square. The curve of $p = 0.5$ bends inwards, so a straight walk between two points of its inside leaves it. This is the picture of the broken triangle inequality, and it is not the ball of a norm.

- With a ball we can finally say what bounded means in $\mathbb{R}^n$. A set $A \subseteq \mathbb{R}^n$ is **bounded** if there is a number M with $\|\vec{x}\| \leq M$ for all $\vec{x} \in A$, meaning the set fits into a ball of radius M.

Remark: Any two norms $\|\cdot\|_a$ and $\|\cdot\|_b$ on $\mathbb{R}^n$ are **equivalent**, in the sense that there are numbers $\alpha, \beta > 0$ with;

$$\alpha \|\vec{x}\|_a \leq \|\vec{x}\|_b \leq \beta \|\vec{x}\|_a \qquad \text{for all } \vec{x} \in \mathbb{R}^n$$

- Meaning we can always fit a ball of one norm into a larger ball of the other one. So; a set that is bounded in one norm is bounded in all of them, and points that get close to each other in one norm get close in all of them. In $\mathbb{R}^n$ we can say bounded or close without naming the norm, only the numbers differ.

- For our three norms the numbers are;

$$\|\vec{x}\|_\infty \leq \|\vec{x}\|_2 \leq \|\vec{x}\|_1 \leq n \|\vec{x}\|_\infty$$

  which we can read from the graph, since a smaller norm has a larger ball. For $\vec{x} = (3, -4)^T$ this is $4 \leq 5 \leq 7 \leq 8$.

Worked Examples

Example: Find the point of the line $x_1 + 2x_2 = 4$ that is closest to the origin when the length is measured with;

- On the line we have $x_1 = 4 - 2x_2$, so every point of it is $(4 - 2x_2,\ x_2)$ and we are left with the single variable x_2.

a) the L_2 norm,

- A squared length has the same minimizer as the length, so we minimize;

$$\begin{aligned} g(x_2) &= (4 - 2x_2)^2 + x_2^2 \\ &= 16 - 16x_2 + 5x_2^2 \end{aligned}$$

- Setting $g'(x_2) = -16 + 10x_2 = 0$ we have $x_2 = 8/5$ and $x_1 = 4 - 16/5 = 4/5$.
- So; the closest point is $(4/5,\ 8/5)$ with the length $(16/25 + 64/25)^{1/2} = (16/5)^{1/2} \approx 1.79$.

b) the L_1 norm,

- Now we minimize $|4 - 2x_2| + |x_2|$. The absolute values change their form at $x_2 = 0$ and at $x_2 = 2$, so we look at the three ranges;

$$|4 - 2x_2| + |x_2| = \begin{cases} 4 - 3x_2 & \text{for } x_2 < 0 \\ 4 - x_2 & \text{for } 0 \le x_2 \le 2 \\ 3x_2 - 4 & \text{for } x_2 > 2 \end{cases}$$

- It falls until $x_2 = 2$ and rises after it. So; the closest point is $(0, 2)$ with the length 2.

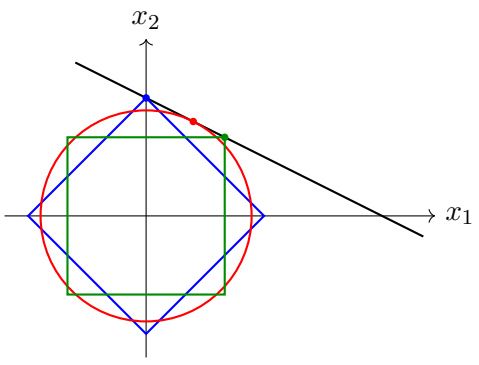
c) the L_∞ norm.

- Now we minimize $\max\{|4 - 2x_2|,\ |x_2|\}$. Between $x_2 = 0$ and $x_2 = 2$ the first number falls and the second one rises, so the larger one of them is lowest where they are equal;

$$4 - 2x_2 = x_2 \quad \Rightarrow \quad x_2 = \frac{4}{3} \quad \Rightarrow \quad x_1 = 4 - \frac{8}{3} = \frac{4}{3}$$

So; the closest point is $(4/3,\ 4/3)$ with the length $4/3 \approx 1.33$.

- These are three different answers to the same question. On the sketch each one is the point where the ball of its norm, grown from the origin, first touches the line, the red disc, the blue standing square and the green full square.
- Notice that the L_1 answer has a zero component, because its ball touches the line with a corner and the corners are on the axes. This is why the Lasso penalty of the Least Squares Data Fitting pages sets some of the parameters exactly to zero.



Your Turn

a) Find $\|\vec{x}\|_1$, $\|\vec{x}\|_2$ and $\|\vec{x}\|_\infty$ for $\vec{x} = (2, -1, -2)^T$ and check the chain of inequalities between them.

b) Describe the set $\{\vec{x} \in \mathbb{R}^2 : \|\vec{x} - (2, 1)^T\|_\infty \leq 1\}$ with its corners.

c) Describe the set $\{\vec{x} \in \mathbb{R}^2 : \|\vec{x}\|_1 \leq 2\}$ with its corners.

d) Show with $\vec{x} = (4, 0)^T$ and $\vec{y} = (0, 9)^T$ that $p = 1/2$ breaks the triangle inequality.

0.1.7 Inner Product

- A norm gives lengths but not directions. To say that a step points downhill we need the angle between the step and the gradient, and angles come from a product of two vectors.

Definition: The **inner product** of $\vec{x}, \vec{y} \in \mathbb{R}^n$, also called the dot product, is the number;

$$\langle \vec{x}, \vec{y} \rangle = \vec{x}^T \vec{y} = \begin{bmatrix} x_1 & \dots & x_n \end{bmatrix} \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \sum_{i=1}^{n} x_i \, y_i$$

- We also write it as $\vec{x} \cdot \vec{y}$. The dot product has three properties, and every product with the same three behaves like it, so we take them as the definition of an inner product on any vector space;

Definition: An **inner product** on a vector space V is a function $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{R}$ that gives a number for every pair of vectors such that, for all $\vec{x}, \vec{y}, \vec{z} \in V$ and all numbers α, β;

1) $\langle \vec{x}, \vec{x} \rangle \geq 0$, and $\langle \vec{x}, \vec{x} \rangle = 0$ only for $\vec{x} = \vec{0}$ (positivity), meaning every vector except the zero vector has a positive length.

2) $\langle \vec{x}, \vec{y} \rangle = \langle \vec{y}, \vec{x} \rangle$ (symmetry), meaning the order of the two vectors does not matter.

3) $\langle \alpha\vec{x} + \beta\vec{y}, \vec{z} \rangle = \alpha\langle \vec{x}, \vec{z} \rangle + \beta\langle \vec{y}, \vec{z} \rangle$ (linearity in the first slot), meaning sums and multiples can be taken out of the product.

- With the symmetry the linearity holds in the second slot too, so we can open a product of two sums term by term in both slots. This is why the product is also called a bilinear form.

Definition: A vector space V together with an inner product on it, written as the pair $(V, \langle \cdot, \cdot \rangle)$, is called an **inner product space**.

- The dot product is our example. The order of $\vec{x}$ and $\vec{y}$ in $\sum_i x_i \, y_i$ does not matter, sums and multiples come out of the sum, and $\langle \vec{x}, \vec{x} \rangle = \sum_i x_i^2$ is a sum of squares, which is zero only when every x_i is zero. So; $(\mathbb{R}^n, \vec{x}^T \vec{y})$ is an inner product space, and in this course the inner product of $\mathbb{R}^n$ is the dot product unless said otherwise. It is not the only one, we will meet an inner product of functions in the Gram-Schmidt pages.

Remark: In $\mathbb{C}^n$ the numbers are complex and the symmetry turns into the conjugate symmetry $\langle \vec{x}, \vec{y} \rangle = \overline{\langle \vec{y}, \vec{x} \rangle}$. With $\vec{y} = \vec{x}$ it says that $\langle \vec{x}, \vec{x} \rangle$ is equal to its own conjugate, so it stays a real number and the positivity still makes sense. The dot product becomes;

$$\langle \vec{x}, \vec{y} \rangle = \vec{y}^{\dagger} \vec{x} = \sum_{i=1}^{n} x_i \, \bar{y}_i$$

where $\vec{y}^{\dagger} = \bar{\vec{y}}^{T}$ is the conjugate transpose, the conjugate of every entry and then the transpose. The linearity stays in the first slot, and in the second slot the numbers come out conjugated;

$$\begin{aligned} \langle \vec{x}, \alpha \vec{y} + \vec{z} \rangle &= \overline{\langle \alpha \vec{y} + \vec{z}, \vec{x} \rangle} \\ &= \overline{\alpha \langle \vec{y}, \vec{x} \rangle + \langle \vec{z}, \vec{x} \rangle} \\ &= \bar{\alpha} \, \overline{\langle \vec{y}, \vec{x} \rangle} + \overline{\langle \vec{z}, \vec{x} \rangle} \\ &= \bar{\alpha} \langle \vec{x}, \vec{y} \rangle + \langle \vec{x}, \vec{z} \rangle \end{aligned}$$

which is called conjugate linearity. Physics books put the conjugate on the first slot instead, $\langle \vec{x} | \vec{y} \rangle = \vec{x}^{\dagger} \vec{y}$, so there the two slots swap their roles.

- Geometrically we can write the inner product as the product of the two lengths and the cosine of the angle θ between the vectors;

$$\langle \vec{x}, \vec{y} \rangle = \|\vec{x}\|_2 \, \|\vec{y}\|_2 \cos\theta \quad \Rightarrow \quad \cos\theta = \frac{\langle \vec{x}, \vec{y} \rangle}{\|\vec{x}\|_2 \, \|\vec{y}\|_2}$$

- This is how we find angles in this course. It takes three sums and no drawing, and it works in any dimension.
- For the fraction to be a cosine we need it to stay between -1 and 1, which is what the following inequality says;

Theorem: For all $\vec{x}, \vec{y} \in \mathbb{R}^n$ we have the **Cauchy Schwarz inequality**;

$$|\langle \vec{x}, \vec{y} \rangle| \leq \|\vec{x}\|_2 \, \|\vec{y}\|_2$$

and the two sides are equal exactly when the vectors are parallel, one a multiple of the other.

Proof: If $\vec{y} = \vec{0}$ both sides are zero and there is nothing to show, so let $\vec{y} \neq \vec{0}$. Then $\langle \vec{y}, \vec{y} \rangle > 0$ by the positivity.

- For any number t the product of $\vec{x} - t\,\vec{y}$ with itself is not negative. Opening it with the linearity in both slots and joining the two middle terms with the symmetry we have;

$$0 \leq \langle \vec{x} - t\,\vec{y}, \ \vec{x} - t\,\vec{y} \rangle = \langle \vec{x}, \vec{x} \rangle - 2t \, \langle \vec{x}, \vec{y} \rangle + t^2 \langle \vec{y}, \vec{y} \rangle$$

- The right hand side is a parabola in t that never goes below zero, so we take the t at its bottom, $t = \langle \vec{x}, \vec{y} \rangle / \langle \vec{y}, \vec{y} \rangle$, where its derivative $-2\langle \vec{x}, \vec{y} \rangle + 2t \langle \vec{y}, \vec{y} \rangle$ is zero. Putting it in we have;

$$\begin{aligned} 0 &\leq \langle \vec{x}, \vec{x} \rangle - 2 \frac{\langle \vec{x}, \vec{y} \rangle^2}{\langle \vec{y}, \vec{y} \rangle} + \frac{\langle \vec{x}, \vec{y} \rangle^2}{\langle \vec{y}, \vec{y} \rangle} \\ &= \langle \vec{x}, \vec{x} \rangle - \frac{\langle \vec{x}, \vec{y} \rangle^2}{\langle \vec{y}, \vec{y} \rangle} \end{aligned}$$

- We multiply with $\langle \vec{y}, \vec{y} \rangle > 0$ and use $\langle \vec{x}, \vec{x} \rangle = \sum_i x_i^2 = \|\vec{x}\|_2^2$, which gives $\langle \vec{x}, \vec{y} \rangle^2 \leq \|\vec{x}\|_2^2 \, \|\vec{y}\|_2^2$. Taking the square root of both sides;

$$\therefore \quad |\langle \vec{x}, \vec{y} \rangle| \leq \|\vec{x}\|_2 \, \|\vec{y}\|_2$$

- The two sides are equal exactly when the first line is an equality, $\langle \vec{x} - t\,\vec{y}, \vec{x} - t\,\vec{y} \rangle = 0$, which by the positivity means $\vec{x} = t\,\vec{y}$. Notice that the proof only used the three properties, so the inequality holds in every inner product space.

Definition: Two vectors are called **orthogonal** if $\langle \vec{x}, \vec{y} \rangle = 0$, written as $\vec{x} \perp \vec{y}$.

- Meaning the angle between them is 90°. From the sign of the inner product we read the rest. It is positive when the angle is below 90° and negative when it is above.

Theorem: If $\vec{x} \perp \vec{y}$ then $\|\vec{x} + \vec{y}\|_2^2 = \|\vec{x}\|_2^2 + \|\vec{y}\|_2^2$, which is the **Pythagorean theorem**.

Proof: We write the squared length as the product of the vector with itself and open it with the linearity;

$$\begin{aligned} \|\vec{x} + \vec{y}\|_2^2 &= \langle \vec{x} + \vec{y},\ \vec{x} + \vec{y} \rangle \\ &= \langle \vec{x}, \vec{x} \rangle + 2\langle \vec{x}, \vec{y} \rangle + \langle \vec{y}, \vec{y} \rangle \end{aligned}$$

and the middle term is zero for orthogonal vectors;

$$\therefore \quad \|\vec{x} + \vec{y}\|_2^2 = \|\vec{x}\|_2^2 + \|\vec{y}\|_2^2$$

The same holds for $\vec{x} - \vec{y}$, because $-\vec{y}$ is orthogonal to $\vec{x}$ as well.

- Notice that we have used twice that the inner product of a vector with itself is the square of its length, $\langle \vec{x}, \vec{x} \rangle = \sum_i x_i^2 = \|\vec{x}\|_2^2$, so we call the Euclidean norm the **norm induced by the inner product**.

- Then every inner product space is a normed space with $\|\vec{x}\| = \langle \vec{x}, \vec{x} \rangle^{1/2}$, and we have seen that every normed space is a metric space with $d(\vec{x}, \vec{y}) = \|\vec{x} - \vec{y}\|$;

$$\text{Inner Product Space} \ \Rightarrow\ \text{Normed Space} \ \Rightarrow\ \text{Metric Space}$$

- A metric only has distances, a norm adds lengths to them and an inner product adds angles. None of the arrows can be turned around. The L_1 and the L_∞ norms for example come from no inner product, which I will not show here.

Example: In the Descent Directions pages we had $\nabla f = (4,2)^T$ at the point $\vec{x}_k = (1,2)^T$, and the step that goes directly to the minimizer was $\vec{p} = (-1,-2)^T$.

a) Find the angle between ∇f and $\vec{p}$.

- The inner product and the two lengths are;

$$\langle \nabla f, \vec{p} \rangle = -4 - 4 = -8 \qquad \|\nabla f\|_2 = (16+4)^{1/2} = 20^{1/2} \qquad \|\vec{p}\|_2 = (1+4)^{1/2} = 5^{1/2}$$

- The product of the lengths is $(20 \times 5)^{1/2} = 100^{1/2} = 10$, so;

$$\cos\theta = \frac{-8}{10} = -\frac{4}{5} \quad \Rightarrow \quad \theta \approx 143^\circ$$

b) Find the angle between ∇f and the steepest descent direction $-\nabla f$, for any gradient $\vec{g} = \nabla f \neq \vec{0}$.

- With the linearity we take the minus out of the product, $\langle \vec{g}, -\vec{g} \rangle = -\langle \vec{g}, \vec{g} \rangle = -\|\vec{g}\|_2^2$, and $-\vec{g}$ has the same length as $\vec{g}$. So;

$$\cos\theta = \frac{-\|\vec{g}\|_2^2}{\|\vec{g}\|_2 \, \|\vec{g}\|_2} = -1 \quad \Rightarrow \quad \theta = 180^\circ$$

- So the numbers of $\nabla f = (4,2)^T$ do not matter here. At every point the steepest descent direction points exactly opposite to the gradient.

c) Which vectors are orthogonal to ∇f?

- We need $\langle \nabla f, \vec{p} \rangle = 4p_1 + 2p_2 = 0$, which is $p_2 = -2p_1$. So; they are the multiples of $(1,-2)^T$, and on the sketch of those pages this is the dashed tangent of the level curve.

- So the condition $\nabla f^T \vec{p} < 0$ of a descent direction says that $\vec{p}$ makes more than 90° with the gradient. The steepest one is at 180°, the direct step is at 143° and the tangent at 90° is where the descent ends.

Worked Examples

Example: Let $\vec{x} = (1, 2, 2)^T$ and $\vec{y} = (3, 0, 4)^T$. Find $\langle \vec{x}, \vec{y} \rangle$, the lengths of the two vectors and the angle between them.

- The inner product is the sum of the products of the components;

$$\langle \vec{x}, \vec{y} \rangle = 1 \times 3 + 2 \times 0 + 2 \times 4 = 3 + 0 + 8 = 11$$

- The lengths are;

$$\|\vec{x}\|_2 = (1 + 4 + 4)^{1/2} = 3 \qquad \|\vec{y}\|_2 = (9 + 0 + 16)^{1/2} = 5$$

- Then the angle is;

$$\cos\theta = \frac{11}{3 \times 5} = \frac{11}{15} \approx 0.73 \quad \Rightarrow \quad \theta \approx 43^\circ$$

- Notice that $11 \leq 15$, as the Cauchy Schwarz inequality says. If the fraction ever comes out above 1 there is a mistake in the calculation.

Your Turn

Let $\vec{x} = (2, 1, -2)^T$, $\vec{y} = (1, 2, 2)^T$ and $\vec{z} = (4, 0, -3)^T$.

a) Find $\langle \vec{x}, \vec{y} \rangle$ and the angle between $\vec{x}$ and $\vec{y}$.

b) Show that $\|\vec{x} + \vec{y}\|_2^2 = \|\vec{x}\|_2^2 + \|\vec{y}\|_2^2$.

c) Find the cosine of the angle between $\vec{x}$ and $\vec{z}$, and the angle.

d) At some point $\vec{x}_k$ the gradient of a function is $\nabla f(\vec{x}_k) = \vec{x}$. Which of the vectors $\vec{y}$ and $-\vec{z}$ are descent directions at that point?

Your Turn

Let $w_1, \ldots, w_n$ be positive numbers and $W = \text{diag}(w_1, \ldots, w_n)$ the diagonal matrix with them on its diagonal. We define the weighted product and the length it gives as;

$$\langle \vec{x}, \vec{y} \rangle_W = \vec{x}^T W \vec{y} = \sum_{i=1}^{n} w_i \, x_i \, y_i \qquad\qquad \|\vec{x}\|_W = \langle \vec{x}, \vec{x} \rangle_W^{1/2}$$

a) Show that $\langle \cdot, \cdot \rangle_W$ has the three properties of an inner product.

b) In $\mathbb{R}^2$ let $W = \text{diag}(1, 3)$. Find the angle between $\vec{x} = (1, 0)^T$ and $\vec{y} = (1, 1)^T$ with this inner product, and compare it with the angle the dot product gives.

c) Show that $(3, 1)^T$ and $(1, -1)^T$ are orthogonal with $W = \text{diag}(1, 3)$ but not with the dot product.

d) Why do the weights have to be positive? Find a nonzero vector that breaks the positivity for $W = \text{diag}(1, -1)$.

0.1.8 Orthogonality and Projection

- In the Basis pages the coordinates of a vector cost us a linear system, $[\vec{x}]_{\mathcal{B}} = A^{-1}\vec{x}$. In a basis whose vectors are orthogonal to each other we get the coordinates for free, one inner product each and no system to solve. We will see that they are the shadows of the vector on the basis vectors, its projections.
- From here on I drop the index of the norm. $\|\vec{x}\| = \langle \vec{x}, \vec{x} \rangle^{1/2}$ is the norm induced by the inner product, which in $\mathbb{R}^n$ is the Euclidean norm $\|\vec{x}\|_2$.

0.1.8.1 Orthogonal Subspaces

Definition: Two subspaces S and T of V are called **orthogonal** to each other if every vector of one is orthogonal to every vector of the other, $\langle \vec{x}, \vec{y} \rangle = 0$ for all $\vec{x} \in S$ and $\vec{y} \in T$.

- In $\mathbb{R}^3$ the x_3-axis is orthogonal to the x_1x_2-plane. If we take a vector $(0, 0, c)^T$ of the axis and a vector $(a, b, 0)^T$ of the plane, we get the inner product $0 + 0 + 0 = 0$.

Definition: Let S be a subspace of V. The set of all the vectors that are orthogonal to every vector of S is called the **orthogonal complement** of S;

$$S^{\perp} = \{\vec{x} \in V : \langle \vec{x}, \vec{y} \rangle = 0 \ \text{ for all } \ \vec{y} \in S\}$$

- Meaning $S^{\perp}$ holds every vector that stands at a right angle to all of S. It is a subspace again. If we take $\vec{x}$ and $\vec{z}$ both orthogonal to a $\vec{y} \in S$, then by the linearity so is any $\alpha\vec{x} + \vec{z}$, since $\langle \alpha\vec{x} + \vec{z}, \vec{y} \rangle = \alpha \times 0 + 0 = 0$.
- For example in $\mathbb{R}^2$ let $S = \text{span}\{(1, 1)^T\}$. We need $\langle \vec{x}, c\,(1, 1)^T \rangle = c\,(x_1 + x_2) = 0$ for every c, so $S^{\perp}$ is the line $x_1 + x_2 = 0$ of the Vector Spaces and Subspaces pages, spanned by $(1, -1)^T$. In $\mathbb{R}^3$ the orthogonal complement of the x_1x_2-plane is the whole x_3-axis.
- We will see in the Four Subspaces pages that the subspaces a matrix makes come in such pairs.

0.1.8.2 Orthogonal Bases

Definition: A set of vectors $\{\vec{x}_1, \ldots, \vec{x}_k\}$ is called **mutually orthogonal**, or an orthogonal set, if $\langle \vec{x}_i, \vec{x}_j \rangle = 0$ for all $i \neq j$. If further every vector has the length $\|\vec{x}_i\| = 1$, the set is called **orthonormal**.

- I will write orthonormal vectors as $\vec{q}_i$. Their inner products are 1 for $i = j$ and 0 for $i \neq j$, which we write short with the Kronecker delta;

$$\langle \vec{q}_i, \vec{q}_j \rangle = \delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

- So orthonormal is orthogonal plus unit length. Every result of these pages holds for both kinds of sets, and the unit length only drops the denominators $\|\vec{x}_i\|^2$ from our formulas. We make an orthogonal set of nonzero vectors orthonormal by dividing every vector by its length, $\vec{q}_i = \vec{x}_i / \|\vec{x}_i\|$.

Lemma: A mutually orthogonal set of nonzero vectors $\vec{x}_1, \ldots, \vec{x}_k$ is linearly independent.

Proof: Let $\sum_j \alpha_j \vec{x}_j = \vec{0}$. We need to show that every α_i is zero.

- We take the inner product of both sides with one of the vectors, $\vec{x}_i$. The right hand side gives $\langle \vec{0}, \vec{x}_i \rangle = 0$. On the left the linearity takes the sum and the numbers out, and since the set is orthogonal every term with $j \neq i$ dies;

$$\Big\langle \sum_j \alpha_j \vec{x}_j,\ \vec{x}_i \Big\rangle = \sum_j \alpha_j \langle \vec{x}_j, \vec{x}_i \rangle = \alpha_i \langle \vec{x}_i, \vec{x}_i \rangle = \alpha_i \, \|\vec{x}_i\|^2 = 0$$

- Now we may divide by $\|\vec{x}_i\|^2$, because $\vec{x}_i \neq \vec{0}$ and by the positivity of the inner product $\langle \vec{x}_i, \vec{x}_i \rangle = \|\vec{x}_i\|^2 > 0$. This holds for every i;

$$\therefore \quad \alpha_i = 0 \quad \text{for all} \quad i$$

- The nonzero is needed. The zero vector is orthogonal to everything, so we can add it to an orthogonal set, $\{(1,0)^T, \vec{0}\}$ for example, but then the set is dependent since $0 \times (1,0)^T + 1 \times \vec{0} = \vec{0}$.

- So n mutually orthogonal nonzero vectors of $\mathbb{R}^n$ are independent, and since one more vector would make them dependent they are a basis, which we call an **orthogonal basis**. With unit vectors it is an **orthonormal basis**, like the standard basis.

Proposition: Let $\{\vec{x}_1, \ldots, \vec{x}_n\}$ be an orthogonal basis of $\mathbb{R}^n$. Then every $\vec{x} \in \mathbb{R}^n$ is;

$$\vec{x} = \sum_{i=1}^{n} \alpha_i \, \vec{x}_i \qquad \text{with} \qquad \alpha_i = \frac{\langle \vec{x}, \vec{x}_i \rangle}{\|\vec{x}_i\|^2}$$

and in an orthonormal basis $\{\vec{q}_1, \ldots, \vec{q}_n\}$ simply $\alpha_i = \langle \vec{x}, \vec{q}_i \rangle$. These α_i are also called the **Fourier coefficients** of $\vec{x}$.

Proof: Since the vectors are a basis, $\vec{x} = \sum_i \alpha_i \vec{x}_i$ with unique numbers α_i, and we only need to find them.

- We take the inner product with $\vec{x}_j$, and as in the lemma only the term $i = j$ survives;

$$\langle \vec{x}, \vec{x}_j \rangle = \sum_i \alpha_i \langle \vec{x}_i, \vec{x}_j \rangle = \alpha_j \, \|\vec{x}_j\|^2$$

- Dividing by $\|\vec{x}_j\|^2 \neq 0$ we get;

$$\therefore \quad \alpha_j = \frac{\langle \vec{x}, \vec{x}_j \rangle}{\|\vec{x}_j\|^2}$$

- The length of $\vec{x}_i$ comes in squared, and we see why when we split the term as;

$$\alpha_i \, \vec{x}_i = \frac{\langle \vec{x}, \vec{x}_i \rangle}{\|\vec{x}_i\|} \, \frac{\vec{x}_i}{\|\vec{x}_i\|}$$

  where one $\|\vec{x}_i\|$ belongs to the number in front and the other turns $\vec{x}_i$ into a unit vector.

Example: Let $\vec{x}_1 = (1,1,0)^T$, $\vec{x}_2 = (1,-1,1)^T$ and $\vec{x}_3 = (-1,1,2)^T$.

a) Show that $\mathcal{B} = \{\vec{x}_1, \vec{x}_2, \vec{x}_3\}$ is an orthogonal basis of $\mathbb{R}^3$.

- We take the three inner products;

$$\begin{aligned}\langle \vec{x}_1, \vec{x}_2 \rangle &= 1 - 1 + 0 = 0 \\ \langle \vec{x}_1, \vec{x}_3 \rangle &= -1 + 1 + 0 = 0 \\ \langle \vec{x}_2, \vec{x}_3 \rangle &= -1 - 1 + 2 = 0\end{aligned}$$

- None of the vectors is $\vec{0}$, so by the lemma we have three independent vectors of $\mathbb{R}^3$, an orthogonal basis.

b) Find the coordinates of $\vec{x} = (3,1,4)^T$ in $\mathcal{B}$ and check them.

- We need the squared lengths and the inner products with $\vec{x}$;

$$\begin{aligned}\|\vec{x}_1\|^2 &= 1 + 1 + 0 = 2 & \langle \vec{x}, \vec{x}_1 \rangle &= 3 \times 1 + 1 \times 1 + 4 \times 0 = 4 \\ \|\vec{x}_2\|^2 &= 1 + 1 + 1 = 3 & \langle \vec{x}, \vec{x}_2 \rangle &= 3 \times 1 + 1 \times (-1) + 4 \times 1 = 6 \\ \|\vec{x}_3\|^2 &= 1 + 1 + 4 = 6 & \langle \vec{x}, \vec{x}_3 \rangle &= 3 \times (-1) + 1 \times 1 + 4 \times 2 = 6\end{aligned}$$

- Then by the proposition we get;

$$\alpha_1 = \frac{4}{2} = 2 \qquad \alpha_2 = \frac{6}{3} = 2 \qquad \alpha_3 = \frac{6}{6} = 1$$

so $[\vec{x}]_{\mathcal{B}} = (2,2,1)^T$.

- Let's check;

$$2\,\vec{x}_1 + 2\,\vec{x}_2 + \vec{x}_3 = (2,2,0)^T + (2,-2,2)^T + (-1,1,2)^T = (3,1,4)^T$$

- Notice that with the method of the Basis pages we would have solved $A\,[\vec{x}]_{\mathcal{B}} = \vec{x}$, three equations with three unknowns. Here six short sums and three divisions did the same job.

0.1.8.3 Projection onto a Vector

- Let's draw each piece $\alpha_i \vec{x}_i$ of the proposition. It is the shadow of $\vec{x}$ on the line of $\vec{x}_i$, the part of $\vec{x}$ that points along $\vec{x}_i$.

Definition: The **projection** of a vector $\vec{x}$ onto a nonzero vector $\vec{u}$ is the vector;

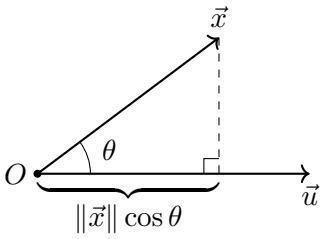
$$\text{proj}_{\vec{u}}\, \vec{x} = \frac{\langle \vec{x}, \vec{u} \rangle}{\|\vec{u}\|^2}\, \vec{u}$$

and the number $\langle \vec{x}, \vec{u} \rangle / \|\vec{u}\|$ is called the **scalar projection** of $\vec{x}$ onto $\vec{u}$.

- With $\langle \vec{x}, \vec{u} \rangle = \|\vec{x}\| \|\vec{u}\| \cos\theta$ of the Inner Product pages we can write the scalar projection as;

$$\frac{\langle \vec{x}, \vec{u} \rangle}{\|\vec{u}\|} = \frac{\|\vec{x}\| \|\vec{u}\| \cos\theta}{\|\vec{u}\|} = \|\vec{x}\| \cos\theta$$

which is the side of the right triangle under the brace.



- We get the projection by multiplying this length with the unit vector along $\vec{u}$;

$$\text{proj}_{\vec{u}}\, \vec{x} = \frac{\langle \vec{x}, \vec{u} \rangle}{\|\vec{u}\|}\, \frac{\vec{u}}{\|\vec{u}\|}$$

which is the split of the square above.

- The scalar projection carries a sign. It is positive when $\theta < 90^\circ$ and negative when $\theta > 90^\circ$, where the shadow falls on the other side of the origin, against $\vec{u}$.

- The dashed drop of the sketch is $\vec{x} - \text{proj}_{\vec{u}}\, \vec{x}$, what is left of $\vec{x}$ after we take away its shadow. Let's show that it is orthogonal to $\vec{u}$;

$$\langle \vec{x} - \text{proj}_{\vec{u}}\, \vec{x},\ \vec{u} \rangle = \langle \vec{x}, \vec{u} \rangle - \frac{\langle \vec{x}, \vec{u} \rangle}{\|\vec{u}\|^2} \langle \vec{u}, \vec{u} \rangle = \langle \vec{x}, \vec{u} \rangle - \langle \vec{x}, \vec{u} \rangle = 0$$

which is why we call this the orthogonal projection. So we can split every vector into a part along $\vec{u}$ and a part orthogonal to it.

- Notice that the number t we chose in the Cauchy Schwarz proof was $\langle \vec{x}, \vec{y} \rangle / \langle \vec{y}, \vec{y} \rangle$. So $t\, \vec{y}$ was the projection of $\vec{x}$ onto $\vec{y}$, and the proof only used that the drop $\vec{x} - t\, \vec{y}$ has a length that is not negative.

Example: Let $\vec{u} = (2,1)^T$.

- In both parts we need $\|\vec{u}\|^2 = 4+1 = 5$, so $\|\vec{u}\| = 5^{1/2}$.

a) Find the projection and the scalar projection of $\vec{x} = (3,4)^T$ onto $\vec{u}$, and show that $\vec{x} - \text{proj}_{\vec{u}}\,\vec{x}$ is orthogonal to $\vec{u}$.

- We start with the inner product, $\langle \vec{x}, \vec{u} \rangle = 3 \times 2 + 4 \times 1 = 6 + 4 = 10$, so;

$$\text{proj}_{\vec{u}}\,\vec{x} = \frac{10}{5}\,(2,1)^T = (4,2)^T \qquad\qquad \frac{\langle \vec{x}, \vec{u} \rangle}{\|\vec{u}\|} = \frac{10}{5^{1/2}} = 2 \times 5^{1/2} \approx 4.47$$

- The drop is $(3,4)^T - (4,2)^T = (-1,2)^T$ and its inner product with $\vec{u}$ is $-1 \times 2 + 2 \times 1 = -2 + 2 = 0$, so it is orthogonal to $\vec{u}$.

- Let's check the scalar projection with the angle. We have $\|\vec{x}\| = (9+16)^{1/2} = 5$ and $\cos\theta = 10/(5 \times 5^{1/2}) = 2/5^{1/2}$, so $\|\vec{x}\| \cos\theta = 5 \times 2/5^{1/2} = 2 \times 5^{1/2}$, the same number.

b) Find the projection and the scalar projection of $\vec{y} = (-3,1)^T$ onto $\vec{u}$, and the angle between $\vec{y}$ and $\vec{u}$.

- Here $\langle \vec{y}, \vec{u} \rangle = -3 \times 2 + 1 \times 1 = -6 + 1 = -5$, so;

$$\text{proj}_{\vec{u}}\,\vec{y} = \frac{-5}{5}\,(2,1)^T = (-2,-1)^T \qquad\qquad \frac{\langle \vec{y}, \vec{u} \rangle}{\|\vec{u}\|} = \frac{-5}{5^{1/2}} = -5^{1/2} \approx -2.24$$

- The scalar projection is negative, so the angle is above $90°$. With $\|\vec{y}\| = (9+1)^{1/2} = 10^{1/2}$ we have;

$$\cos\theta = \frac{-5}{10^{1/2} \times 5^{1/2}} = \frac{-5}{5 \times 2^{1/2}} = -\frac{1}{2^{1/2}} \quad \Rightarrow \quad \theta = 135°$$

and the shadow $(-2,-1)^T$ points against $\vec{u}$.

0.1.8.4 The Projection Matrix

- For a unit vector $\vec{q}$ the denominator $\|\vec{q}\|^2$ is 1, and the projection is just $\text{proj}_{\vec{q}}\,\vec{x} = \langle \vec{x}, \vec{q} \rangle\, \vec{q} = (\vec{q}^T \vec{x})\, \vec{q}$. We want to write it as a matrix times $\vec{x}$, because in the Symmetric Matrices pages a whole matrix turns out to be a sum of such matrices.

- We can put the number $\vec{q}^T \vec{x}$ on either side of the vector $\vec{q}$, and then move the brackets since matrix products are associative;

$$(\vec{q}^T \vec{x})\, \vec{q} = \vec{q}\,(\vec{q}^T \vec{x}) = (\vec{q}\,\vec{q}^T)\, \vec{x}$$

Definition: For a unit vector $\vec{q} \in \mathbb{R}^n$ the matrix $P = \vec{q}\,\vec{q}^T$ is called the **projection matrix** onto $\vec{q}$, and $P\vec{x} = (\vec{q}^T \vec{x})\, \vec{q}$ is the projection of $\vec{x}$ onto $\vec{q}$.

- Let's see what kind of object $\vec{q}\,\vec{q}^T$ is. $\vec{q}$ is a column, an $n \times 1$ matrix, and $\vec{q}^T$ is a row, a $1 \times n$ matrix. In $\vec{q}^T \vec{q}$ the row comes first, $1 \times n$ times $n \times 1$ gives a 1×1 matrix, a number, and this is the inner product. In $\vec{q}\,\vec{q}^T$ the column comes first, $n \times 1$ times $1 \times n$ gives an $n \times n$ matrix, called the **outer product**. For $n = 2$ it is;

$$\vec{q}\,\vec{q}^T = \begin{bmatrix} q_1 \\ q_2 \end{bmatrix} \begin{bmatrix} q_1 & q_2 \end{bmatrix} = \begin{bmatrix} q_1^2 & q_1 q_2 \\ q_2 q_1 & q_2^2 \end{bmatrix}$$

  so the entry in the row i and the column j is $q_i q_j$, and P is symmetric.

- Projecting twice changes nothing. The shadow $P\vec{x}$ already lies on the line of $\vec{q}$, and the shadow of a vector that lies on the line is the vector itself, so projecting the shadow again gives the same shadow. In matrices this is $P^2 = P$, and with $\vec{q}^T \vec{q} = 1$ we show it in one line;

$$P^2 = \vec{q}\,\vec{q}^T \vec{q}\,\vec{q}^T = \vec{q}\,(\vec{q}^T \vec{q})\, \vec{q}^T = \vec{q}\,\vec{q}^T = P$$

- What P throws away, $\vec{x} - P\vec{x} = (I - P)\vec{x}$, is the dashed drop of the projection sketch. So $I - P$ projects onto the orthogonal complement of $\vec{q}$, all the vectors orthogonal to $\vec{q}$, as we can check;

$$\vec{q}^T (I - P)\vec{x} = \vec{q}^T \vec{x} - (\vec{q}^T \vec{q})(\vec{q}^T \vec{x}) = \vec{q}^T \vec{x} - \vec{q}^T \vec{x} = 0$$

Example: Let $\vec{q} = (3/5, 4/5)^T$ and $\vec{x} = (2, 1)^T$.

a) Show that $\vec{q}$ is a unit vector and write out $P = \vec{q}\,\vec{q}^T$.

- We have $\vec{q}^T\vec{q} = (3/5)^2 + (4/5)^2 = 9/25 + 16/25 = 1$, so $\vec{q}$ is a unit vector. The outer product is;

$$P = \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix} \begin{bmatrix} 3/5 & 4/5 \end{bmatrix} = \begin{bmatrix} 9/25 & 12/25 \\ 12/25 & 16/25 \end{bmatrix} = \frac{1}{25}\begin{bmatrix} 9 & 12 \\ 12 & 16 \end{bmatrix}$$

b) Find $P\vec{x}$ and compare it with $(\vec{q}^T\vec{x})\,\vec{q}$.

- We multiply the matrix with $\vec{x}$;

$$P\vec{x} = \frac{1}{25}\begin{bmatrix} 9 \times 2 + 12 \times 1 \\ 12 \times 2 + 16 \times 1 \end{bmatrix} = \frac{1}{25}\begin{bmatrix} 30 \\ 40 \end{bmatrix} = \begin{bmatrix} 6/5 \\ 8/5 \end{bmatrix}$$

- The other way we have $\vec{q}^T\vec{x} = 3/5 \times 2 + 4/5 \times 1 = 6/5 + 4/5 = 2$ and $2\,\vec{q} = (6/5, 8/5)^T$, the same vector. The matrix does nothing more than the inner product and the multiple.

c) Show that $P^2 = P$.

- With the $1/25$ of each factor in front;

$$P^2 = \frac{1}{625}\begin{bmatrix} 9 & 12 \\ 12 & 16 \end{bmatrix}\begin{bmatrix} 9 & 12 \\ 12 & 16 \end{bmatrix} = \frac{1}{625}\begin{bmatrix} 81+144 & 108+192 \\ 108+192 & 144+256 \end{bmatrix} = \frac{1}{625}\begin{bmatrix} 225 & 300 \\ 300 & 400 \end{bmatrix}$$

and since $225 = 9 \times 25$, $300 = 12 \times 25$ and $400 = 16 \times 25$, we get back $1/25$ times the matrix of part a. So; $P^2 = P$.

d) Find $(I - P)\vec{x}$ and show that it is orthogonal to $\vec{q}$.

- Using $P\vec{x}$ of part b we have;

$$(I - P)\vec{x} = \vec{x} - P\vec{x} = (2, 1)^T - (6/5, 8/5)^T = (4/5, -3/5)^T$$

- Its inner product with $\vec{q}$ is $3/5 \times 4/5 + 4/5 \times (-3/5) = 12/25 - 12/25 = 0$, so it is orthogonal to $\vec{q}$. Notice that $\vec{x} = P\vec{x} + (I - P)\vec{x}$ is the split of $\vec{x}$ into its shadow and its drop.

Worked Examples

Example: Let $\vec{u} = (1,1,1)^T$ and $\vec{q} = \vec{u}/\|\vec{u}\|$ its unit vector.

a) Write out $P = \vec{q}\,\vec{q}^T$ and $I - P$.

- We have $\|\vec{u}\| = (1+1+1)^{1/2} = 3^{1/2}$, so $\vec{q} = 3^{-1/2}(1,1,1)^T$. In the outer product the two roots meet and give $3^{-1/2} \times 3^{-1/2} = 1/3$;

$$P = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \qquad I - P = \frac{1}{3}\begin{bmatrix} 3-1 & -1 & -1 \\ -1 & 3-1 & -1 \\ -1 & -1 & 3-1 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$$

where for $I - P$ we wrote the identity as $1/3$ times the matrix with 3 on its diagonal. So a root in the unit vector does not mean a root in P.

b) Split any $\vec{x} = (x_1, x_2, x_3)^T$ into $P\vec{x}$ and $(I - P)\vec{x}$, and show that the second piece is orthogonal to $\vec{u}$.

- Every row of P adds up the components of $\vec{x}$. With their mean $\bar{x} = (x_1 + x_2 + x_3)/3$ we have;

$$P\vec{x} = \frac{1}{3}\begin{bmatrix} x_1 + x_2 + x_3 \\ x_1 + x_2 + x_3 \\ x_1 + x_2 + x_3 \end{bmatrix} = \begin{bmatrix} \bar{x} \\ \bar{x} \\ \bar{x} \end{bmatrix}$$

- And with the matrix $I - P$ we get;

$$(I - P)\vec{x} = \frac{1}{3}\begin{bmatrix} 2x_1 - x_2 - x_3 \\ -x_1 + 2x_2 - x_3 \\ -x_1 - x_2 + 2x_3 \end{bmatrix} = \begin{bmatrix} x_1 - \bar{x} \\ x_2 - \bar{x} \\ x_3 - \bar{x} \end{bmatrix}$$

since in the first row $\frac{1}{3}(2x_1 - x_2 - x_3) = x_1 - \frac{1}{3}(x_1 + x_2 + x_3) = x_1 - \bar{x}$, and the same in the other two.

- The two pieces give $\vec{x}$ back, $\bar{x} + (x_i - \bar{x}) = x_i$ in every slot. The inner product of the second piece with $\vec{u}$ is;

$$(x_1 - \bar{x}) + (x_2 - \bar{x}) + (x_3 - \bar{x}) = (x_1 + x_2 + x_3) - 3\bar{x} = 0$$

so it is orthogonal to $\vec{u}$.

c) Check that $\|\vec{x}\|^2 = \|P\vec{x}\|^2 + \|(I - P)\vec{x}\|^2$, and read it as a statement about the mean.

- With the two pieces of part b the three squared lengths are;

$$\|\vec{x}\|^2 = \sum_{i=1}^{3} x_i^2 \qquad \|P\vec{x}\|^2 = 3\bar{x}^2 \qquad \|(I - P)\vec{x}\|^2 = \sum_{i=1}^{3}(x_i - \bar{x})^2$$

- We open the last one and use $\sum_i x_i = 3\bar{x}$;

$$\begin{aligned}\sum_{i=1}^{3}(x_i-\bar{x})^2 &= \sum_{i=1}^{3}x_i^2 - 2\bar{x}\sum_{i=1}^{3}x_i + 3\bar{x}^2 \\ &= \sum_{i=1}^{3}x_i^2 - 6\bar{x}^2 + 3\bar{x}^2 \\ &= \sum_{i=1}^{3}x_i^2 - 3\bar{x}^2\end{aligned}$$

  so adding $\|P\vec{x}\|^2 = 3\bar{x}^2$ gives $\|\vec{x}\|^2$. This is the Pythagorean theorem for the shadow and the drop, which are orthogonal.

- Dividing it by 3 we have;

$$\frac{1}{3}\sum_{i=1}^{3}x_i^2 = \bar{x}^2 + \frac{1}{3}\sum_{i=1}^{3}(x_i-\bar{x})^2$$

  Meaning the mean of the squares is the square of the mean plus the variance of the components. So the variance split of statistics is the Pythagorean theorem along $\vec{u} = (1,1,1)^T$.

- Notice what the two matrices do. $P\vec{x}$ puts the mean of the components into every slot, and $(I-P)\vec{x}$ takes that mean away from every component, so its components add up to zero. The vectors with $x_1 + x_2 + x_3 = 0$ are the plane V of the Basis and Dimension worked examples, and it is the orthogonal complement of $\vec{u}$, since $\langle \vec{x}, \vec{u} \rangle = x_1 + x_2 + x_3$.

Your Turn

Let $\{\vec{x}_1, \ldots, \vec{x}_n\}$ be an orthogonal basis of $\mathbb{R}^n$, and let a vector $\vec{x}$ be written in two ways;

$$\vec{x} = \sum_{i=1}^{n} \alpha_i \, \vec{x}_i = \sum_{i=1}^{n} \beta_i \, \vec{x}_i$$

a) Take the inner product of both expansions with $\vec{x}_j$ and show that $\alpha_j = \beta_j$ for every j.

b) Show the same in a second way. Write $\vec{x} - \vec{x} = \vec{0}$ as a combination of the $\vec{x}_i$ and open its squared length $\|\vec{x} - \vec{x}\|^2 = 0$.

c) Where did you use that the $\vec{x}_i$ are orthogonal, and where that they are nonzero?

Your Turn

Let $\vec{u} = (1, 1, 2)^T$ and $\vec{x} = (3, 1, 4)^T$.

a) Find the projection and the scalar projection of $\vec{x}$ onto $\vec{u}$.

b) Write out the projection matrix P onto $\vec{u}$, using the unit vector along $\vec{u}$, and check that $P\vec{x}$ is the projection of part a.

c) Find $(I - P)\vec{x}$, show that it is orthogonal to $\vec{u}$, and check that $\|\vec{x}\|^2 = \|P\vec{x}\|^2 + \|(I - P)\vec{x}\|^2$.

d) Find a nonzero vector $\vec{y}$ with $P\vec{y} = \vec{0}$, and show that $P\vec{u} = \vec{u}$.

0.1.9 Gram-Schmidt and QR

- We have seen in the previous pages how easy everything is in an orthonormal basis, but most bases we meet are not orthogonal, the columns of a data matrix for example. With the Gram-Schmidt process we turn any linearly independent set into an orthonormal one that spans the same space, and the QR decomposition is the same process written as a matrix product.

0.1.9.1 The Gram-Schmidt Process

- We start with linearly independent vectors $\vec{x}_1, \ldots, \vec{x}_k$ and build orthogonal vectors $\vec{z}_1, \ldots, \vec{z}_k$ one by one. Every step does the same thing. We take the new vector and take away from it its shadows on the vectors already done, and what is left is orthogonal to all of them.

- The first vector has nothing to be orthogonal to, so we keep it;

$$\vec{z}_1 = \vec{x}_1$$

- From $\vec{x}_2$ we take away its shadow on $\vec{z}_1$;

$$\vec{z}_2 = \vec{x}_2 - \text{proj}_{\vec{z}_1} \vec{x}_2 = \vec{x}_2 - \frac{\langle \vec{x}_2, \vec{z}_1 \rangle}{\|\vec{z}_1\|^2} \vec{z}_1$$

  which is the dashed drop of the projection pages, so as we have shown there it is orthogonal to $\vec{z}_1$;

$$\langle \vec{z}_2, \vec{z}_1 \rangle = \langle \vec{x}_2, \vec{z}_1 \rangle - \frac{\langle \vec{x}_2, \vec{z}_1 \rangle}{\|\vec{z}_1\|^2} \langle \vec{z}_1, \vec{z}_1 \rangle = \langle \vec{x}_2, \vec{z}_1 \rangle - \langle \vec{x}_2, \vec{z}_1 \rangle = 0$$

- From $\vec{x}_3$ we take away both of its shadows;

$$\vec{z}_3 = \vec{x}_3 - \text{proj}_{\vec{z}_1} \vec{x}_3 - \text{proj}_{\vec{z}_2} \vec{x}_3$$

  We show that it is orthogonal to $\vec{z}_1$ the same way. In $\langle \vec{z}_3, \vec{z}_1 \rangle$ the shadow on $\vec{z}_1$ cancels $\langle \vec{x}_3, \vec{z}_1 \rangle$ as above, and the shadow on $\vec{z}_2$ gives nothing since $\vec{z}_2 \perp \vec{z}_1$. With the roles of $\vec{z}_1$ and $\vec{z}_2$ swapped we get that it is orthogonal to $\vec{z}_2$ too.

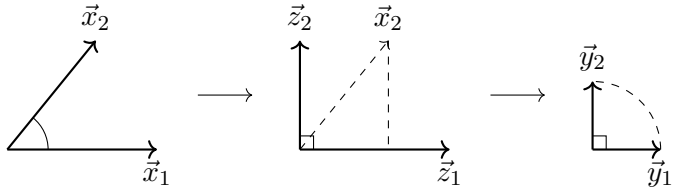
- Every next step works the same way, so we write the general step and the normalization at the end as;

$$\vec{z}_k = \vec{x}_k - \sum_{j=1}^{k-1} \frac{\langle \vec{x}_k, \vec{z}_j \rangle}{\|\vec{z}_j\|^2} \vec{z}_j \qquad\qquad \vec{y}_k = \frac{\vec{z}_k}{\|\vec{z}_k\|}$$

  where the $\vec{z}_k$ are orthogonal and the $\vec{y}_k$ are orthonormal.

- Each $\vec{z}_k$ is $\vec{x}_k$ minus a combination of $\vec{z}_1, \ldots, \vec{z}_{k-1}$, which are themselves combinations of $\vec{x}_1, \ldots, \vec{x}_{k-1}$. So at every step the first k new vectors span the same space as the first k old ones.

- We show the process for two vectors in the plane in the following sketch. $\vec{x}_1$ stays, $\vec{x}_2$ loses its shadow along $\vec{x}_1$ and stands up straight as $\vec{z}_2$, and then both are cut to unit length on the dashed unit circle;



- If the $\vec{x}_i$ are dependent, some $\vec{x}_k$ is a combination of the vectors before it. Then all of it is shadows, $\vec{z}_k = \vec{0}$ and we can not divide by its length. So Gram-Schmidt also finds the dependent vectors, and we drop such an $\vec{x}_k$ and go on.

Example: Let $\vec{x}_1 = (1,2,2)^T$, $\vec{x}_2 = (0,3,6)^T$ and $\vec{x}_3 = (6,3,3)^T$. Apply the Gram-Schmidt process to them, and find the orthogonal vectors $\vec{z}_1, \vec{z}_2, \vec{z}_3$ and the orthonormal vectors $\vec{y}_1, \vec{y}_2, \vec{y}_3$.

- The first vector stays, $\vec{z}_1 = (1,2,2)^T$ with $\|\vec{z}_1\|^2 = 1+4+4 = 9$.

- For $\vec{z}_2$ we need the shadow of $\vec{x}_2$ on $\vec{z}_1$. The inner product is $\langle \vec{x}_2, \vec{z}_1 \rangle = 0\times 1 + 3\times 2 + 6\times 2 = 0+6+12 = 18$, so;

$$\vec{z}_2 = (0,3,6)^T - \frac{18}{9}(1,2,2)^T = (0,3,6)^T - (2,4,4)^T = (-2,-1,2)^T$$

  with $\|\vec{z}_2\|^2 = 4+1+4 = 9$. Let's check, $\langle \vec{z}_2, \vec{z}_1 \rangle = -2-2+4 = 0$.

- For $\vec{z}_3$ we need the shadows of $\vec{x}_3$ on $\vec{z}_1$ and on $\vec{z}_2$;

$$\langle \vec{x}_3, \vec{z}_1 \rangle = 6\times 1 + 3\times 2 + 3\times 2 = 6+6+6 = 18$$
$$\langle \vec{x}_3, \vec{z}_2 \rangle = 6\times(-2) + 3\times(-1) + 3\times 2 = -12-3+6 = -9$$

- Then we get;

$$\begin{aligned}\vec{z}_3 &= (6,3,3)^T - \frac{18}{9}(1,2,2)^T - \frac{-9}{9}(-2,-1,2)^T \\ &= (6,3,3)^T - (2,4,4)^T + (-2,-1,2)^T \\ &= (2,-2,1)^T\end{aligned}$$

  with $\|\vec{z}_3\|^2 = 4+4+1 = 9$. Let's check again, $\langle \vec{z}_3, \vec{z}_1 \rangle = 2-4+2 = 0$ and $\langle \vec{z}_3, \vec{z}_2 \rangle = -4+2+2 = 0$.

- Notice that the shadow on $\vec{z}_2$ had a negative number in front. $\vec{x}_3$ makes more than 90° with $\vec{z}_2$, so taking its shadow away added $\vec{z}_2$.

- All three lengths are $9^{1/2} = 3$, so we divide each vector by 3;

$$\vec{y}_1 = \frac{1}{3}(1,2,2)^T \qquad \vec{y}_2 = \frac{1}{3}(-2,-1,2)^T \qquad \vec{y}_3 = \frac{1}{3}(2,-2,1)^T$$

Remark: Nothing in the process needs columns of numbers, we only need an inner product. For continuous functions on $[-1,1]$ the integral;

$$\langle f,g\rangle = \int_{-1}^{1} f(x)\,g(x)\,dx$$

has the three properties of the Inner Product pages, and the vectors are now the functions themselves. For Gram-Schmidt on $1, x, x^2$ we need the integrals;

$$\langle 1,1\rangle = \int_{-1}^{1} dx = 2 \qquad \langle x,1\rangle = \int_{-1}^{1} x\,dx = 0$$
$$\langle x^2,1\rangle = \langle x,x\rangle = \int_{-1}^{1} x^2\,dx = \frac{2}{3} \qquad \langle x^2,x\rangle = \int_{-1}^{1} x^3\,dx = 0$$

So we keep $z_1 = 1$, then $z_2 = x$ since x is already orthogonal to 1, and;

$$z_3 = x^2 - \frac{\langle x^2,1\rangle}{\langle 1,1\rangle}\,1 - \frac{\langle x^2,x\rangle}{\langle x,x\rangle}\,x = x^2 - \frac{2/3}{2} - 0 = x^2 - \frac{1}{3}$$

These are the first Legendre polynomials $P_0 = 1$, $P_1 = x$ and $P_2 = (3x^2-1)/2 = \frac{3}{2}(x^2 - 1/3)$, up to their scaling.

0.1.9.2 QR Decomposition

- Now we run Gram-Schmidt on the columns of a square matrix $A = [\,\vec{a}_1 \mid \cdots \mid \vec{a}_n\,]$ with linearly independent columns, and call its orthonormal vectors $\vec{q}_1, \ldots, \vec{q}_n$. Gram-Schmidt built them out of the columns, and with the QR decomposition we write the columns back out of them.

Definition: The **QR decomposition** of a square matrix A with linearly independent columns is $A = QR$, where $Q = [\,\vec{q}_1 \mid \cdots \mid \vec{q}_n\,]$ holds the orthonormal vectors that Gram-Schmidt makes of the columns $\vec{a}_j$, and R is the upper triangular matrix;

$$R = \begin{bmatrix} \|\vec{z}_1\| & \langle \vec{a}_2,\vec{q}_1\rangle & \langle \vec{a}_3,\vec{q}_1\rangle & \cdots & \langle \vec{a}_n,\vec{q}_1\rangle \\ 0 & \|\vec{z}_2\| & \langle \vec{a}_3,\vec{q}_2\rangle & \cdots & \langle \vec{a}_n,\vec{q}_2\rangle \\ 0 & 0 & \|\vec{z}_3\| & \cdots & \langle \vec{a}_n,\vec{q}_3\rangle \\ \vdots & \vdots & & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \|\vec{z}_n\| \end{bmatrix}$$

- Meaning the lengths $r_{jj} = \|\vec{z}_j\|$ sit on the diagonal, the inner products $r_{ij} = \langle \vec{a}_j,\vec{q}_i\rangle$ with $i < j$ above it and zeros below it. Let's see where this comes from.

- Since $\vec{z}_i = \|\vec{z}_i\|\,\vec{q}_i$, we can write every shadow of Gram-Schmidt with the $\vec{q}_i$;

$$\frac{\langle \vec{a}_j,\vec{z}_i\rangle}{\|\vec{z}_i\|^2}\,\vec{z}_i = \Big\langle \vec{a}_j, \frac{\vec{z}_i}{\|\vec{z}_i\|}\Big\rangle \frac{\vec{z}_i}{\|\vec{z}_i\|} = \langle \vec{a}_j,\vec{q}_i\rangle\,\vec{q}_i$$

- Then we solve the general step of Gram-Schmidt for the column $\vec{a}_j$;

$$\vec{a}_j = \vec{z}_j + \sum_{i=1}^{j-1} \langle \vec{a}_j,\vec{q}_i\rangle\,\vec{q}_i = \|\vec{z}_j\|\,\vec{q}_j + \sum_{i=1}^{j-1} \langle \vec{a}_j,\vec{q}_i\rangle\,\vec{q}_i$$

- Taking the inner product of this with $\vec{q}_j$ every term of the sum dies, so we get $\langle \vec{a}_j, \vec{q}_j \rangle = \|\vec{z}_j\|$ and all the numbers in front are inner products with the $\vec{q}_i$;

$$\vec{a}_j = \sum_{i=1}^{j} \langle \vec{a}_j, \vec{q}_i \rangle \, \vec{q}_i$$

  This is the proposition of the Orthogonality pages for $\vec{a}_j$, and its coordinates on $\vec{q}_{j+1}, \ldots, \vec{q}_n$ are zero because $\vec{a}_j$ only needs the first j of the $\vec{q}_i$. These zeros are the zeros below the diagonal.

- A combination of the columns of Q is Q times the vector of its numbers, as in the Basis pages. So the column $\vec{a}_j$ is Q times the column j of R, for example;

$$\vec{a}_2 = r_{12}\,\vec{q}_1 + r_{22}\,\vec{q}_2 = [\,\vec{q}_1 \mid \cdots \mid \vec{q}_n\,] \begin{bmatrix} r_{12} \\ r_{22} \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

  and putting all the columns side by side we have $A = QR$.

Example: Let A be the matrix whose columns are the vectors of the Gram-Schmidt example;

$$A = \begin{bmatrix} 1 & 0 & 6 \\ 2 & 3 & 3 \\ 2 & 6 & 3 \end{bmatrix}$$

where Gram-Schmidt gave $\vec{z}_1 = (1,2,2)^T$, $\vec{z}_2 = (-2,-1,2)^T$ and $\vec{z}_3 = (2,-2,1)^T$, each of length 3. Find the QR decomposition of A and check that $QR = A$.

- We put the unit vectors $\vec{q}_j = \vec{z}_j/3$ into the columns of Q;

$$Q = \frac{1}{3}\begin{bmatrix} 1 & -2 & 2 \\ 2 & -1 & -2 \\ 2 & 2 & 1 \end{bmatrix}$$

- The diagonal of R holds the lengths, $r_{11} = r_{22} = r_{33} = 3$. Above the diagonal we have;

$$\begin{aligned} r_{12} &= \langle \vec{a}_2, \vec{q}_1 \rangle = \frac{1}{3}(0 \times 1 + 3 \times 2 + 6 \times 2) = \frac{18}{3} = 6 \\ r_{13} &= \langle \vec{a}_3, \vec{q}_1 \rangle = \frac{1}{3}(6 \times 1 + 3 \times 2 + 3 \times 2) = \frac{18}{3} = 6 \\ r_{23} &= \langle \vec{a}_3, \vec{q}_2 \rangle = \frac{1}{3}(6 \times (-2) + 3 \times (-1) + 3 \times 2) = \frac{-9}{3} = -3 \end{aligned}$$

  which are the inner products we found in the Gram-Schmidt example, divided by the length 3. So;

$$R = \begin{bmatrix} 3 & 6 & 6 \\ 0 & 3 & -3 \\ 0 & 0 & 3 \end{bmatrix}$$

- Let's check, with the $1/3$ of Q in front;

$$QR = \frac{1}{3}\begin{bmatrix} 3 & 6-6 & 6+6+6 \\ 6 & 12-3 & 12+3-6 \\ 6 & 12+6 & 12-6+3 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 3 & 0 & 18 \\ 6 & 9 & 9 \\ 6 & 18 & 9 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 6 \\ 2 & 3 & 3 \\ 2 & 6 & 3 \end{bmatrix} = A$$

- If you check this with a computer, `numpy.linalg.qr` may give some columns of Q and the same rows of R with flipped signs. Both are right, since the two minus signs cancel in the product. Gram-Schmidt is the choice with a positive diagonal.

0.1.9.3 Orthogonal Matrices

Definition: A square matrix Q whose columns are orthonormal is called an **orthogonal matrix**. Its name says orthogonal but the columns are orthonormal, this is just the convention.

- In Q^TQ the rows of Q^T are the columns of Q, so we get the inner products of the columns with each other;

$$Q^TQ = \begin{bmatrix} \vec{q}_1^T \\ \vdots \\ \vec{q}_n^T \end{bmatrix} \begin{bmatrix} \vec{q}_1 & \cdots & \vec{q}_n \end{bmatrix} = \begin{bmatrix} \vec{q}_1^T\vec{q}_1 & \vec{q}_1^T\vec{q}_2 & \cdots & \vec{q}_1^T\vec{q}_n \\ \vec{q}_2^T\vec{q}_1 & \vec{q}_2^T\vec{q}_2 & \cdots & \vec{q}_2^T\vec{q}_n \\ \vdots & & \ddots & \vdots \\ \vec{q}_n^T\vec{q}_1 & \vec{q}_n^T\vec{q}_2 & \cdots & \vec{q}_n^T\vec{q}_n \end{bmatrix} = I$$

  since the entries are δ_{ij}, ones on the diagonal and zeros everywhere else.

- So Q^T is a left inverse of Q, and we can show that for a square matrix a left inverse is also a right inverse. I will call them L and M here, since R is taken. Let $LA = I$ for a square A. Then $\det L \det A = \det I = 1$, so $\det A \neq 0$, A is invertible and $M = A^{-1}$ is a right inverse, $AM = I$. Then;

$$L = LI = L(AM) = (LA)M = IM = M$$

- For an orthogonal matrix this means;

$$Q^TQ = QQ^T = I \quad \Rightarrow \quad Q^{-1} = Q^T$$

  so the inverse of an orthogonal matrix costs nothing, it is a transpose.

- QQ^T is the table of the inner products of the rows of Q, in the same way as Q^TQ was for its columns. So $QQ^T = I$ says that the rows of Q are orthonormal as well.

- We can also multiply two matrices column by row, as the sum of each column of the first times the matching row of the second. For QQ^T we get a sum of outer products;

$$QQ^T = \begin{bmatrix} \vec{q}_1 & \cdots & \vec{q}_n \end{bmatrix} \begin{bmatrix} \vec{q}_1^T \\ \vdots \\ \vec{q}_n^T \end{bmatrix} = \sum_{i=1}^{n} \vec{q}_i\,\vec{q}_i^T = I$$

  so the projection matrices onto all n directions add up to the identity. Applied to a vector it says $\vec{x} = \sum_i (\vec{q}_i^T\vec{x})\,\vec{q}_i$, the vector built back from its shadows, which is the proposition of the Orthogonality pages for an orthonormal basis.

- Let's see that Q does not change lengths;

$$\|Q\vec{x}\|^2 = (Q\vec{x})^T(Q\vec{x}) = \vec{x}^TQ^TQ\,\vec{x} = \vec{x}^T\vec{x} = \|\vec{x}\|^2$$

  and the same steps give $\langle Q\vec{x}, Q\vec{y}\rangle = \langle \vec{x}, \vec{y}\rangle$, so the angles stay too. Meaning an orthogonal matrix only rotates or reflects, it never stretches.

Example: Let Q be the matrix below, whose first column is the unit vector $\vec{q}$ of the projection matrix example, where we found the $\vec{q}\,\vec{q}^{\,T}$ beside it;

$$Q = \begin{bmatrix} 3/5 & -4/5 \\ 4/5 & 3/5 \end{bmatrix} \qquad\qquad \vec{q}\,\vec{q}^{\,T} = \frac{1}{25}\begin{bmatrix} 9 & 12 \\ 12 & 16 \end{bmatrix}$$

a) Show that Q is an orthogonal matrix and that its rows are orthonormal too.

- The columns are $\vec{q}_1 = (3/5, 4/5)^T$ and $\vec{q}_2 = (-4/5, 3/5)^T$, and we take their inner products;

$$\begin{aligned} \vec{q}_1^{\,T}\vec{q}_1 &= 9/25 + 16/25 = 1 \\ \vec{q}_2^{\,T}\vec{q}_2 &= 16/25 + 9/25 = 1 \\ \vec{q}_1^{\,T}\vec{q}_2 &= -12/25 + 12/25 = 0 \end{aligned}$$

  so $Q^TQ = I$ and Q is an orthogonal matrix.

- We do the same for the rows $(3/5, -4/5)$ and $(4/5, 3/5)$. Their squared lengths are $9/25 + 16/25 = 1$ and $16/25 + 9/25 = 1$, and their inner product is $12/25 - 12/25 = 0$, so they are orthonormal as well.

b) Show that $\vec{q}_1\vec{q}_1^{\,T} + \vec{q}_2\vec{q}_2^{\,T} = I$ for the columns $\vec{q}_1$ and $\vec{q}_2$ of Q.

- The first outer product is the $\vec{q}\,\vec{q}^{\,T}$ we already have. The second one is;

$$\vec{q}_2\vec{q}_2^{\,T} = \begin{bmatrix} -4/5 \\ 3/5 \end{bmatrix}\begin{bmatrix} -4/5 & 3/5 \end{bmatrix} = \frac{1}{25}\begin{bmatrix} 16 & -12 \\ -12 & 9 \end{bmatrix}$$

- Adding the two we get;

$$\vec{q}_1\vec{q}_1^{\,T} + \vec{q}_2\vec{q}_2^{\,T} = \frac{1}{25}\begin{bmatrix} 9+16 & 12-12 \\ 12-12 & 16+9 \end{bmatrix} = \frac{1}{25}\begin{bmatrix} 25 & 0 \\ 0 & 25 \end{bmatrix} = I$$

- Notice that $\vec{q}_2\vec{q}_2^{\,T}$ is the $I - P$ of the projection matrix example. In the plane the vectors orthogonal to $\vec{q}$ are the line of $\vec{q}_2$, so projecting onto the complement of $\vec{q}$ is projecting onto $\vec{q}_2$.

c) For $\vec{x} = (1, 2)^T$ find $Q\vec{x}$ and compare its length with the length of $\vec{x}$.

- We multiply;

$$Q\vec{x} = \begin{bmatrix} 3/5 \times 1 - 4/5 \times 2 \\ 4/5 \times 1 + 3/5 \times 2 \end{bmatrix} = \begin{bmatrix} 3/5 - 8/5 \\ 4/5 + 6/5 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

- The lengths are $\|Q\vec{x}\|^2 = 1 + 4 = 5$ and $\|\vec{x}\|^2 = 1 + 4 = 5$, the same. This Q is the rotation by the angle with $\cos\theta = 3/5$ and $\sin\theta = 4/5$, about $53°$.

Remark: In $\mathbb{C}^n$ the transpose turns into the conjugate transpose. A square complex matrix with orthonormal columns is called a **unitary matrix**, and it satisfies $Q^\dagger Q = QQ^\dagger = I$, so $Q^{-1} = Q^\dagger$. Everything of these pages holds with ^{T} replaced by †.

0.1.9.4 Solving with QR

- To solve $A\vec{x} = \vec{b}$ we write $A = QR$ and multiply both sides from the left with Q^T. Since $Q^TQ = I$;

$$QR\,\vec{x} = \vec{b} \quad \Rightarrow \quad Q^TQR\,\vec{x} = Q^T\vec{b} \quad \Rightarrow \quad R\,\vec{x} = Q^T\vec{b}$$

- R is upper triangular, so the last row of this system has one unknown, the row above it two, and so on. We solve it from the bottom up, every row giving one new unknown, which is called back substitution. No inverse is needed, only a transpose.

- Notice that if we apply the same step to A itself we get $R = Q^TA$, so $r_{ij} = \vec{q}_i^T\vec{a}_j$, the inner products we built R from.

- For a tall X with more rows than columns and independent columns Gram-Schmidt works the same. Q has the shape of X with $Q^TQ = I$ (but $QQ^T \neq I$, since Q is not square). R is square with the positive lengths $\|\vec{z}_j\|$ on its diagonal, so $\det R$, the product of its diagonal, is not zero and R is invertible. The normal equations of the Least Squares Data Fitting topic then turn into a triangular system, after we cancel the invertible R^T;

$$X^TX\vec{\alpha} = X^T\vec{y} \quad \Rightarrow \quad R^TQ^TQR\,\vec{\alpha} = R^TQ^T\vec{y} \quad \Rightarrow \quad R\,\vec{\alpha} = Q^T\vec{y}$$

  which we solve without ever forming X^TX, the numerically better way.

Example: With the QR decomposition of the previous examples;

$$A = \begin{bmatrix} 1 & 0 & 6 \\ 2 & 3 & 3 \\ 2 & 6 & 3 \end{bmatrix} \qquad Q = \frac{1}{3}\begin{bmatrix} 1 & -2 & 2 \\ 2 & -1 & -2 \\ 2 & 2 & 1 \end{bmatrix} \qquad R = \begin{bmatrix} 3 & 6 & 6 \\ 0 & 3 & -3 \\ 0 & 0 & 3 \end{bmatrix}$$

solve $A\vec{x} = \vec{b}$ for $\vec{b} = (8, 4, 1)^T$.

- We first need $Q^T\vec{b}$. The rows of Q^T are the columns of Q;

$$Q^T\vec{b} = \frac{1}{3}\begin{bmatrix} 1 & 2 & 2 \\ -2 & -1 & 2 \\ 2 & -2 & 1 \end{bmatrix}\begin{bmatrix} 8 \\ 4 \\ 1 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 8+8+2 \\ -16-4+2 \\ 16-8+1 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 18 \\ -18 \\ 9 \end{bmatrix} = \begin{bmatrix} 6 \\ -6 \\ 3 \end{bmatrix}$$

- So the system $R\vec{x} = Q^T\vec{b}$ is;

$$\begin{aligned} 3x_1 + 6x_2 + 6x_3 &= 6 \\ 3x_2 - 3x_3 &= -6 \\ 3x_3 &= 3 \end{aligned}$$

- From the bottom up. The last row gives $x_3 = 1$. The middle row is then $3x_2 - 3 = -6$, so $x_2 = -1$. The first row is $3x_1 - 6 + 6 = 6$, so $x_1 = 2$ and;

$$\vec{x} = (2, -1, 1)^T$$

- Let's check with A, $A\vec{x} = (2 + 0 + 6,\ 4 - 3 + 3,\ 4 - 6 + 3)^T = (8, 4, 1)^T = \vec{b}$.

Worked Examples

Example: Let A be the tall matrix;

$$A = \begin{bmatrix} 1 & 3 \\ 2 & 3 \\ -2 & 0 \end{bmatrix}$$

a) Apply the Gram-Schmidt process to the columns of A and write out Q.

- The columns are $\vec{a}_1 = (1, 2, -2)^T$ and $\vec{a}_2 = (3, 3, 0)^T$. The first one stays, $\vec{z}_1 = \vec{a}_1$ with $\|\vec{z}_1\|^2 = 1 + 4 + 4 = 9$.
- The shadow of $\vec{a}_2$ needs $\langle \vec{a}_2, \vec{z}_1 \rangle = 3 \times 1 + 3 \times 2 + 0 \times (-2) = 3 + 6 + 0 = 9$, so;

$$\vec{z}_2 = (3, 3, 0)^T - \frac{9}{9}(1, 2, -2)^T = (2, 1, 2)^T$$

  with $\|\vec{z}_2\|^2 = 4 + 1 + 4 = 9$. Let's check, $\langle \vec{z}_2, \vec{z}_1 \rangle = 2 + 2 - 4 = 0$.
- Both lengths are 3, so the columns of Q are $\vec{q}_1 = (1, 2, -2)^T/3$ and $\vec{q}_2 = (2, 1, 2)^T/3$;

$$Q = \frac{1}{3}\begin{bmatrix} 1 & 2 \\ 2 & 1 \\ -2 & 2 \end{bmatrix}$$

  which has the shape of A, 3×2.

b) Find R from the lengths and the inner products, and check that $QR = A$.

- The diagonal of R holds the lengths, $r_{11} = r_{22} = 3$, and above it there is only one inner product;

$$r_{12} = \langle \vec{a}_2, \vec{q}_1 \rangle = \frac{1}{3}(3 \times 1 + 3 \times 2 + 0 \times (-2)) = \frac{9}{3} = 3 \qquad \Rightarrow \qquad R = \begin{bmatrix} 3 & 3 \\ 0 & 3 \end{bmatrix}$$

- R is square, 2×2. Let's check, with the $1/3$ of Q in front;

$$QR = \frac{1}{3}\begin{bmatrix} 1 \times 3 & 1 \times 3 + 2 \times 3 \\ 2 \times 3 & 2 \times 3 + 1 \times 3 \\ -2 \times 3 & -2 \times 3 + 2 \times 3 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 3 & 9 \\ 6 & 9 \\ -6 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 2 & 3 \\ -2 & 0 \end{bmatrix} = A$$

c) Show that $Q^TQ = I$, and find the first column of QQ^T.

- Q^TQ is the table of the inner products of the two columns. With the $1/9$ in front its diagonal is $(1 + 4 + 4)/9 = 1$ and $(4 + 1 + 4)/9 = 1$, and both entries off the diagonal are $(2 + 2 - 4)/9 = 0$, so $Q^TQ = I$.
- QQ^T is 3×3, the table of the inner products of the rows of Q. Its first column holds the products of every row with the first row $(1, 2)/3$, which is $(1 \times 1 + 2 \times 2,\ 2 \times 1 + 1 \times 2,\ -2 \times 1 + 2 \times 2)^T/9 = (5, 4, 2)^T/9$. This is not $(1, 0, 0)^T$, so $QQ^T \neq I$. Three orthonormal rows in $\mathbb{R}^2$ would be independent by the lemma of the Orthogonality pages, which is impossible in dimension 2, so the rows of a tall Q are never orthonormal.

Your Turn

Let;

$$A = \begin{bmatrix} 3 & 7 & -1 \\ 0 & 0 & 5 \\ 4 & 1 & 7 \end{bmatrix}$$

a) Apply the Gram-Schmidt process to the columns of A. Find the orthogonal vectors $\vec{z}_1, \vec{z}_2, \vec{z}_3$ and the columns $\vec{q}_1, \vec{q}_2, \vec{q}_3$ of Q.

b) Find R from the lengths and the inner products.

c) Check that $QR = A$.

Your Turn

a) Let Q_1 and Q_2 be orthogonal $n \times n$ matrices. Show that $Q_1 Q_2$ is an orthogonal matrix too.

b) Show that $\det Q = 1$ or $\det Q = -1$ for every orthogonal matrix Q. You may use $\det(AB) = \det A \det B$ and $\det A^T = \det A$.

c) Check both results with;

$$Q_1 = \frac{1}{13}\begin{bmatrix} 5 & -12 \\ 12 & 5 \end{bmatrix} \qquad Q_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Find $Q_1 Q_2$, show that its columns are orthonormal, and find $\det Q_1$, $\det Q_2$ and $\det(Q_1 Q_2)$.

0.1.10 The Four Subspaces

- A matrix $A \in \mathbb{R}^{m \times n}$ takes a vector $\vec{x} \in \mathbb{R}^n$ and returns the vector $A\vec{x} \in \mathbb{R}^m$, so it is a map from $\mathbb{R}^n$ to $\mathbb{R}^m$. Most of what we will do comes down to solving $A\vec{x} = \vec{b}$, and before solving it we want to know two things, whether it can be solved at all and, if it can, how many solutions it has.
- Four subspaces of A answer both, two of them in $\mathbb{R}^n$ and two in $\mathbb{R}^m$. The Singular Value Decomposition pages are built on them too.
- We will go slowly and carry one small matrix through the whole subtopic;

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}$$

It has $m = 2$ rows and $n = 3$ columns, so it takes vectors of $\mathbb{R}^3$ to vectors of $\mathbb{R}^2$.

0.1.10.1 Column Space and Null Space

- Let's multiply our A with any $\vec{x}$ and look at the result column by column;

$$A\vec{x} = \begin{bmatrix} x_1 + 2x_2 + 3x_3 \\ 2x_1 + 4x_2 + 6x_3 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 4 \end{bmatrix} + x_3 \begin{bmatrix} 3 \\ 6 \end{bmatrix}$$

- Meaning $A\vec{x}$ is a linear combination of the columns of A, and the entries of $\vec{x}$ are its weights. This holds for every matrix. With the columns $\vec{a}_1, \dots, \vec{a}_n$ we have $A\vec{x} = x_1\vec{a}_1 + \cdots + x_n\vec{a}_n$, which is what we used for the coordinates in the Basis and Dimension pages.

Definition: The **column space** of $A \in \mathbb{R}^{m \times n}$, also called the **range** of A, is the set of all the vectors that A can return;

$$R(A) = \{A\vec{x} : \vec{x} \in \mathbb{R}^n\} \subseteq \mathbb{R}^m$$

- By the line above the vectors $A\vec{x}$ are exactly the linear combinations of the columns, so $R(A)$ is the span of the columns of A. That is why the range and the column space are the same set. Being a span, it is a subspace of $\mathbb{R}^m$.
- For our A we see that the second column is 2 times the first and the third is 3 times the first. So every combination is a multiple of $(1, 2)^T$;

$$A\vec{x} = (x_1 + 2x_2 + 3x_3) \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \Rightarrow \quad R(A) = \text{span}\left\{(1, 2)^T\right\}$$

which is a line through the origin of $\mathbb{R}^2$. So three columns give us only one direction.

- Then we have the answer to the first question. $A\vec{x} = \vec{b}$ can be solved exactly when $\vec{b} \in R(A)$, meaning $\vec{b}$ is one of the vectors that A can build.

Definition: The **null space** of $A \in \mathbb{R}^{m \times n}$ is the set of the vectors that A sends to zero;

$$N(A) = \left\{\vec{x} \in \mathbb{R}^n : A\vec{x} = \vec{0}\right\} \subseteq \mathbb{R}^n$$

- It is a subspace of $\mathbb{R}^n$, since $A\vec{x} = \vec{0}$ and $A\vec{y} = \vec{0}$ give $A(\vec{x} + \vec{y}) = \vec{0}$ and $A(\alpha\,\vec{x}) = \vec{0}$. Notice where it lives. The null space holds inputs $\vec{x}$, so it is in $\mathbb{R}^n$, while the column space holds outputs and is in $\mathbb{R}^m$.
- For our A the second row of $A\vec{x} = \vec{0}$ is twice the first, so we have only one equation;

$$x_1 + 2x_2 + 3x_3 = 0$$

- One equation for three unknowns leaves two of them free. Let's take $x_2 = s$ and $x_3 = t$, then $x_1 = -2s - 3t$ and;

$$\vec{x} = \begin{bmatrix} -2s - 3t \\ s \\ t \end{bmatrix} = s \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} \quad \Rightarrow \quad N(A) = \text{span}\left\{(-2,1,0)^T,\ (-3,0,1)^T\right\}$$

  which is a plane through the origin of $\mathbb{R}^3$.
- Now the second question. If $\vec{x}_0$ is one solution of $A\vec{x} = \vec{b}$, then adding any $\vec{v} \in N(A)$ gives another one, since $A(\vec{x}_0 + \vec{v}) = \vec{b} + \vec{0} = \vec{b}$. The other way around, two solutions differ by a vector of $N(A)$, because $A(\vec{x} - \vec{x}_0) = \vec{b} - \vec{b} = \vec{0}$. So; all the solutions are;

$$\vec{x}_0 + N(A)$$

  the affine space of the Vector Spaces and Subspaces pages. Hence; a solvable system has exactly one solution when $N(A) = \{\vec{0}\}$ and infinitely many otherwise.

Example: Let A be the matrix of these pages, with $R(A) = \text{span}\{(1,2)^T\}$ and $N(A) = \text{span}\{(-2,1,0)^T, (-3,0,1)^T\}$ as found above;

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}$$

a) Decide whether $A\vec{x} = (2,4)^T$ can be solved, and if it can, find all of its solutions.

- We have $(2,4)^T = 2\,(1,2)^T \in R(A)$, so it can be solved. One solution we can read off. The first column of A is $(1,2)^T$, so 2 times the first column does it, $\vec{x}_0 = (2,0,0)^T$.
- Then all the solutions are $\vec{x}_0$ plus the null space;

$$\vec{x} = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} + s \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}$$

- Let's check one of them. With $s = t = 1$ we have $\vec{x} = (-3,1,1)^T$ and $A\vec{x} = (-3+2+3,\ -6+4+6)^T = (2,4)^T$.

b) Decide whether $A\vec{x} = (-1,3)^T$ can be solved.

- Every vector of $R(A)$ is a multiple of $(1,2)^T$, so its second entry is twice its first. Here $2 \times (-1) = -2 \neq 3$, so $(-1,3)^T \notin R(A)$ and the system has no solution.
- Notice that in b) the best we can do is to come as close to $\vec{b}$ as possible, which is a least squares problem. We will see below how $\vec{b}$ splits into the part that A can reach and the part it can not.

0.1.10.2 Row Space and Left Null Space

- The transpose A^T is A with its rows turned into columns. If a_{ij} is the entry of A in the row i and the column j, then A^T has a_{ji} there, and $A^T \in \mathbb{R}^{n\times m}$ is a map back from $\mathbb{R}^m$ to $\mathbb{R}^n$. Its column space and its null space are the other two subspaces.

Definition: For $A \in \mathbb{R}^{m\times n}$ the **row space** $R(A^T) \subseteq \mathbb{R}^n$ is the span of the rows of A, and the **left null space** is;

$$N(A^T) = \left\{ \vec{z} \in \mathbb{R}^m : A^T\vec{z} = \vec{0} \right\} \subseteq \mathbb{R}^m$$

- The name left comes from transposing $A^T\vec{z} = \vec{0}$, which we can write as $\vec{z}^T A = \vec{0}^T$, so $\vec{z}$ multiplies A from the left.
- So the four subspaces of $A \in \mathbb{R}^{m\times n}$ are;

  1) $R(A) \subseteq \mathbb{R}^m$ the column space, everything A can return
  2) $N(A) \subseteq \mathbb{R}^n$ the null space, everything A sends to $\vec{0}$
  3) $R(A^T) \subseteq \mathbb{R}^n$ the row space, the span of the rows
  4) $N(A^T) \subseteq \mathbb{R}^m$ the left null space, everything A^T sends to $\vec{0}$

  The row space and the null space live where $\vec{x}$ lives, in $\mathbb{R}^n$. The column space and the left null space live where $A\vec{x}$ lives, in $\mathbb{R}^m$.

- For our A the rows are $(1,2,3)$ and $(2,4,6)$, and the second is twice the first. So the row space is the line $R(A^T) = \text{span}\{(1,2,3)^T\}$ in $\mathbb{R}^3$.
- For the left null space we solve $A^T\vec{z} = \vec{0}$;

$$A^T\vec{z} = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = (z_1 + 2z_2) \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \vec{0}$$

so $z_1 = -2z_2$. With $z_2 = 1$ we have $N(A^T) = \text{span}\{(-2,1)^T\}$, a line in $\mathbb{R}^2$.

0.1.10.3 Rank

Definition: The **column rank** of A is $\dim R(A)$, the number of independent columns, and the **row rank** of A is $\dim R(A^T)$, the number of independent rows.

Theorem: For every matrix the column rank and the row rank are equal. Their common value is called the **rank** of A, written as $r = \text{rank}\, A$.

- I will not prove this one, and it is a surprise. The columns of our A live in $\mathbb{R}^2$ and its rows in $\mathbb{R}^3$, and nothing seems to connect the two counts. For our A both of them are 1, we found a single direction each time. We will see it happen on a matrix where the count is not 1 in the example below.
- Since $R(A)$ is a subspace of $\mathbb{R}^m$ its dimension is at most m, and since it is spanned by n columns its dimension is at most n (in the Basis and Dimension pages we have seen that more than $\dim V$ vectors are always dependent). So;

$$r = \text{rank}\, A \leq \min\{m, n\}$$

- The rank also counts the null space. Without proof as well I present the **rank nullity theorem**;

Theorem: For $A \in \mathbb{R}^{m\times n}$ with rank r the null space has the dimension;

$$\dim N(A) = n - r$$

- In words, of the n directions of $\mathbb{R}^n$, r of them are carried into independent directions of $\mathbb{R}^m$ and the other $n - r$ are crushed to $\vec{0}$.
- For our A it gives $\dim N(A) = 3 - 1 = 2$, and indeed we found the null space with two vectors. Applied to A^T, which has m columns and the same rank, it gives $\dim N(A^T) = m - r = 2 - 1 = 1$, our single vector $(-2,1)^T$.
- So; $A\vec{x} = \vec{b}$ has at most one solution exactly when $r = n$, meaning all of the columns are independent.

Example: Let;

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

a) Find the column rank of B.

- The columns are $\vec{b}_1 = (1,4,7)^T$, $\vec{b}_2 = (2,5,8)^T$ and $\vec{b}_3 = (3,6,9)^T$. Going from one column to the next we add $(1,1,1)^T$ each time;

$$\vec{b}_2 - \vec{b}_1 = (1,1,1)^T = \vec{b}_3 - \vec{b}_2 \quad \Rightarrow \quad \vec{b}_1 - 2\,\vec{b}_2 + \vec{b}_3 = \vec{0}$$

- So the third column is a combination of the first two, $\vec{b}_3 = 2\,\vec{b}_2 - \vec{b}_1$. The first two are not multiples of each other, so they are independent and the column rank is 2.

b) Find the row rank of B.

- The rows are $(1,2,3)$, $(4,5,6)$ and $(7,8,9)$. This time we add $(3,3,3)$ from one row to the next, so in the same way;

$$(1,2,3) - 2\,(4,5,6) + (7,8,9) = (1-8+7,\ 2-10+8,\ 3-12+9) = (0,0,0)$$

The first two rows are not multiples of each other, so the row rank is 2 as well. Two different calculations and the same count, as the theorem says.

c) Find $N(B)$ and check the rank nullity theorem.

- Part a) already gave us a vector of the null space. The combination $\vec{b}_1 - 2\,\vec{b}_2 + \vec{b}_3 = \vec{0}$ is $B\vec{x}$ with $\vec{x} = (1,-2,1)^T$. To get all of $N(B)$ we take $B\vec{x} = \vec{0}$ and subtract 4 times the first row from the second and 7 times the first row from the third;

$$\begin{aligned} x_1 + 2x_2 + 3x_3 &= 0 \\ -3x_2 - 6x_3 &= 0 \\ -6x_2 - 12x_3 &= 0 \end{aligned}$$

- The last two rows both say $x_2 = -2x_3$. With $x_3 = t$ we have $x_2 = -2t$ and $x_1 = 4t - 3t = t$, so;

$$N(B) = \text{span}\left\{(1,-2,1)^T\right\}$$

- Its dimension is 1, and rank nullity gives $\dim N(B) = n - r = 3 - 2 = 1$. The found result is consistent with the theorem.

0.1.10.4 Orthogonal Decomposition

- Let's go back to the four spaces of our A and take inner products. The row $(1,2,3)$ is orthogonal to both vectors $(-2,1,0)^T$ and $(-3,0,1)^T$ of the null space, $-2+2+0=0$ and $-3+0+3=0$. The column $(1,2)^T$ is orthogonal to the vector $(-2,1)^T$ of the left null space, $-2+2=0$. This is no coincidence.
- For the first pair there is a plain reason. The ith entry of $A\vec{x}$ is the ith row of A times $\vec{x}$, so $A\vec{x}=\vec{0}$ says that $\vec{x}$ has inner product zero with every row. Then it is orthogonal to every combination of the rows too, which is the whole row space.

Theorem: For every $A \in \mathbb{R}^{m\times n}$ the null spaces are the orthogonal complements of the row and the column space;

$$N(A) = R(A^T)^\perp \qquad \text{and} \qquad N(A^T) = R(A)^\perp$$

Proof: We show the second one. A vector $\vec{z} \in \mathbb{R}^m$ is in $R(A)^\perp$ when it is orthogonal to every $A\vec{x}$;

- Moving the transpose over with $(A^T\vec{z})^T = \vec{z}^T A$ we have;

$$\begin{aligned} \vec{z}^T(A\vec{x}) = 0 \ \text{ for all } \ \vec{x}\in\mathbb{R}^n \ &\iff \ (A^T\vec{z})^T\vec{x} = 0 \ \text{ for all } \ \vec{x}\in\mathbb{R}^n \\ &\iff \ A^T\vec{z} = \vec{0} \end{aligned}$$

  where in the last step we take $\vec{x} = A^T\vec{z}$ itself, which gives $\|A^T\vec{z}\|^2 = 0$. The other way around is immediate.

- So $R(A)^\perp$ and $N(A^T)$ hold the same vectors. The first one is the same statement for A^T in place of A, since $(A^T)^T = A$;

$$\therefore \quad N(A^T) = R(A)^\perp \quad \text{and} \quad N(A) = R(A^T)^\perp$$

- Without proof I present that a subspace and its orthogonal complement together fill the whole space, so we can write;

$$\mathbb{R}^n = R(A^T) \oplus N(A) \qquad \text{and} \qquad \mathbb{R}^m = R(A) \oplus N(A^T)$$

- The symbol $\oplus$ is the **orthogonal direct sum**. $\mathbb{R}^m = R(A) \oplus N(A^T)$ means that every $\vec{b} \in \mathbb{R}^m$ splits in exactly one way into a piece of the column space and a piece of the left null space, and the two pieces are orthogonal;

$$\vec{b} = \vec{q} + \vec{r} \qquad \text{with} \qquad \vec{q}\in R(A), \ \ \vec{r}\in N(A^T), \ \ \vec{q}\perp\vec{r}$$

  where $\vec{q}$ is the projection of $\vec{b}$ onto $R(A)$ and $\vec{r} = \vec{b} - \vec{q}$ is what is left.

- The dimensions agree with this. With rank nullity for A^T we have $\dim R(A) + \dim N(A^T) = r + (m-r) = m$, and in the same way $r + (n-r) = n$ for $\mathbb{R}^n$.

Example: Let A be the matrix of these pages, with $R(A) = \text{span}\{(1,2)^T\}$ and $N(A^T) = \text{span}\{(-2,1)^T\}$. Split $\vec{b} = (-1,3)^T$ into $\vec{b} = \vec{q} + \vec{r}$ with $\vec{q} \in R(A)$ and $\vec{r} \in N(A^T)$.

- The column space is the line along $\vec{u} = (1,2)^T$, so $\vec{q}$ is the projection of $\vec{b}$ onto $\vec{u}$ of the Orthogonality and Projection pages;

$$\vec{q} = \frac{\langle \vec{b}, \vec{u} \rangle}{\|\vec{u}\|^2} \vec{u} = \frac{-1+6}{1+4} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \frac{5}{5} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

- The rest is;

$$\vec{r} = \vec{b} - \vec{q} = \begin{bmatrix} -1-1 \\ 3-2 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

- Let's check. $\vec{r} = (-2,1)^T$ is the vector of $N(A^T)$ itself, and $\langle \vec{q}, \vec{r} \rangle = -2+2 = 0$. So;

$$\begin{bmatrix} -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

- This is the $\vec{b}$ of the earlier example that $A\vec{x} = \vec{b}$ could not reach. The best A can do is $\vec{q}$, and $\vec{r}$ is the part of $\vec{b}$ that no $\vec{x}$ will ever produce.

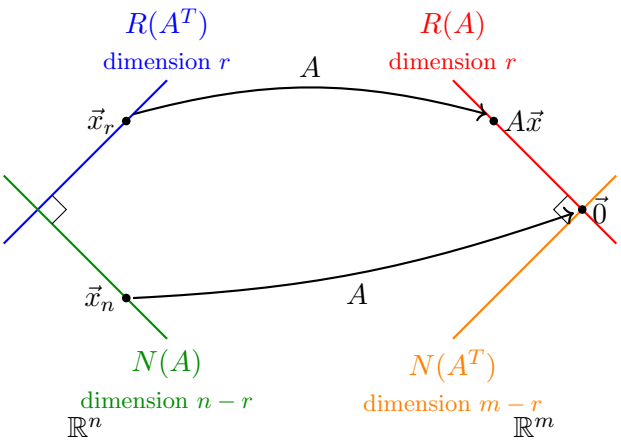
- This split is what happens in the Least Squares Data Fitting topic. There the normal equations said $X^T(\vec{y} - X\vec{\alpha}) = \vec{0}$, so the residual vector of the best fit is in $N(X^T)$. The best fit $X\vec{\alpha}$ is the piece $\vec{q}$ of $\vec{y}$ in the column space and the residual is the piece $\vec{r}$.

- Now we can also see what A does to an $\vec{x}$. With $\mathbb{R}^n = R(A^T) \oplus N(A)$ we split $\vec{x} = \vec{x}_r + \vec{x}_n$ into a row space piece and a null space piece, and since $A\vec{x}_n = \vec{0}$;

$$A\vec{x} = A\vec{x}_r + A\vec{x}_n = A\vec{x}_r$$

- For our A take $\vec{x} = (3,1,3)^T = (1,2,3)^T + (2,-1,0)^T$, where the second piece is $-(-2,1,0)^T \in N(A)$. Then $A\vec{x} = A(1,2,3)^T = (1+4+9,\ 2+8+18)^T = (14,28)^T$, and the null space piece added nothing.

- So; A only sees the row space piece of $\vec{x}$ and carries it into the column space, while the null space piece dies. The left null space is the part of $\mathbb{R}^m$ that A never reaches. All of this is shown in the following sketch, where every space is drawn as a line whatever its dimension;



Worked Examples

Example: Let;

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 5 & 1 \end{bmatrix} \qquad \text{and} \qquad B = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & -1 \\ 3 & 3 & 0 \end{bmatrix}$$

- Before any calculation one should write down where each subspace lives. For $A \in \mathbb{R}^{2\times 3}$ the null space and the row space are in $\mathbb{R}^3$, while the column space and the left null space are in $\mathbb{R}^2$. For $B \in \mathbb{R}^{3\times 3}$ all four are in $\mathbb{R}^3$. A basis vector with the wrong number of entries is the first sign of a mix up.

a) Find the rank of A and a basis of each of its four subspaces.

- We start with the null space, because rank nullity then gives us the rank for free. $A\vec{x} = \vec{0}$ is;

$$\begin{aligned} x_1 + 2x_2 &= 0 \\ 2x_1 + 5x_2 + x_3 &= 0 \end{aligned}$$

- Subtracting 2 times the first row from the second we have $x_2 + x_3 = 0$. So we have two equations for three unknowns and one of them is free. With $x_3 = t$ we get $x_2 = -t$ and $x_1 = -2x_2 = 2t$;

$$N(A) = \text{span}\left\{(2, -1, 1)^T\right\}$$

- Then rank nullity gives $r = n - \dim N(A) = 3 - 1 = 2$.

- The row space is the span of the rows. The rows $(1, 2, 0)$ and $(2, 5, 1)$ are not multiples of each other, so both of them stay;

$$R(A^T) = \text{span}\left\{(1, 2, 0)^T,\ (2, 5, 1)^T\right\}$$

and its dimension 2 is the rank again, as the theorem says.

- The column space has the dimension $r = 2$ as well, and it is a subspace of $\mathbb{R}^2$. Two independent vectors of $\mathbb{R}^2$ are already a basis of $\mathbb{R}^2$, so $R(A)$ is the whole plane. As its basis we can take the first two columns, which are not multiples of each other;

$$R(A) = \text{span}\left\{(1, 2)^T,\ (2, 5)^T\right\} = \mathbb{R}^2$$

- For the left null space we solve $A^T\vec{z} = \vec{0}$, which is $z_1(1, 2, 0)^T + z_2(2, 5, 1)^T = \vec{0}$. The third entry says $z_2 = 0$ and then the first says $z_1 = 0$. So; $N(A^T) = \{\vec{0}\}$, and its dimension is $m - r = 2 - 2 = 0$.

- Notice what this means for $A\vec{x} = \vec{b}$. Since $R(A) = \mathbb{R}^2$ every $\vec{b}$ can be reached, and since $N(A) \neq \{\vec{0}\}$ there are always infinitely many solutions.

b) Find the rank of B and a basis of each of its four subspaces.

- Again we start with $B\vec{x} = \vec{0}$. We subtract 2 times the first row from the second and 3 times the first row from the third;

$$\begin{aligned} x_1 + 2x_2 + x_3 &= 0 \\ -3x_2 - 3x_3 &= 0 \\ -3x_2 - 3x_3 &= 0 \end{aligned}$$

- The last two rows are the same, so we have two equations for three unknowns and one free variable. With $x_3 = t$ we get $x_2 = -t$ and $x_1 = -2x_2 - x_3 = 2t - t = t$;

$$N(B) = \text{span}\left\{(1, -1, 1)^T\right\}$$

- So $r = 3 - 1 = 2$. Let's see it in the rows. The third row is the sum of the first two, $(1 + 2,\ 2 + 1,\ 1 - 1) = (3, 3, 0)$, while the first two are not multiples of each other;

$$R(B^T) = \text{span}\left\{(1, 2, 1)^T,\ (2, 1, -1)^T\right\}$$

- For the columns the null space vector already tells us which combination dies. $B(1, -1, 1)^T = \vec{0}$ says that the first column minus the second plus the third is $\vec{0}$, so the third column is the second minus the first, $(2 - 1,\ 1 - 2,\ 3 - 3)^T = (1, -1, 0)^T$. The first two columns are not multiples of each other, so they are a basis;

$$R(B) = \text{span}\left\{(1, 2, 3)^T,\ (2, 1, 3)^T\right\}$$

which is a plane through the origin of $\mathbb{R}^3$.

- For the left null space we solve $B^T\vec{z}=\vec{0}$, whose rows are the columns of B;

$$\begin{aligned} z_1+2z_2+3z_3&=0\\ 2z_1+z_2+3z_3&=0\\ z_1-z_2&=0 \end{aligned}$$

- The third row gives $z_2=z_1$, and then the first one is $3z_1+3z_3=0$, so $z_3=-z_1$. With $z_1=1$ we have;

$$N(B^T)=\text{span}\left\{(1,1,-1)^T\right\}$$

- Notice that this is the row relation we found above, the first row plus the second minus the third is zero, written as $\vec{z}^TB=\vec{0}^T$. So the null space collects the combinations of the columns that die, and the left null space the combinations of the rows.

c) For both matrices check that the null space is orthogonal to the row space and the left null space to the column space, and that the dimensions add up to n in $\mathbb{R}^n$ and to m in $\mathbb{R}^m$.

- For A we take the inner products of the null space vector with the two rows;

$$\langle(1,2,0)^T,(2,-1,1)^T\rangle=2-2+0=0 \qquad \langle(2,5,1)^T,(2,-1,1)^T\rangle=4-5+1=0$$

and $N(A^T)=\{\vec{0}\}$ is orthogonal to everything. The dimensions are $2+1=3$ in $\mathbb{R}^3$ and $2+0=2$ in $\mathbb{R}^2$.

- For B we have the rows against the null space on the left and the columns against the left null space on the right;

$$\begin{aligned} \langle(1,2,1)^T,(1,-1,1)^T\rangle&=1-2+1=0 & \langle(1,2,3)^T,(1,1,-1)^T\rangle&=1+2-3=0\\ \langle(2,1,-1)^T,(1,-1,1)^T\rangle&=2-1-1=0 & \langle(2,1,3)^T,(1,1,-1)^T\rangle&=2+1-3=0 \end{aligned}$$

- The dimensions are $2+1=3$ in both copies of $\mathbb{R}^3$. The found results are consistent with the theorems.

Your Turn

Let;

$$M = \begin{bmatrix} 1 & 1 & 2 & 1 \\ 1 & 2 & 3 & 0 \\ 2 & 3 & 5 & 1 \end{bmatrix}$$

a) Say in which space each of the four subspaces of M lives, and find the rank of M.

b) Find bases of $N(M)$ and $R(M^T)$, and check that every vector of one basis is orthogonal to every vector of the other.

c) Find bases of $R(M)$ and $N(M^T)$, and check the same for them.

d) Check that $\dim R(M^T) + \dim N(M) = 4$ and $\dim R(M) + \dim N(M^T) = 3$.

Your Turn

Let;

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 2 \end{bmatrix} \qquad \vec{b}_1 = (1,3,4)^T \qquad \vec{b}_2 = (3,0,3)^T$$

a) Find a basis of $N(A^T)$.

b) Decide whether $\vec{b}_1 \in R(A)$. If it is, solve $A\vec{x} = \vec{b}_1$ and say how many solutions there are.

c) Show that $\vec{b}_2 \notin R(A)$, and split it into $\vec{b}_2 = \vec{q} + \vec{r}$ with $\vec{q} \in R(A)$ and $\vec{r} \in N(A^T)$.

d) Find the $\vec{x}$ with $A\vec{x} = \vec{q}$, and check that $\|\vec{b}_2\|^2 = \|\vec{q}\|^2 + \|\vec{r}\|^2$.

0.1.11 Eigenvalues and Diagonalization

- A matrix usually turns a vector into a new direction, but for most square matrices there are a few directions that it only stretches. Along such a direction the matrix acts like a single number. If we can build a basis out of these directions, the whole matrix acts as a list of numbers, one for each direction, and hard things like the powers A^k become powers of numbers.

Definition: Let $A \in \mathbb{R}^{n \times n}$. A vector $\vec{x} \neq \vec{0}$ with;

$$A\vec{x} = \lambda\,\vec{x}$$

is called an **eigenvector** of A, and the number λ is called the **eigenvalue** that belongs to it.

- Meaning A only stretches $\vec{x}$ by the factor λ, and a negative λ also flips it. We leave the zero vector out, because $A\vec{0} = \lambda\,\vec{0}$ holds for every λ and would tell us nothing.

- To find them we move everything to one side, $(A - \lambda I)\,\vec{x} = \vec{0}$. So an eigenvector is a nonzero vector of the null space of $A - \lambda I$. By the Four Subspaces pages such a vector exists exactly when the columns of $A - \lambda I$ are dependent, its rank is below n, and for a square matrix this happens exactly when its determinant is zero;

$$\lambda \text{ is an eigenvalue of } A \quad \Longleftrightarrow \quad \det(A - \lambda I) = 0$$

Definition: The polynomial $p(\lambda) = \det(A - \lambda I)$ is called the **characteristic polynomial** of A. For an eigenvalue λ the null space;

$$\mathcal{E}_\lambda = N(A - \lambda I)$$

is called the **eigenspace** of λ.

- The eigenspace holds all the eigenvectors of λ together with $\vec{0}$. Being a null space it is a subspace, so any nonzero multiple of an eigenvector is an eigenvector too, and we usually pick the one with the nicest numbers.

- $p(\lambda)$ is a polynomial of degree n, so it has n roots when we count them with their multiplicities, but some of them may be complex. The rotation by $90°$, the matrix with the rows $(0, -1)$ and $(1, 0)$, turns every direction of the plane, so there is no real direction that it only stretches. Indeed its $p(\lambda) = \lambda^2 + 1$ has the roots $\lambda = \pm i$.

Example: Find the eigenvalues and the eigenspaces of;

$$A = \begin{bmatrix} 4 & -5 \\ 2 & -3 \end{bmatrix}$$

- First we need the characteristic polynomial;

$$\begin{aligned} p(\lambda) = \det \begin{bmatrix} 4-\lambda & -5 \\ 2 & -3-\lambda \end{bmatrix} &= (4-\lambda)(-3-\lambda) - (-5) \times 2 \\ &= \lambda^2 - \lambda - 12 + 10 \\ &= \lambda^2 - \lambda - 2 \\ &= (\lambda - 2)(\lambda + 1) \end{aligned}$$

  so the eigenvalues are $\lambda_1 = 2$ and $\lambda_2 = -1$.

- For $\lambda_1 = 2$ we solve $(A - 2I)\,\vec{x} = \vec{0}$ with $\vec{x} = (a, b)^T$;

$$\begin{bmatrix} 2 & -5 \\ 2 & -5 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \Rightarrow \quad 2a = 5b$$

- Both rows say the same thing, so we are free to choose one of a and b. Choosing $b = 1$ would give $a = 5/2$, so to avoid the fraction I choose $b = 2$ and then $a = 5$;

$$\mathcal{E}_2 = \text{span}\left\{(5, 2)^T\right\}$$

- For $\lambda_2 = -1$ we have $A - (-1)I = A + I$;

$$\begin{bmatrix} 5 & -5 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \Rightarrow \quad a = b \quad \Rightarrow \quad \mathcal{E}_{-1} = \text{span}\left\{(1, 1)^T\right\}$$

- Notice that in both cases the rows of $A - \lambda I$ came out as multiples of each other. For a 2×2 matrix they have to, we chose λ so that the matrix is singular. If they do not, there is a mistake in the eigenvalue.

0.1.11.1 Diagonalization

- Let's put n eigenvectors $\vec{x}_1, \ldots, \vec{x}_n$ into the columns of a matrix X and their eigenvalues into the diagonal matrix $\Lambda = \text{diag}(\lambda_1, \ldots, \lambda_n)$. Multiplying column by column;

$$AX = \left[A\vec{x}_1 \mid \cdots \mid A\vec{x}_n\right] = \left[\lambda_1\vec{x}_1 \mid \cdots \mid \lambda_n\vec{x}_n\right] = X\Lambda$$

  where in the last step multiplying X from the right with a diagonal matrix scales each column of X with its diagonal entry.

- If the n eigenvectors are linearly independent, X has an inverse and we can multiply $AX = X\Lambda$ from the right with X^{-1};

$$A = X\Lambda X^{-1}$$

  We call this the **diagonalization** of A.

- Let's read it from the right. $X^{-1}\vec{x}$ is the coordinates of $\vec{x}$ in the basis of the eigenvectors, the $[\vec{x}]_{\mathcal{B}} = A^{-1}\vec{x}$ of the Basis and Dimension pages with the eigenvectors as the columns. Then Λ multiplies each coordinate with its own eigenvalue, and X builds the vector back from its coordinates. So in the basis of its eigenvectors we can think of A as just the list $\lambda_1, \dots, \lambda_n$.
- The powers come for free, since the inner $X^{-1}X$ pairs cancel when we multiply A with itself;

$$A^k = X\Lambda X^{-1}\, X\Lambda X^{-1} \cdots X\Lambda X^{-1} = X\Lambda^k X^{-1} \qquad \text{with} \qquad \Lambda^k = \operatorname{diag}(\lambda_1^k, \dots, \lambda_n^k)$$

Example: Let A be the matrix of the previous example, with the rows $(4, -5)$ and $(2, -3)$, the eigenvalues $\lambda_1 = 2$ and $\lambda_2 = -1$ and the eigenvectors $(5, 2)^T$ and $(1, 1)^T$.

a) Write X, Λ and X^{-1}.

- We put the eigenvectors into the columns in the order of their eigenvalues;

$$X = \begin{bmatrix} 5 & 1 \\ 2 & 1 \end{bmatrix} \qquad \Lambda = \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix}$$

- The determinant of X is $5 \times 1 - 1 \times 2 = 3$. For a 2×2 matrix the inverse swaps the diagonal entries, flips the signs of the other two and divides by the determinant, so;

$$X^{-1} = \frac{1}{3}\begin{bmatrix} 1 & -1 \\ -2 & 5 \end{bmatrix}$$

b) Show $AX = X\Lambda$ column by column.

- The first column of AX is A times the first eigenvector, and the second is A times the second;

$$A\begin{bmatrix} 5 \\ 2 \end{bmatrix} = \begin{bmatrix} 20 - 10 \\ 10 - 6 \end{bmatrix} = \begin{bmatrix} 10 \\ 4 \end{bmatrix} = 2\begin{bmatrix} 5 \\ 2 \end{bmatrix} \qquad A\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 - 5 \\ 2 - 3 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \end{bmatrix} = -1\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

- On the other side $X\Lambda$ scales the first column of X by 2 and the second by -1;

$$X\Lambda = \begin{bmatrix} 5 \times 2 & 1 \times (-1) \\ 2 \times 2 & 1 \times (-1) \end{bmatrix} = \begin{bmatrix} 10 & -1 \\ 4 & -1 \end{bmatrix}$$

which has the same columns as AX.

c) Multiply out $X\Lambda X^{-1}$ and check that it gives A.

- Using $X\Lambda$ with the rows $(10, -1)$ and $(4, -1)$ from part b) and X^{-1} from part a);

$$\begin{aligned} X\Lambda X^{-1} &= \frac{1}{3}\begin{bmatrix} 10 & -1 \\ 4 & -1 \end{bmatrix}\begin{bmatrix} 1 & -1 \\ -2 & 5 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 10 + 2 & -10 - 5 \\ 4 + 2 & -4 - 5 \end{bmatrix} \\ &= \frac{1}{3}\begin{bmatrix} 12 & -15 \\ 6 & -9 \end{bmatrix} = \begin{bmatrix} 4 & -5 \\ 2 & -3 \end{bmatrix} = A \end{aligned}$$

Example: For the same $A = X\Lambda X^{-1}$ of the previous example, with the rows $(4,-5)$ and $(2,-3)$, the eigenvalues $\lambda_1 = 2$ and $\lambda_2 = -1$ and the eigenvectors $(5,2)^T$ and $(1,1)^T$, let $\vec{x} = (1,0)^T$.

a) Write $\vec{x}$ as a combination of the eigenvectors $(5,2)^T$ and $(1,1)^T$.

- The coefficients are the coordinates of $\vec{x}$ in the eigenvector basis;

$$X^{-1}\vec{x} = \frac{1}{3}\begin{bmatrix} 1 & -1 \\ -2 & 5 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1/3 \\ -2/3 \end{bmatrix} \quad \Rightarrow \quad \vec{x} = \frac{1}{3}\begin{bmatrix} 5 \\ 2 \end{bmatrix} - \frac{2}{3}\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

- Let's check, $(5/3 - 2/3,\ 2/3 - 2/3)^T = (1,0)^T$.

b) Find $A\vec{x}$ from this combination and check it against $A\vec{x}$ calculated directly.

- A multiplies each eigenvector with its eigenvalue, so each coefficient gets multiplied with its eigenvalue;

$$A\vec{x} = \frac{1}{3} \times 2\begin{bmatrix} 5 \\ 2 \end{bmatrix} - \frac{2}{3} \times (-1)\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 10/3 + 2/3 \\ 4/3 + 2/3 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

- If we calculate it directly, $A(1,0)^T$ is the first column of A, which is $(4,2)^T$. Which is the same result.

c) Find $A^k\vec{x}$ for any k, and $A^{10}\vec{x}$.

- Applying A k times multiplies the coefficients with λ_i^k;

$$A^k\vec{x} = \frac{1}{3} \times 2^k\begin{bmatrix} 5 \\ 2 \end{bmatrix} - \frac{2}{3} \times (-1)^k\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

- For $k = 10$ we have $2^{10} = 1024$ and $(-1)^{10} = 1$, so;

$$A^{10}\vec{x} = \frac{1024}{3}\begin{bmatrix} 5 \\ 2 \end{bmatrix} - \frac{2}{3}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5120/3 - 2/3 \\ 2048/3 - 2/3 \end{bmatrix} = \begin{bmatrix} 5118/3 \\ 2046/3 \end{bmatrix} = \begin{bmatrix} 1706 \\ 682 \end{bmatrix}$$

- Without the eigenvectors we would have needed ten multiplications with A. Notice also that for a large k the 2^k term wins, so $A^k\vec{x}$ turns more and more along $(5,2)^T$, the eigenvector with the larger $|\lambda|$.

0.1.11.2 When It Works

- $A = X\Lambda X^{-1}$ needs n independent eigenvectors, otherwise X has no inverse. Each eigenvalue brings as many independent eigenvectors as the dimension of its eigenspace, so we have to count them.

Definition: Let λ_i be an eigenvalue of A. Its **algebraic multiplicity** a_i is the power of $(\lambda - \lambda_i)$ in $p(\lambda)$, and its **geometric multiplicity** is $g_i = \dim \mathcal{E}_{\lambda_i}$.

- Meaning a_i counts how many times λ_i is a root of the polynomial, and g_i counts how many independent eigenvectors it really has. Without proof I present;

$$1 \leq g_i \leq a_i$$

- The a_i add up to n, the degree of p. Eigenvectors of different eigenvalues are always linearly independent (also without proof), so the bases of all the eigenspaces together give $\sum_i g_i$ independent eigenvectors. We have n of them exactly when $g_i = a_i$ for every eigenvalue. Then A is called **diagonalizable**, and this is the case where everything is fine.
- If all of the n eigenvalues are different, every $a_i = 1$ and so $1 \le g_i \le 1$ forces $g_i = 1$. Hence; a matrix with distinct eigenvalues is always diagonalizable, as our A with 2 and -1 was.

Example: Let;

$$J = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \qquad \text{and} \qquad I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

a) Find the eigenvalues of J with their algebraic and geometric multiplicities, and decide whether J is diagonalizable.

- $J - \lambda I$ is triangular, so its determinant is the product of its diagonal;

$$p(\lambda) = (1-\lambda)(1-\lambda) = (\lambda - 1)^2$$

so $\lambda = 1$ is the only eigenvalue, with $a = 2$.

- For its eigenspace we solve $(J - I)\,\vec{x} = \vec{0}$;

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} b \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \Rightarrow \quad b = 0 \quad \Rightarrow \quad \mathcal{E}_1 = \text{span}\left\{(1,0)^T\right\}$$

- So $g = 1 < 2 = a$. There is only one eigen direction and we can not fill the two columns of X, so J is not diagonalizable. Geometrically J slides every point along the x_1-axis by its x_2, a shear, and only the x_1-axis keeps its direction.

b) Do the same for I.

- For I we have $p(\lambda) = (1-\lambda)^2$ as well, so again $\lambda = 1$ with $a = 2$. But $I - I$ is the zero matrix and every vector solves $(I - I)\,\vec{x} = \vec{0}$, so $\mathcal{E}_1 = \mathbb{R}^2$ and $g = 2 = a$. I is diagonalizable, it is already diagonal with $X = I$.
- Notice that the two matrices have the same characteristic polynomial, so $p(\lambda)$ alone can not tell us whether a matrix is diagonalizable. We have to look at the eigenspaces.

- For a matrix that is not diagonalizable the best we can do is the **Jordan form**, a matrix with the eigenvalues on its diagonal and some 1's right above it, like J itself. We will not need it in this course.

0.1.11.3 Determinant and Trace

- Two numbers that we can read off A without solving anything already know the eigenvalues. With the eigenvalues counted with their algebraic multiplicities;

$$\det A = \prod_{i=1}^{n} \lambda_i \qquad \text{and} \qquad \text{tr}\, A = \sum_{i=1}^{n} \lambda_i$$

where the trace $\text{tr}\, A = a_{11} + \cdots + a_{nn}$ is the sum of the diagonal entries.

- The first one comes from $p(\lambda)$. Over the complex numbers it splits into its roots, $p(\lambda) = \det(A - \lambda I) = (\lambda_1 - \lambda)(\lambda_2 - \lambda)\cdots(\lambda_n - \lambda)$, and putting $\lambda = 0$ we have $\det A = \lambda_1\lambda_2\cdots\lambda_n$. The trace I will give without proof in general.
- For a 2×2 matrix with the rows (a, b) and (c, d) we can see both at once. Opening the determinant and opening the product of the roots;

$$p(\lambda) = (a - \lambda)(d - \lambda) - bc = \lambda^2 - (a + d)\,\lambda + (ad - bc)$$
$$p(\lambda) = (\lambda - \lambda_1)(\lambda - \lambda_2) = \lambda^2 - (\lambda_1 + \lambda_2)\,\lambda + \lambda_1\lambda_2$$

  and comparing the coefficients gives $\operatorname{tr} A = a + d = \lambda_1 + \lambda_2$ and $\det A = ad - bc = \lambda_1\lambda_2$.
- So if you ever forget how to get $p(\lambda)$ of a 2×2 matrix, just remember;

$$p(\lambda) = \lambda^2 - (\operatorname{tr} A)\,\lambda + \det A$$

Example: Check $\det A = \prod_i \lambda_i$ and $\operatorname{tr} A = \sum_i \lambda_i$ for the matrix A with the rows $(4, -5)$ and $(2, -3)$ and the eigenvalues 2 and -1, and for the shear J with the rows $(1, 1)$ and $(0, 1)$ and the eigenvalue 1 of algebraic multiplicity 2.

- For A we have;

$$\det A = 4 \times (-3) - (-5) \times 2 = -12 + 10 = -2 = 2 \times (-1)$$
$$\operatorname{tr} A = 4 + (-3) = 1 = 2 + (-1)$$

  and indeed $p(\lambda) = \lambda^2 - \lambda - 2$ of the first example is $\lambda^2 - (\operatorname{tr} A)\,\lambda + \det A$.
- For J the double eigenvalue counts twice, $\det J = 1\times1 - 1\times0 = 1 = 1\times1$ and $\operatorname{tr} J = 1 + 1 = 2$.
- Then a matrix is invertible exactly when $\det A \neq 0$, which is exactly when none of its eigenvalues is zero. This fits the Four Subspaces pages. $\lambda = 0$ is an eigenvalue when $A\vec{x} = 0\,\vec{x} = \vec{0}$ has a nonzero solution, so $\mathcal{E}_0 = N(A)$, and a nonzero null space is exactly what stops A from being invertible.

Worked Examples

Example: Let;

$$T_1 = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 3 \end{bmatrix} \qquad \text{and} \qquad T_2 = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 3 \end{bmatrix}$$

a) Find the eigenvalues of T_1 and of T_2 with their algebraic multiplicities.

- Both matrices are triangular, so $T_i - \lambda I$ is triangular too and its determinant is the product of its diagonal. The two diagonals are the same, so for both matrices;

$$p(\lambda) = (2-\lambda)(2-\lambda)(3-\lambda) = (2-\lambda)^2(3-\lambda)$$

  and we have $\lambda = 2$ with $a = 2$ and $\lambda = 3$ with $a = 1$.

- Only the repeated eigenvalue can make trouble. For $\lambda = 3$ we have $1 \leq g \leq a = 1$, so $g = 1$ whatever the other entries are.

b) Find the eigenspaces of T_1 with their geometric multiplicities, and decide whether T_1 is diagonalizable.

- For $\lambda = 2$ we solve $(T_1 - 2I)\,\vec{x} = \vec{0}$;

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} b \\ c \\ c \end{bmatrix} = \vec{0} \quad \Rightarrow \quad b = c = 0 \quad \Rightarrow \quad \mathcal{E}_2 = \text{span}\left\{(1,0,0)^T\right\}$$

  so $g = 1 < 2 = a$.

- For $\lambda = 3$ the rows say $-a + b = 0$ and $-b + c = 0$;

$$\begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \vec{0} \quad \Rightarrow \quad a = b = c \quad \Rightarrow \quad \mathcal{E}_3 = \text{span}\left\{(1,1,1)^T\right\}$$

- Together we have $1 + 1 = 2$ independent eigenvectors, one short of the 3 columns of X. Hence; T_1 is not diagonalizable.

c) Do the same for T_2, and if it is diagonalizable write X and Λ.

- For T_2 and $\lambda = 2$ the first row of $T_2 - 2I$ is all zeros;

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \vec{0} \quad \Rightarrow \quad c = 0, \;\; a \text{ and } b \text{ free} \quad \Rightarrow \quad \mathcal{E}_2 = \text{span}\left\{(1,0,0)^T,\ (0,1,0)^T\right\}$$

so here $g = 2 = a$.

- For $\lambda = 3$;

$$\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \vec{0} \quad \Rightarrow \quad a = 0, \;\; b = c \quad \Rightarrow \quad \mathcal{E}_3 = \text{span}\left\{(0,1,1)^T\right\}$$

- So; we have $2 + 1 = 3$ independent eigenvectors and T_2 is diagonalizable, with;

$$X = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \qquad \Lambda = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

- Let's check the last column, $T_2(0,1,1)^T = (0,\ 2+1,\ 3)^T = 3\,(0,1,1)^T$.

- Notice that the two matrices differ in a single entry and have the same characteristic polynomial, like the shear J with the rows $(1,1)$ and $(0,1)$ and the identity I of the previous pages, which both have $p(\lambda) = (\lambda - 1)^2$. The 1 that joins the two 2's of the diagonal is what costs T_1 an eigenvector, while the 1 next to the 3 does no harm.

Your Turn

Let;

$$A = \begin{bmatrix} 2 & 1 \\ 4 & -1 \end{bmatrix} \qquad \vec{x} = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$$

a) Find the eigenvalues and the eigenvectors of A.

b) Check that $\det A$ and $\operatorname{tr} A$ agree with the eigenvalues.

c) Write X, Λ and X^{-1}, and check that $X\Lambda X^{-1} = A$.

d) Write $\vec{x}$ as a combination of the eigenvectors, and find $A^5\vec{x}$.

Your Turn

Let;

$$M = \begin{bmatrix} 4 & 0 & 1 \\ 2 & 3 & 2 \\ -1 & 0 & 2 \end{bmatrix} \qquad \text{and} \qquad K = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{bmatrix}$$

a) Find the eigenvalues of M with their algebraic and geometric multiplicities, and decide whether M is diagonalizable.

b) Do the same for K, and if it is diagonalizable write X and Λ.

0.1.12 Symmetric Matrices

- The Hessian of every function we will minimize is symmetric (the Quadratic Functions pages), so these are the matrices we will meet the most. Luckily they are the nicest case of the previous pages. A symmetric matrix is always diagonalizable, its eigenvalues are real and its eigenvectors can be chosen orthonormal, so the X^{-1} of the diagonalization becomes a plain transpose.

Definition: A matrix $A \in \mathbb{R}^{n\times n}$ is called **symmetric** if $A^T = A$, meaning $a_{ij} = a_{ji}$ for all i and j.

- In other words the entries are mirrored across the diagonal, as in the H of the Quadratic Functions pages with the rows $(8, -2)$ and $(-2, 2)$.

- Everything we need about them is in one theorem, called the **spectral theorem**;

Theorem: Let $A \in \mathbb{R}^{n\times n}$ be symmetric. Then;

1) A has n real eigenvalues $\lambda_1, \ldots, \lambda_n$, counted with their algebraic multiplicities.

2) Its eigenvectors can be chosen orthonormal, $\vec{q}_1, \ldots, \vec{q}_n$ with $A\vec{q}_i = \lambda_i \vec{q}_i$.

3) With the orthogonal matrix $Q = [\vec{q}_1 \mid \cdots \mid \vec{q}_n]$ and $\Lambda = \text{diag}(\lambda_1, \ldots, \lambda_n)$;

$$Q^T A Q = \Lambda \qquad \text{and} \qquad A = Q\Lambda Q^T = \sum_{i=1}^{n} \lambda_i \, \vec{q}_i \vec{q}_i^{\,T}$$

- This Q is not the Q of QR. It is the X of the previous pages, the matrix with the eigenvectors in its columns, but its columns are orthonormal, so it is an orthogonal matrix too and $Q^{-1} = Q^T$ (the Gram-Schmidt and QR pages). So; the diagonalization $A = X\Lambda X^{-1}$ becomes $A = Q\Lambda Q^T$, with no inverse to calculate. Multiplying it from the left with Q^T and from the right with Q, the $Q^T Q = I$ pairs cancel and we have $Q^T A Q = \Lambda$.

- We get the sum by multiplying $Q\Lambda = [\lambda_1 \vec{q}_1 \mid \cdots \mid \lambda_n \vec{q}_n]$ with the rows $\vec{q}_i^{\,T}$ of Q^T, column times row. It is the same opening as $QQ^T = \sum_i \vec{q}_i \vec{q}_i^{\,T} = I$ of the Gram-Schmidt and QR pages, only now every term carries its λ_i.

- $Q^T A Q = \Lambda$ is not the row echelon form of A. Row reduction changes the eigenvalues. For example if we subtract half of the first row of the matrix with the rows $(2, 1)$ and $(1, 2)$ from its second row, we get the rows $(2, 1)$ and $(0, 3/2)$, a triangular matrix with the eigenvalues 2 and $3/2$, while the original one has 1 and 3 (its trace is 4 and its determinant 3). $Q^T A Q = Q^{-1} A Q$ is a change of basis instead, and it keeps the eigenvalues, since $A\vec{x} = \lambda \vec{x}$ gives $(Q^{-1}AQ)(Q^{-1}\vec{x}) = Q^{-1}A\vec{x} = \lambda\, Q^{-1}\vec{x}$.

Example: Let;

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

a) Find its eigenvalues and its normalized eigenvectors.

- With the trace 4 and the determinant $4-1=3$ we can write the characteristic polynomial at once;

$$p(\lambda) = \lambda^2 - 4\lambda + 3 = (\lambda - 3)(\lambda - 1)$$

so the eigenvalues are $\lambda_1 = 3$ and $\lambda_2 = 1$, both real as the theorem says.

- For $\lambda_1 = 3$ we have $A - 3I$ with the rows $(-1, 1)$ and $(1, -1)$, so $a = b$ and the eigenvector is along $(1, 1)^T$. For $\lambda_2 = 1$ the matrix $A - I$ has the rows $(1, 1)$ and $(1, 1)$, so $a = -b$ and the eigenvector is along $(-1, 1)^T$. Both have the length $2^{1/2}$, so the normalized eigenvectors are;

$$\vec{q}_1 = \frac{1}{2^{1/2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \qquad \vec{q}_2 = \frac{1}{2^{1/2}} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

- Notice that $\vec{q}_1^T \vec{q}_2 = \frac{1}{2}(-1 + 1) = 0$. We did not choose them orthogonal, they came out that way.

b) Write Q and check that $Q^T A Q = \Lambda$.

- We put the eigenvectors into the columns of Q;

$$Q = \frac{1}{2^{1/2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \qquad Q^T = \frac{1}{2^{1/2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$

- The two factors $1/2^{1/2}$ give $1/2$ in front, and we first multiply A with the columns, which only scales them by 3 and by 1;

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2+1 & -2+1 \\ 1+2 & -1+2 \end{bmatrix} = \begin{bmatrix} 3 & -1 \\ 3 & 1 \end{bmatrix}$$

- Then we multiply with the rows of Q^T;

$$Q^T A Q = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 3 & -1 \\ 3 & 1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 3+3 & -1+1 \\ -3+3 & 1+1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 6 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} = \Lambda$$

c) Write out the sum $\sum_i \lambda_i \, \vec{q}_i \vec{q}_i^T$ and check that it gives A.

- Each term is an eigenvalue times a column times a row;

$$\lambda_1 \, \vec{q}_1 \vec{q}_1^T = 3 \times \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 3/2 & 3/2 \\ 3/2 & 3/2 \end{bmatrix}$$

$$\lambda_2 \, \vec{q}_2 \vec{q}_2^T = 1 \times \frac{1}{2} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 1/2 & -1/2 \\ -1/2 & 1/2 \end{bmatrix}$$

- Adding the two;

$$\begin{bmatrix} 3/2 + 1/2 & 3/2 - 1/2 \\ 3/2 - 1/2 & 3/2 + 1/2 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} = A$$

0.1.12.1 Why It Holds

- A real matrix can have complex eigenvalues, as the rotation by 90° of the previous pages did, and then its eigenvectors are complex too. So to show that a symmetric matrix has none, we first have to allow $\lambda \in \mathbb{C}$ and $\vec{x} \in \mathbb{C}^n$, and then show that λ is real after all.

Proof: We prove 1) of the theorem. Let $A\vec{x} = \lambda\,\vec{x}$ with $\vec{x} \neq \vec{0}$.

- We multiply it from the left with $\bar{\vec{x}}^T$, the conjugate transpose $\vec{x}^\dagger$;

$$\bar{\vec{x}}^T A\vec{x} = \lambda\,\bar{\vec{x}}^T\vec{x}$$

  where $\bar{\vec{x}}^T\vec{x} = \sum_i \bar{x}_i x_i = \sum_i |x_i|^2$ is a real number and positive since $\vec{x} \neq \vec{0}$.

- Now we take the complex conjugate of $A\vec{x} = \lambda\,\vec{x}$. A is real so it stays as it is, $A\bar{\vec{x}} = \bar{\lambda}\,\bar{\vec{x}}$. Transposing this and using $A^T = A$ we have $\bar{\vec{x}}^T A = \bar{\lambda}\,\bar{\vec{x}}^T$, and multiplying from the right with $\vec{x}$;

$$\bar{\vec{x}}^T A\vec{x} = \bar{\lambda}\,\bar{\vec{x}}^T\vec{x}$$

- The left hand sides are the same number, so $\lambda\,\bar{\vec{x}}^T\vec{x} = \bar{\lambda}\,\bar{\vec{x}}^T\vec{x}$, and we can divide by $\bar{\vec{x}}^T\vec{x} > 0$;

$$\therefore \quad \lambda = \bar{\lambda} \quad \Rightarrow \quad \lambda \in \mathbb{R}$$

- Then $A - \lambda I$ is a real matrix, and its null space has real vectors, so we can take the eigenvectors real as well.

Proposition: Eigenvectors of a symmetric matrix that belong to different eigenvalues are orthogonal.

Proof: Let $A\vec{q_i} = \lambda_i\vec{q_i}$ and $A\vec{q_j} = \lambda_j\vec{q_j}$ with $\lambda_i \neq \lambda_j$. We calculate the number $\vec{q_i}^T A\vec{q_j}$ in two ways;

- Letting A act to the right, and letting it act to the left with $A^T = A$;

$$\vec{q_i}^T A\vec{q_j} = \vec{q_i}^T(\lambda_j\vec{q_j}) = \lambda_j\,\vec{q_i}^T\vec{q_j}$$
$$\vec{q_i}^T A\vec{q_j} = (A^T\vec{q_i})^T\vec{q_j} = (A\vec{q_i})^T\vec{q_j} = \lambda_i\,\vec{q_i}^T\vec{q_j}$$

- Subtracting the two we have $(\lambda_i - \lambda_j)\,\vec{q_i}^T\vec{q_j} = 0$, and since $\lambda_i \neq \lambda_j$;

$$\therefore \quad \vec{q_i}^T\vec{q_j} = 0$$

- So the eigenvectors of different eigenvalues are orthogonal whether we want it or not, and dividing each by its length makes them orthonormal. Only inside a repeated eigenvalue do we have a choice. The rest of the theorem I present without proof. A repeated eigenvalue of a symmetric matrix always has $g_i = a_i$, so its eigenspace has enough independent eigenvectors, and inside it we choose an orthonormal basis with Gram-Schmidt. For I for example every vector is an eigenvector of $\lambda = 1$, and any orthonormal basis will do.

- Everything here holds for a complex matrix with $A^\dagger = A$, which is called **Hermitian**. In the proof of the real eigenvalues the conjugate transpose of $A\vec{x} = \lambda\,\vec{x}$ gives $\vec{x}^\dagger A^\dagger = \bar{\lambda}\,\vec{x}^\dagger$ at once, and with $A^\dagger = A$ the rest goes the same way. Q becomes a unitary matrix, $Q^\dagger Q = I$, with $A = Q\Lambda Q^\dagger$.

0.1.12.2 A Sum of Projections

- Each $\vec{q}_i\vec{q}_i^{\,T}$ in $A = \sum_i \lambda_i\,\vec{q}_i\vec{q}_i^{\,T}$ is the projection matrix onto $\vec{q}_i$ of the Orthogonality and Projection pages. So; a symmetric matrix is a sum of projections onto its eigen directions, each one stretched by its eigenvalue. For $A = I$ all $\lambda_i = 1$ and the sum is $\sum_i \vec{q}_i\vec{q}_i^{\,T} = I$ again.

- Let's apply it to a vector, $\vec{y} = A\vec{x}$;

$$\vec{y} = A\vec{x} = \sum_{i=1}^{n} \lambda_i\,\vec{q}_i\vec{q}_i^{\,T}\vec{x} = \sum_{i=1}^{n} \lambda_i\,\left(\vec{q}_i^{\,T}\vec{x}\right)\vec{q}_i = \sum_{i=1}^{n} \alpha_i\,\vec{q}_i \qquad \text{with} \qquad \alpha_i = \lambda_i\,\vec{q}_i^{\,T}\vec{x}$$

  where in the second step we used that $\vec{q}_i^{\,T}\vec{x}$ is a number and can move to the front.

- Meaning we split $\vec{x}$ into its pieces along the eigen directions, $\vec{q}_i^{\,T}\vec{x}$ being the size of each piece, stretch each piece by its own λ_i and add them up again.

- The α_i are also the coordinates of $\vec{y}$ in the orthonormal basis, $\alpha_i = \langle\vec{y},\vec{q}_i\rangle$, which are the Fourier coefficients of $\vec{y}$ of the Orthogonality and Projection pages. Since $\|\vec{q}_i\| = 1$ there is no denominator. Let's check it with $\vec{q}_i^{\,T}A = (A^T\vec{q}_i)^T = (A\vec{q}_i)^T = \lambda_i\vec{q}_i^{\,T}$;

$$\langle\vec{y},\vec{q}_i\rangle = \vec{q}_i^{\,T}\vec{y} = \vec{q}_i^{\,T}A\vec{x} = \lambda_i\,\vec{q}_i^{\,T}\vec{x} = \alpha_i$$

Remark: This is the finite dimensional version of expanding a function in the eigenfunctions of a Hermitian operator, as we expand a wave function in the energy eigenstates of the Hamiltonian, $\psi = \sum_n c_n\psi_n$ with $c_n = \langle\psi,\psi_n\rangle$, the same form as $\alpha_i = \langle\vec{y},\vec{q}_i\rangle$. The two proofs above only used $A^\dagger = A$ and the inner product, so with the integral as the inner product they go through for the operator as well. That is why its eigenvalues are real and its eigenfunctions of different eigenvalues are orthogonal.

Example: Let A be any matrix with the rows (p,r) and (r,p), like the matrix of the previous example with $p=2$ and $r=1$. All of them have the same eigenvectors, since;

$$A\begin{bmatrix}1\\1\end{bmatrix}=\begin{bmatrix}p+r\\r+p\end{bmatrix}=(p+r)\begin{bmatrix}1\\1\end{bmatrix}\qquad A\begin{bmatrix}-1\\1\end{bmatrix}=\begin{bmatrix}-p+r\\-r+p\end{bmatrix}=(p-r)\begin{bmatrix}-1\\1\end{bmatrix}$$

so the eigenvalues are $\lambda_1=p+r$ and $\lambda_2=p-r$ and the orthonormal eigenvectors are $\vec{q}_1=(1,1)^T/2^{1/2}$ and $\vec{q}_2=(-1,1)^T/2^{1/2}$, whatever p and r are.

a) For $\vec{x}=(1,0)^T$ find $\vec{y}=A\vec{x}$ as $\sum_i \alpha_i\,\vec{q}_i$ with $\alpha_i=\lambda_i\,\vec{q}_i^{\,T}\vec{x}$, and check that $\alpha_i=\langle\vec{y},\vec{q}_i\rangle$.

- First we need the sizes of the pieces of $\vec{x}$ along the eigenvectors;

$$\vec{q}_1^{\,T}\vec{x}=\frac{1}{2^{1/2}}(1+0)=\frac{1}{2^{1/2}}\qquad \vec{q}_2^{\,T}\vec{x}=\frac{1}{2^{1/2}}(-1+0)=-\frac{1}{2^{1/2}}$$

- Stretching them by the eigenvalues we have;

$$\alpha_1=(p+r)\,\vec{q}_1^{\,T}\vec{x}=\frac{p+r}{2^{1/2}}\qquad \alpha_2=(p-r)\,\vec{q}_2^{\,T}\vec{x}=-\frac{p-r}{2^{1/2}}$$

- Then;

$$\vec{y}=\frac{p+r}{2^{1/2}}\times\frac{1}{2^{1/2}}\begin{bmatrix}1\\1\end{bmatrix}-\frac{p-r}{2^{1/2}}\times\frac{1}{2^{1/2}}\begin{bmatrix}-1\\1\end{bmatrix}=\begin{bmatrix}(p+r)/2+(p-r)/2\\(p+r)/2-(p-r)/2\end{bmatrix}=\begin{bmatrix}p\\r\end{bmatrix}$$

  which is the first column of A, as $A(1,0)^T$ should be.

- For the check we take the inner products of $\vec{y}=(p,r)^T$ with the eigenvectors, $\langle\vec{y},\vec{q}_1\rangle=(p+r)/2^{1/2}=\alpha_1$ and $\langle\vec{y},\vec{q}_2\rangle=(-p+r)/2^{1/2}=-(p-r)/2^{1/2}=\alpha_2$.

b) Find where A sends the unit circle, the vectors $\vec{x}$ with $\|\vec{x}\| = 1$, for $p \neq \pm r$.

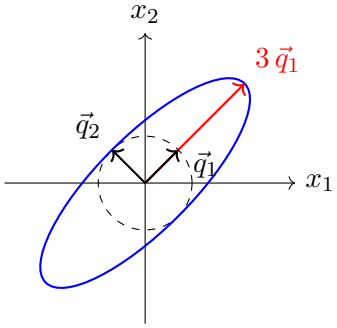
- In the orthonormal basis every unit vector is $\vec{x} = \cos t\, \vec{q}_1 + \sin t\, \vec{q}_2$, since its length squared is $\cos^2 t + \sin^2 t = 1$. Each piece gets stretched by its eigenvalue;

$$A\vec{x} = \lambda_1 \cos t\, \vec{q}_1 + \lambda_2 \sin t\, \vec{q}_2$$

- So the coordinates of $A\vec{x}$ along $\vec{q}_1$ and $\vec{q}_2$ are $y_1 = \lambda_1 \cos t$ and $y_2 = \lambda_2 \sin t$. With $p \neq \pm r$ both eigenvalues are nonzero and we can divide by them;

$$\left(\frac{y_1}{\lambda_1}\right)^2 + \left(\frac{y_2}{\lambda_2}\right)^2 = \cos^2 t + \sin^2 t = 1$$

- This is an ellipse with the half axis $|p + r|$ along $\vec{q}_1$ and $|p - r|$ along $\vec{q}_2$. The sketch shows it in blue for $p = 2$ and $r = 1$, with the half axes 3 and 1. The eigenvectors are the axes of the ellipse and the absolute values of the eigenvalues are their lengths. For $p = \pm r$ one eigenvalue is zero and the ellipse gets squashed into a segment.



Worked Examples

Example: Let;

$$S = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{bmatrix}$$

a) Find the eigenvalues of S.

- We open $\det(S - \lambda I)$ along the first row;

$$\begin{aligned} p(\lambda) &= (2-\lambda)\left[(3-\lambda)(2-\lambda) - 1\right] - 1 \times \left[1 \times (2-\lambda) - 1 \times 0\right] \\ &= (2-\lambda)\left[(3-\lambda)(2-\lambda) - 1 - 1\right] \\ &= (2-\lambda)\left(\lambda^2 - 5\lambda + 4\right) \\ &= (2-\lambda)(\lambda-1)(\lambda-4) \end{aligned}$$

  where in the second step we took the common factor $2 - \lambda$ out. So the eigenvalues are $\lambda_1 = 4$, $\lambda_2 = 2$ and $\lambda_3 = 1$, all real as the spectral theorem says, and all different.

- Let's check them with the trace and the determinant, $2 + 3 + 2 = 7 = 4 + 2 + 1$ and $\det S = 2 \times (6 - 1) - 1 \times 2 = 8 = 4 \times 2 \times 1$.

b) Find its normalized eigenvectors, and check that $Q^T Q = I$.

- For $\lambda_1 = 4$ we solve $(S - 4I)\,\vec{x} = \vec{0}$ with $\vec{x} = (a, b, c)^T$;

$$\begin{bmatrix} -2 & 1 & 0 \\ 1 & -1 & 1 \\ 0 & 1 & -2 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \vec{0} \quad \Rightarrow \quad b = 2a, \;\; b = 2c \quad \Rightarrow \quad (1, 2, 1)^T$$

  where the first and the third row gave the two conditions, and the second row checks them, $1 - 2 + 1 = 0$.

- For $\lambda_2 = 2$;

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \vec{0} \quad \Rightarrow \quad b = 0, \quad a + c = 0 \quad \Rightarrow \quad (1, 0, -1)^T$$

- For $\lambda_3 = 1$;

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \vec{0} \quad \Rightarrow \quad a = -b, \quad c = -b \quad \Rightarrow \quad (1, -1, 1)^T$$

  where I chose $b = -1$ to keep the first entry positive.

- Their lengths are $(1+4+1)^{1/2} = 6^{1/2}$, $(1+0+1)^{1/2} = 2^{1/2}$ and $(1+1+1)^{1/2} = 3^{1/2}$, so the normalized eigenvectors are;

$$\vec{q}_1 = \frac{1}{6^{1/2}} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \qquad \vec{q}_2 = \frac{1}{2^{1/2}} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \qquad \vec{q}_3 = \frac{1}{3^{1/2}} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

- The entries of $Q^T Q$ are the inner products $\vec{q}_i^T \vec{q}_j$ of the columns of Q. The diagonal ones are 1 since we normalized, and for the others it is enough to take the vectors before the division;

$$\begin{aligned} \langle (1,2,1)^T, (1,0,-1)^T \rangle &= 1 + 0 - 1 = 0 \\ \langle (1,2,1)^T, (1,-1,1)^T \rangle &= 1 - 2 + 1 = 0 \\ \langle (1,0,-1)^T, (1,-1,1)^T \rangle &= 1 + 0 - 1 = 0 \end{aligned}$$

  so $Q^T Q = I$. We did not have to make them orthogonal. The three eigenvalues are different, so the eigenvectors came out orthogonal by themselves.

c) Write the spectral decomposition $S = Q\Lambda Q^T$ and the sum $\sum_i \lambda_i \, \vec{q}_i \vec{q}_i^{\,T}$, and check that the sum gives S.

- We put the eigenvectors into the columns of Q in the order of their eigenvalues, and then $S = Q\Lambda Q^T$ with;

$$Q = \begin{bmatrix} 1/6^{1/2} & 1/2^{1/2} & 1/3^{1/2} \\ 2/6^{1/2} & 0 & -1/3^{1/2} \\ 1/6^{1/2} & -1/2^{1/2} & 1/3^{1/2} \end{bmatrix} \qquad \Lambda = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- In the sum the roots square away, since $\vec{q}_1 \vec{q}_1^{\,T} = \frac{1}{6}\,(1,2,1)^T(1,2,1)$ and so on. So the three terms are matrices of fractions, and over the common denominator 3 they are;

$$\lambda_1 \, \vec{q}_1 \vec{q}_1^{\,T} = 4 \times \frac{1}{6} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2 & 4 & 2 \\ 4 & 8 & 4 \\ 2 & 4 & 2 \end{bmatrix}$$

$$\lambda_2 \, \vec{q}_2 \vec{q}_2^{\,T} = 2 \times \frac{1}{2} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & 1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 3 & 0 & -3 \\ 0 & 0 & 0 \\ -3 & 0 & 3 \end{bmatrix}$$

$$\lambda_3 \, \vec{q}_3 \vec{q}_3^{\,T} = 1 \times \frac{1}{3} \begin{bmatrix} 1 & -1 & 1 \\ -1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & -1 & 1 \\ -1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$$

- Adding the three we have;

$$\frac{1}{3} \begin{bmatrix} 2+3+1 & 4+0-1 & 2-3+1 \\ 4+0-1 & 8+0+1 & 4+0-1 \\ 2-3+1 & 4+0-1 & 2+3+1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 6 & 3 & 0 \\ 3 & 9 & 3 \\ 0 & 3 & 6 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{bmatrix} = S$$

- Notice that every term is symmetric on its own, as every $\vec{q}\,\vec{q}^{\,T}$ is. One should keep the eigenvalues and the eigenvectors in the same order in Λ and in Q, otherwise each λ_i stretches the wrong direction.

Your Turn

Let;

$$A = \begin{bmatrix} 3 & 2 \\ 2 & 0 \end{bmatrix}$$

a) Find the eigenvalues and the normalized eigenvectors of A.

b) Write Q and Λ, and check that $Q^T Q = I$.

c) Write A as the sum $\sum_i \lambda_i \, \vec{q}_i \vec{q}_i^{\,T}$, and check that it gives A.

d) Let $\vec{u}$ and $\vec{v}$ be any two vectors with $A\vec{u} = 4\,\vec{u}$ and $A\vec{v} = -\vec{v}$. Without using the entries of A, show that $\vec{u} \perp \vec{v}$ by calculating $\vec{u}^T A \vec{v}$ in two ways.

0.1.13 Definite Matrices

- In the Quadratic Functions pages we have seen that whether a stationary point $\vec{x}^*$ is a minimizer comes down to the sign of $\vec{z}^T H \vec{z}$. The definition of positive definite asks for this sign for every $\vec{z} \neq \vec{0}$, and we can not try infinitely many vectors one by one.
- So; we need tests that we can actually compute from the numbers of the matrix. We will see three of them, the eigenvalues, the minors and the Cholesky decomposition, but first we need the language of quadratic forms.

0.1.13.1 Quadratic Forms

Definition: For $A \in \mathbb{R}^{n \times n}$ the function;

$$f(\vec{x}) = \vec{x}^T A \vec{x} = \sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij}\, x_i x_j$$

is called a **quadratic form**, also a pure quadratic form.

- Meaning every term has degree two. A square x_i^2 comes with the coefficient a_{ii}, and a product $x_i x_j$ with $i \neq j$ appears twice in the sum, so it comes with $a_{ij} + a_{ji}$.
- If we add a linear part and a number we have $\vec{x}^T A \vec{x} + \vec{c}^T \vec{x} + \alpha$, which is the quadratic function of the Quadratic Functions pages (there with a half in front of the form).

Example: Let A be the matrix below and $H = (A + A^T)/2$ its symmetric version. Show that $\vec{x}^T A \vec{x}$ and $\vec{x}^T H \vec{x}$ are the same function of $\vec{x} \in \mathbb{R}^2$.

$$A = \begin{bmatrix} 1 & 0 \\ 2 & 4 \end{bmatrix}$$

- First let's form the symmetric version;

$$H = \frac{1}{2}\left(\begin{bmatrix} 1 & 0 \\ 2 & 4 \end{bmatrix} + \begin{bmatrix} 1 & 2 \\ 0 & 4 \end{bmatrix} \right) = \frac{1}{2} \begin{bmatrix} 2 & 2 \\ 2 & 8 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 4 \end{bmatrix}$$

- Then we multiply the form of A out, first $A\vec{x}$ and then the row in front;

$$\vec{x}^T A \vec{x} = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} x_1 \\ 2x_1 + 4x_2 \end{bmatrix} = x_1^2 + 2x_1 x_2 + 4x_2^2$$

- And the same for H;

$$\begin{aligned} \vec{x}^T H \vec{x} &= \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} x_1 + x_2 \\ x_1 + 4x_2 \end{bmatrix} \\ &= x_1^2 + x_1 x_2 + x_1 x_2 + 4x_2^2 \\ &= x_1^2 + 2x_1 x_2 + 4x_2^2 \end{aligned}$$

- So they are the same function. Notice that the coefficient of $x_1 x_2$ is $a_{12} + a_{21} = 0 + 2 = 2$ in both. The matrix A puts all of it into one entry, H shares it equally as 1 and 1.

- This always works. As in the Quadratic Functions pages, $\vec{x}^T A\vec{x}$ is a number and a number is equal to its own transpose, so $\vec{x}^T A\vec{x} = (\vec{x}^T A\vec{x})^T = \vec{x}^T A^T \vec{x}$. Taking half of each we have;

$$\vec{x}^T A\vec{x} = \frac{1}{2}\,\vec{x}^T A\vec{x} + \frac{1}{2}\,\vec{x}^T A^T \vec{x} = \vec{x}^T H\vec{x}$$

- So a quadratic form only sees H, and we can always swap A for H without changing the form at all. This is what "without loss of generality A is symmetric" means. By assuming a symmetric matrix we lose no quadratic form, so from now on the matrix of a quadratic form is always symmetric.

0.1.13.2 The Five Classes

Definition: A symmetric matrix $A \in \mathbb{R}^{n\times n}$ is called;

1) **positive definite** (pd) if $\vec{x}^T A\vec{x} > 0$ for all $\vec{x} \neq \vec{0}$,

2) **positive semidefinite** (psd) if $\vec{x}^T A\vec{x} \geq 0$ for all $\vec{x}$,

3) **negative definite** (nd) if $\vec{x}^T A\vec{x} < 0$ for all $\vec{x} \neq \vec{0}$,

4) **negative semidefinite** (nsd) if $\vec{x}^T A\vec{x} \leq 0$ for all $\vec{x}$,

5) **indefinite** if $\vec{x}^T A\vec{x} > 0$ for some $\vec{x}$ and $\vec{y}^T A\vec{y} < 0$ for some $\vec{y}$.

- Meaning an indefinite matrix takes both signs. Its quadratic function curves up in some directions and down in others, the saddle point of the Quadratic Functions pages. A semidefinite matrix is not indefinite, its form only touches zero and never crosses it.

- Notice that a pd matrix is also psd. Also A is nd exactly when $-A$ is pd, so whatever we learn for the positive side we get for the negative side for free.

- For a diagonal matrix $D = \text{diag}(d_1, d_2)$, with d_1 and d_2 on its diagonal and zeros elsewhere, the form has no mixed terms, $\vec{x}^T D\vec{x} = d_1x_1^2 + d_2x_2^2$. So we can read the class by eye;

  1) $\text{diag}(2,3)$ gives $2x_1^2 + 3x_2^2$, positive unless $\vec{x} = \vec{0}$, so pd.

  2) $\text{diag}(2,0)$ gives $2x_1^2 \geq 0$, but it is zero at $\vec{x} = (0,1)^T$, so psd and not pd.

  3) $\text{diag}(-1,-4)$ gives $-x_1^2 - 4x_2^2$, negative unless $\vec{x} = \vec{0}$, so nd.

  4) $\text{diag}(-1,0)$ gives $-x_1^2 \leq 0$, zero at $\vec{x} = (0,1)^T$, so nsd and not nd.

  5) $\text{diag}(1,-1)$ gives $x_1^2 - x_2^2$, which is 1 at $(1,0)^T$ and -1 at $(0,1)^T$, so indefinite.

0.1.13.3 The Eigenvalue Test

- The $\vec{x}$ in the definition is every nonzero vector of $\mathbb{R}^n$, which is exactly why we can not check it directly. What we can compute are the eigenvalues, and they turn every symmetric matrix into the diagonal case above.

- By the spectral theorem of the Symmetric Matrices pages we have $A = Q\Lambda Q^T$, where the columns of Q are orthonormal eigenvectors $\vec{q}_i$ of A and $\Lambda = \text{diag}(\lambda_1, \ldots, \lambda_n)$. This is the orthogonal Q of eigenvectors, not the Q of QR, but it has the same property $Q^TQ = QQ^T = I$. Let's put it into the form and write $\vec{y} = Q^T\vec{x}$, the coordinates of $\vec{x}$ in the eigenvectors;

$$\vec{x}^T A\vec{x} = \vec{x}^T Q\Lambda Q^T\vec{x} = (Q^T\vec{x})^T\,\Lambda\,(Q^T\vec{x}) = \vec{y}^T\Lambda\,\vec{y} = \sum_{i=1}^{n} \lambda_i\, y_i^2$$

- Since Q is invertible, $\vec{y} \neq \vec{0}$ exactly when $\vec{x} \neq \vec{0}$. So the form of A takes the same values as the diagonal form $\sum \lambda_i y_i^2$, and we read the class from the signs of the λ_i as we did by eye.

Theorem: Let $A \in \mathbb{R}^{n \times n}$ be symmetric with the eigenvalues $\lambda_1, \ldots, \lambda_n$. Then A is;

1) pd if and only if $\lambda_i > 0$ for all i,

2) psd if and only if $\lambda_i \geq 0$ for all i,

3) nd if and only if $\lambda_i < 0$ for all i,

4) nsd if and only if $\lambda_i \leq 0$ for all i,

5) indefinite if and only if $\lambda_i > 0$ for some i and $\lambda_j < 0$ for some j.

- The way back is even shorter. If A is pd we put one of its own eigenvectors into the form, $\vec{q}_i^T A \vec{q}_i = \lambda_i \, \vec{q}_i^T \vec{q}_i = \lambda_i > 0$, and the same works for the other classes.

- So this is why one computes eigenvalues to classify a Hessian. We never try vectors, the eigenvectors are the only directions we need.

- A pd matrix has $\det A = \prod_i \lambda_i > 0$ by the Eigenvalues pages, so it is invertible and the stationary point $\vec{x}^* = -H^{-1}\vec{c}$ of the Quadratic Functions pages exists and is unique.

Example: Let H be the matrix of the Quadratic Functions pages and B the matrix below.

$$H = \begin{bmatrix} 8 & -2 \\ -2 & 2 \end{bmatrix} \qquad\qquad B = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$$

a) Classify H with its eigenvalues.

- The characteristic polynomial is;

$$\det(H - \lambda I) = (8 - \lambda)(2 - \lambda) - (-2)(-2) = \lambda^2 - 10\lambda + 16 - 4 = \lambda^2 - 10\lambda + 12$$

- With the quadratic formula we have;

$$\lambda = \frac{10 \pm (100 - 48)^{1/2}}{2} = \frac{10 \pm 2\sqrt{13}}{2} = 5 \pm \sqrt{13}$$

- Since $\sqrt{13} < \sqrt{25} = 5$ both eigenvalues are positive, $\lambda_1 = 5 + \sqrt{13} \approx 8.61$ and $\lambda_2 = 5 - \sqrt{13} \approx 1.39$, so H is pd. Which is the same result we obtained with the sum of squares $6z_1^2 + 2(z_1 - z_2)^2$ in those pages.

b) Classify B with its eigenvalues and find one vector with a positive form and one with a negative form.

- Here we have;

$$\det(B-\lambda I) = (1-\lambda)^2 - 4 = 0 \quad \Rightarrow \quad 1-\lambda = \pm 2 \quad \Rightarrow \quad \lambda_1 = 3, \;\; \lambda_2 = -1$$

so one eigenvalue is positive and one is negative, and B is indefinite.

- The vectors we look for are the eigenvectors. For $\lambda_1 = 3$ the rows of $B - 3I$ are $(-2, 2)$ and $(2, -2)$, so $x_1 = x_2$ and $\vec{x} = (1, 1)^T$. For $\lambda_2 = -1$ the rows of $B + I$ are $(2, 2)$ twice, so $y_1 = -y_2$ and $\vec{y} = (1, -1)^T$. Then;

$$\vec{x}^T B \vec{x} = 1 + 2 \times 2 \times 1 \times 1 + 1 = 6 > 0$$
$$\vec{y}^T B \vec{y} = 1 + 2 \times 2 \times 1 \times (-1) + 1 = -2 < 0$$

- Notice that $6 = 3 \times 2$ and $-2 = -1 \times 2$. Along an eigenvector the form is $\lambda_i \|\vec{x}\|_2^2$, the sum $\sum \lambda_i y_i^2$ with a single y_i left in it.

0.1.13.4 Sylvester's Criterion

- The eigenvalues of a 3×3 matrix already need the roots of a cubic. Determinants are much cheaper, and there is a test that only needs them.

Definition: The **leading principal minors** of $A = (a_{ij}) \in \mathbb{R}^{n \times n}$ are the determinants of its upper left $k \times k$ blocks;

$$P_k = \det \begin{bmatrix} a_{11} & \cdots & a_{1k} \\ \vdots & \ddots & \vdots \\ a_{k1} & \cdots & a_{kk} \end{bmatrix} \qquad k = 1, \ldots, n$$

- So $P_1 = a_{11}$, $P_2 = a_{11}a_{22} - a_{12}a_{21}$ and so on up to $P_n = \det A$.

Theorem: (Sylvester's criterion) Let $A \in \mathbb{R}^{n \times n}$ be symmetric. Then;

1) A is pd if and only if $P_k > 0$ for all $k = 1, \ldots, n$,

2) A is nd if and only if the signs alternate starting negative, $P_1 < 0$, $P_2 > 0$, $P_3 < 0$ and so on, meaning $(-1)^k P_k > 0$,

3) if $\det A \neq 0$ and neither of the two patterns holds, A is indefinite,

4) if $\det A = 0$, A is neither pd nor nd, but it may be semidefinite.

- The pattern of 2) comes from $-A$. A is nd exactly when $-A$ is pd, and multiplying a $k \times k$ block by -1 multiplies its determinant by $(-1)^k$. So the minors of $-A$ are $(-1)^k P_k$ and 1) asks them to be positive.

- For 3), $\det A = \prod_i \lambda_i \neq 0$ means no eigenvalue is zero, so A can not be semidefinite without being definite. If it is not pd and not nd, only indefinite is left.

- For a 2×2 matrix it is even shorter. We have $\lambda_1 + \lambda_2 = \operatorname{tr} A$ and $\lambda_1 \lambda_2 = \det A$, and two numbers are both positive exactly when their sum and their product are. So a symmetric

2×2 matrix is pd if and only if $\operatorname{tr} A > 0$ and $\det A > 0$. For H of the previous example, with the rows $(8, -2)$ and $(-2, 2)$, $\operatorname{tr} H = 8 + 2 = 10$ and $\det H = 16 - 4 = 12$, pd once again.

- One should be careful with 4). The leading minors can not detect semidefiniteness, nonnegative minors prove nothing. For example the matrix B with the rows $(2, 2, 2)$, $(2, 2, 2)$ and $(2, 2, -1)$ has $P_1 = 2$, $P_2 = 4 - 4 = 0$ and $P_3 = 0$ (the first two rows are equal), all nonnegative. But $\vec{e}_1^T B \vec{e}_1 = b_{11} = 2 > 0$ and $\vec{e}_3^T B \vec{e}_3 = b_{33} = -1 < 0$, so B is indefinite. With a zero determinant we go back to the eigenvalues.

- The test is for symmetric matrices only. For a nonsymmetric A we first take its symmetric version H, which has the same form. The matrix with the rows $(1, -4)$ and $(0, 1)$ has the minors 1 and 1, which would say pd. However; its H has the rows $(1, -2)$ and $(-2, 1)$ with $P_2 = 1 - 4 = -3 < 0$, so it is indefinite, and indeed $\vec{x} = (1, 1)^T$ gives $1 - 4 + 1 = -2 < 0$.

Example: Classify the three matrices below with Sylvester's criterion and check the result with their eigenvalues.

$$A_1 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \qquad A_2 = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \qquad A_3 = \begin{bmatrix} 1 & -2 & 4 \\ -2 & 2 & 0 \\ 4 & 0 & -7 \end{bmatrix}$$

a) A_1

- The first two minors are $P_1 = 2$ and $P_2 = 2 \times 2 - (-1)(-1) = 3$. For P_3 we expand along the first row;

$$P_3 = 2\,(4 - 1) - (-1)\big((-1) \times 2 - (-1) \times 0\big) + 0 = 6 - 2 = 4$$

- All three are positive, so A_1 is pd.

- For the eigenvalues, for the sake of our mental health let's write $\mu = 2 - \lambda$ on the diagonal and expand the same way;

$$\begin{aligned} \det(A_1 - \lambda I) &= \mu(\mu^2 - 1) - (-1)\big((-1)\,\mu - 0\big) \\ &= \mu^3 - \mu - \mu = \mu(\mu^2 - 2) \end{aligned}$$

- So $\mu = 0$ or $\mu = \pm\sqrt{2}$, which is $\lambda = 2$ and $\lambda = 2 \mp \sqrt{2}$. The eigenvalues $2 - \sqrt{2} \approx 0.59$, 2 and $2 + \sqrt{2} \approx 3.41$ are all positive, consistent with Sylvester.

b) A_2

- The upper left 2×2 block is the same as in part a, so $P_1 = 2$ and $P_2 = 3$. Then;

$$P_3 = 2\,(4-1) - (-1)\big((-1)\times 2 - (-1)(-1)\big) + (-1)\big((-1)(-1) - 2\times(-1)\big) = 6 - 3 - 3 = 0$$

- By 4) A_2 is not definite, and Sylvester tells us nothing more.
- Every row of A_2 sums to zero, so $A_2(1,1,1)^T = \vec{0}$ and $\lambda = 0$ is an eigenvalue, which is where the zero determinant comes from. With $\mu = 2 - \lambda$ once again;

$$\begin{aligned}\det(A_2 - \lambda I) &= \mu(\mu^2 - 1) - (-1)(-\mu - 1) + (-1)(1 + \mu) \\ &= \mu^3 - 3\mu - 2 \\ &= (\mu + 1)^2(\mu - 2)\end{aligned}$$

- So $\mu = -1$ twice and $\mu = 2$, which is $\lambda = 3, 3, 0$. All of them are ≥ 0 and one is zero, so A_2 is psd.
- Notice that we can also see it directly, $\vec{x}^T A_2 \vec{x} = (x_1 - x_2)^2 + (x_2 - x_3)^2 + (x_1 - x_3)^2 \geq 0$, which is zero exactly along $(1,1,1)^T$.

c) A_3

- The minors are $P_1 = 1$, $P_2 = 1 \times 2 - (-2)(-2) = -2$ and;

$$\begin{aligned}P_3 &= 1\big(2\times(-7) - 0\big) - (-2)\big((-2)(-7) - 0\times 4\big) + 4\big((-2)\times 0 - 2\times 4\big) \\ &= -14 + 28 - 32 = -18\end{aligned}$$

- Here $\det A_3 \neq 0$. $P_1 > 0$ rules out nd, which needs $P_1 < 0$, and $P_2 < 0$ rules out pd. So by 3) A_3 is indefinite.
- The diagonal already shows both signs, $\vec{e}_1^T A_3 \vec{e}_1 = 1 > 0$ and $\vec{e}_3^T A_3 \vec{e}_3 = -7 < 0$. A numerical solver gives the eigenvalues $\lambda \approx -8.71$, 0.49 and 4.22, one negative and two positive, with the product $\approx -18 = \det A_3$.

0.1.13.5 Cholesky Decomposition

- Sylvester needs n determinants. There is a cheaper test that at the same time gives us a way to solve the linear systems of a pd matrix.

Theorem: A symmetric $A \in \mathbb{R}^{n\times n}$ is pd if and only if it can be written as;

$$A = LL^T$$

where L is lower triangular with positive diagonal entries. This is the **Cholesky decomposition** of A. Equivalently $A = \tilde{L}D\tilde{L}^T$, where $\tilde{L}$ is lower triangular with ones on its diagonal and D is diagonal with positive entries, and then $L = \tilde{L}D^{1/2}$.

- One direction is two lines. If $A = LL^T$ then;

$$\vec{x}^T A\vec{x} = \vec{x}^T LL^T\vec{x} = (L^T\vec{x})^T(L^T\vec{x}) = \|L^T\vec{x}\|_2^2 \geq 0$$

and since L^T is triangular with a positive diagonal, $\det L^T > 0$ and $L^T\vec{x} \neq \vec{0}$ for $\vec{x} \neq \vec{0}$. The other direction I state without proof, it is the calculation below never getting stuck.

Example: Find the Cholesky decomposition $A = LL^T$ and the form $\tilde{L}D\tilde{L}^T$ of;

$$A = \begin{bmatrix} 4 & -2 & 2 \\ -2 & 2 & -3 \\ 2 & -3 & 14 \end{bmatrix}$$

- We write L with unknown entries and multiply LL^T out;

$$LL^T = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} l_{11} & l_{21} & l_{31} \\ 0 & l_{22} & l_{32} \\ 0 & 0 & l_{33} \end{bmatrix} = \begin{bmatrix} l_{11}^2 & l_{11}l_{21} & l_{11}l_{31} \\ l_{21}l_{11} & l_{21}^2 + l_{22}^2 & l_{21}l_{31} + l_{22}l_{32} \\ l_{31}l_{11} & l_{31}l_{21} + l_{32}l_{22} & l_{31}^2 + l_{32}^2 + l_{33}^2 \end{bmatrix}$$

- Both sides are symmetric, so we only compare the lower half with A. Going column by column every equation has only one new unknown;

$$\begin{aligned} l_{11}^2 = 4 \quad &\Rightarrow \quad l_{11} = 2 \\ l_{21}\, l_{11} = -2 \quad &\Rightarrow \quad l_{21} = -2/2 = -1 \\ l_{31}\, l_{11} = 2 \quad &\Rightarrow \quad l_{31} = 2/2 = 1 \\ l_{21}^2 + l_{22}^2 = 2 \quad &\Rightarrow \quad l_{22} = (2-1)^{1/2} = 1 \\ l_{31}\, l_{21} + l_{32}\, l_{22} = -3 \quad &\Rightarrow \quad l_{32} = \big(-3 - 1 \times (-1)\big)/1 = -2 \\ l_{31}^2 + l_{32}^2 + l_{33}^2 = 14 \quad &\Rightarrow \quad l_{33} = (14 - 1 - 4)^{1/2} = 3 \end{aligned}$$

- For the other form we divide every column of L by its diagonal entry, which pulls $D^{1/2} = \text{diag}(2, 1, 3)$ out to the right. Then we have;

$$L = \begin{bmatrix} 2 & 0 & 0 \\ -1 & 1 & 0 \\ 1 & -2 & 3 \end{bmatrix} \qquad \tilde{L} = \begin{bmatrix} 1 & 0 & 0 \\ -1/2 & 1 & 0 \\ 1/2 & -2 & 1 \end{bmatrix} \qquad D = \text{diag}(4, 1, 9)$$

- Notice that the leading minors of A are $P_1 = 4$, $P_2 = 8 - 4 = 4$ and $P_3 = 36$, which are 2^2, $(2 \times 1)^2$ and $(2 \times 1 \times 3)^2$. Sylvester and Cholesky look at the same numbers.

- The pattern is the same for any n. Going column by column, a diagonal entry is the root of what is left of a_{jj}, and an entry below it is what is left of a_{ij} divided by l_{jj};

$$l_{jj} = \Big(a_{jj} - \sum_{k<j} l_{jk}^2\Big)^{1/2} \qquad\qquad l_{ij} = \frac{1}{l_{jj}}\Big(a_{ij} - \sum_{k<j} l_{ik}\, l_{jk}\Big) \quad \text{for} \quad i > j$$

- If what is left under a root is ≤ 0 the decomposition does not exist and A is not pd. For A_3 of the Sylvester examples, with the rows $(1, -2, 4)$, $(-2, 2, 0)$ and $(4, 0, -7)$, we get $l_{11} = 1$

and $l_{21} = -2/1 = -2$, and then $l_{22}^2 = 2 - (-2)^2 = -2 < 0$, so we stop. So Cholesky is also a pd test, at about half the work of the usual elimination, and it is the test that algorithms actually use. Newton's method will use it later to check the Hessian at every step.

- The second use is the system $A\vec{x}^* = \vec{b}$ of a stationary point, $\vec{x}^* = A^{-1}\vec{b}$. The inverse is never computed. With $A = LL^T$ we write $\vec{y} = L^T\vec{x}$ and the system splits into two triangular ones;

$$L\vec{y} = \vec{b} \qquad \text{then} \qquad L^T\vec{x} = \vec{y}$$

The first is solved from the top down and the second from the bottom up, every row giving one new unknown.

Example: Solve $A\vec{x} = \vec{b}$ with $\vec{b} = (10, -10, 33)^T$ and the matrix A of the previous example, using its Cholesky factor L.

$$A = \begin{bmatrix} 4 & -2 & 2 \\ -2 & 2 & -3 \\ 2 & -3 & 14 \end{bmatrix} \qquad L = \begin{bmatrix} 2 & 0 & 0 \\ -1 & 1 & 0 \\ 1 & -2 & 3 \end{bmatrix}$$

- First $L\vec{y} = \vec{b}$ from the top down;

$$\begin{aligned} 2y_1 = 10 \quad &\Rightarrow \quad y_1 = 5 \\ -y_1 + y_2 = -10 \quad &\Rightarrow \quad y_2 = -10 + 5 = -5 \\ y_1 - 2y_2 + 3y_3 = 33 \quad &\Rightarrow \quad y_3 = (33 - 5 - 10)/3 = 6 \end{aligned}$$

- Then $L^T\vec{x} = \vec{y}$ from the bottom up, where the rows of L^T are $(2, -1, 1)$, $(0, 1, -2)$ and $(0, 0, 3)$;

$$\begin{aligned} 3x_3 = 6 \quad &\Rightarrow \quad x_3 = 2 \\ x_2 - 2x_3 = -5 \quad &\Rightarrow \quad x_2 = -5 + 4 = -1 \\ 2x_1 - x_2 + x_3 = 5 \quad &\Rightarrow \quad x_1 = (5 - 1 - 2)/2 = 1 \end{aligned}$$

- So $\vec{x} = (1, -1, 2)^T$. Let's check it with A itself;

$$A\vec{x} = \begin{bmatrix} 4 + 2 + 4 \\ -2 - 2 - 6 \\ 2 + 3 + 28 \end{bmatrix} = \begin{bmatrix} 10 \\ -10 \\ 33 \end{bmatrix} = \vec{b}$$

Worked Examples

Example: Classify the four matrices below with both the eigenvalue test and Sylvester's criterion.

$$A = \begin{bmatrix} 5 & 2 \\ 2 & 2 \end{bmatrix} \qquad B = \begin{bmatrix} 2 & 7 \\ -1 & 2 \end{bmatrix} \qquad C = \begin{bmatrix} -3 & 0 & 1 \\ 0 & -3 & 1 \\ 1 & 1 & -2 \end{bmatrix} \qquad D = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

a) A

- A is symmetric, so we can go straight to the tests. Its leading principal minors are;

$$P_1 = 5 \qquad P_2 = 5 \times 2 - 2 \times 2 = 10 - 4 = 6$$

  and both are positive, so by Sylvester A is pd.

- For the eigenvalues we use that the trace $5 + 2 = 7$ is their sum and the determinant 6 is their product, so we can write the characteristic polynomial at once;

$$\lambda^2 - 7\lambda + 6 = (\lambda - 6)(\lambda - 1) = 0 \quad \Rightarrow \quad \lambda_1 = 6, \;\; \lambda_2 = 1$$

  Both are positive, so the eigenvalue test also says pd.

b) B, and find one $\vec{x}$ with $\vec{x}^T B \vec{x} < 0$.

- B is not symmetric, so first we form its symmetric version;

$$H = \frac{1}{2}\left(\begin{bmatrix} 2 & 7 \\ -1 & 2 \end{bmatrix} + \begin{bmatrix} 2 & -1 \\ 7 & 2 \end{bmatrix} \right) = \frac{1}{2} \begin{bmatrix} 4 & 6 \\ 6 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 3 & 2 \end{bmatrix}$$

- The minors of H are $P_1 = 2$ and $P_2 = 2 \times 2 - 3 \times 3 = 4 - 9 = -5$. Here $\det H \neq 0$, $P_1 > 0$ rules out nd and $P_2 < 0$ rules out pd, so by 3) H is indefinite, and with it the form of B.

- With the trace 4 and the determinant -5 the eigenvalues of H are;

$$\lambda^2 - 4\lambda - 5 = (\lambda - 5)(\lambda + 1) = 0 \quad \Rightarrow \quad \lambda_1 = 5, \;\; \lambda_2 = -1$$

  one positive and one negative, indefinite once again.

- For the vector I take the eigenvector of $\lambda_2 = -1$, since along it the form is $\lambda_2 \|\vec{x}\|_2^2 < 0$. The rows of $H + I$ are $(3, 3)$ twice, so $x_1 = -x_2$ and $\vec{x} = (1, -1)^T$. The form of B has the coefficient $b_{12} + b_{21} = 7 - 1 = 6$ for $x_1 x_2$, so;

$$\begin{aligned} \vec{x}^T B \vec{x} &= 2x_1^2 + 6x_1x_2 + 2x_2^2 \\ &= 2 \times 1 + 6 \times 1 \times (-1) + 2 \times 1 \\ &= 2 - 6 + 2 = -2 < 0 \end{aligned}$$

- For comparison $(1, 1)^T$ gives $2 + 6 + 2 = 10 > 0$, so the form of B indeed takes both signs.

- One should never put B itself into the tests. Its own minors are 2 and $2\times2-7\times(-1)=11$, both positive, and they would call it pd. Its own eigenvalues come from $\lambda^2-4\lambda+11$, whose discriminant $16-44$ is negative, so they are not even real.

c) C

- C is symmetric. The first two minors are $P_1=-3$ and $P_2=(-3)(-3)-0\times0=9$, and for P_3 we expand along the first row;

$$\begin{aligned} P_3 &= -3\big((-3)(-2)-1\times1\big)-0+1\big(0\times1-(-3)\times1\big)\\ &= -3\times5+3=-12 \end{aligned}$$

- The signs are $-$, $+$, $-$, alternating and starting negative, so by 2) C is nd.

- For the eigenvalues we expand $\det(C-\lambda I)$ the same way, and the factor $3+\lambda$ shows up in both terms;

$$\begin{aligned} \det(C-\lambda I) &= (-3-\lambda)\big((-3-\lambda)(-2-\lambda)-1\big)+1\big(0-(-3-\lambda)\big)\\ &= -(3+\lambda)\big((3+\lambda)(2+\lambda)-1\big)+(3+\lambda)\\ &= (3+\lambda)\big(2-(\lambda^2+5\lambda+6)\big)\\ &= -(\lambda+3)(\lambda^2+5\lambda+4)\\ &= -(\lambda+3)(\lambda+1)(\lambda+4) \end{aligned}$$

- So $\lambda=-1,-3,-4$, all negative, and the eigenvalue test agrees, C is nd. Notice that $(-1)(-3)(-4)=-12=P_3=\det C$.

- If you have to find the roots of a cubic by hand, look for a common factor before multiplying everything out. Here it saved us from guessing a root.

d) D

- D is symmetric. Its minors are $P_1 = 2$, $P_2 = 2 \times 2 - 0 \times 0 = 4$ and;

$$P_3 = 2\,(2 \times 1 - 1 \times 1) - 0 + 1\,(0 \times 1 - 2 \times 1) = 2 - 2 = 0$$

- Since $\det D = 0$, by 4) D is neither pd nor nd, and Sylvester tells us nothing more. The first two minors being positive does not help us here.
- So we go back to the eigenvalues. Expanding along the first row once again, now the factor $2 - \lambda$ is common;

$$\begin{aligned}
\det(D - \lambda I) &= (2-\lambda)\big((2-\lambda)(1-\lambda) - 1\big) + 1\big(0 - (2-\lambda)\big) \\
&= (2-\lambda)\big((2-\lambda)(1-\lambda) - 2\big) \\
&= (2-\lambda)(\lambda^2 - 3\lambda) \\
&= (2-\lambda)\,\lambda\,(\lambda - 3)
\end{aligned}$$

- The eigenvalues are 3, 2 and 0. All of them are ≥ 0 and one is zero, so D is psd.
- The zero eigenvalue belongs to $\vec{x} = (1, 1, -2)^T$, since $D\vec{x} = (2-2,\ 2-2,\ 1+1-2)^T = \vec{0}$. Along it the form is $\vec{x}^T D\vec{x} = \vec{x}^T\vec{0} = 0$, which is exactly why D is not pd.

Your Turn

Let a be a number and;

$$A_a = \begin{bmatrix} a & 1 & 1 \\ 1 & a & 1 \\ 1 & 1 & a \end{bmatrix}$$

a) Find the leading principal minors P_1, P_2 and P_3 of A_a as functions of a, and factor P_3 (the value $a = 1$ is one of its roots).

b) For which values of a is A_a pd?

c) For which values of a is A_a nd?

d) At $a = 1$ Sylvester's criterion says nothing. Classify A_1 by writing out its quadratic form.

Your Turn

Let;

$$A = \begin{bmatrix} 4 & 2 & -4 \\ 2 & 10 & 1 \\ -4 & 1 & 6 \end{bmatrix} \qquad \vec{b} = \begin{bmatrix} 2 \\ -5 \\ -3 \end{bmatrix}$$

a) Find the Cholesky factor L of A by comparing entries, and check $LL^T = A$.

b) Without expanding any determinant, find $\det A$ from L.

c) Solve $A\vec{x} = \vec{b}$ with the two triangular systems $L\vec{y} = \vec{b}$ and $L^T\vec{x} = \vec{y}$, and check your $\vec{x}$ with A.

0.1.14 Singular Value Decomposition

- The decomposition $A = X\Lambda X^{-1}$ of the Eigenvalues pages needs a square matrix, and only for a symmetric one are the eigenvectors orthonormal. Most matrices we meet are not like that. The data matrix X of the Least Squares Data Fitting topic has many more rows than columns.
- The singular value decomposition works for every $m \times n$ matrix and still gives us orthonormal bases, one on each side. It is behind least squares, image compression and the pseudo inverse, so we will go through it slowly.

0.1.14.1 The Statement

Theorem: Every $A \in \mathbb{R}^{m\times n}$ of rank r can be written as;

$$A = U\Sigma V^T$$

where $U \in \mathbb{R}^{m\times m}$ and $V \in \mathbb{R}^{n\times n}$ are orthogonal, $U^TU = UU^T = I$ and $V^TV = VV^T = I$, and $\Sigma \in \mathbb{R}^{m\times n}$ is zero everywhere except its first r diagonal entries;

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0$$

- We call the σ_i the **singular values** of A, the columns $\vec{u}_i$ of U the left singular vectors and the columns $\vec{v}_i$ of V the right singular vectors.
- The decomposition is unique in the sense that the σ_i are, once we sort them. The vectors are not, for example we can always flip the signs of a pair $\vec{u}_i$, $\vec{v}_i$ together.
- In $\mathbb{C}^{m\times n}$ the same holds with $A = U\Sigma V^\dagger$ and unitary U and V, $U^\dagger U = UU^\dagger = I$, and the σ_i are still real and positive.
- Σ has the shape of A. We fill the largest square diagonal block that fits and write zeros everywhere else. For a tall and a wide matrix of rank 2 we have;

$$\Sigma_{3\times 2} = \begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \\ 0 & 0 \end{bmatrix} \qquad \Sigma_{2\times 3} = \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \end{bmatrix}$$

  so with more rows the extra row of zeros is at the bottom, and with more columns the extra column is on the right.
- If the rank is below $\min\{m, n\}$, the diagonal entries after σ_r are zero as well. In blocks we can always write Σ as;

$$\Sigma = \begin{bmatrix} \Sigma_r & 0 \\ 0 & 0 \end{bmatrix} \qquad \Sigma_r = \operatorname{diag}(\sigma_1, \ldots, \sigma_r)$$

0.1.14.2 The Big Picture

- Now let's see what U and V are. The matrix A takes vectors of $\mathbb{R}^n$ to $\mathbb{R}^m$, and by the Four Subspaces pages each of the two spaces splits into two orthogonal pieces;

$$\mathbb{R}^n = R(A^T) \oplus N(A) \qquad \mathbb{R}^m = R(A) \oplus N(A^T)$$

- For each of the four pieces we pick an orthonormal basis;

  1) $\vec{v}_1, \ldots, \vec{v}_r$ of the row space $R(A^T)$,

  2) $\vec{v}_{r+1}, \ldots, \vec{v}_n$ of the null space $N(A)$,

  3) $\vec{u}_1, \ldots, \vec{u}_r$ of the column space $R(A)$,

  4) $\vec{u}_{r+1}, \ldots, \vec{u}_m$ of the left null space $N(A^T)$.

- The dimensions fit, $\dim R(A^T) = r$ and $\dim N(A) = n - r$. Since the two spaces are orthogonal, 1) and 2) together are n orthonormal vectors of $\mathbb{R}^n$, and we put them as the columns of V. In the same way 3) and 4) are m orthonormal vectors of $\mathbb{R}^m$, and we put them into U. These two splits are exactly why the $\vec{v}$'s and the $\vec{u}$'s can fill square orthogonal matrices.

- What holds it all together is how A acts on these bases, and we choose them so that;

$$A\vec{v}_i = \sigma_i \vec{u}_i \quad \text{for} \quad i \leq r \qquad\qquad A\vec{v}_j = \vec{0} \quad \text{for} \quad j > r$$

  Meaning A takes every $\vec{v}$ of the row space to a stretched $\vec{u}$ of the column space, and it sends the $\vec{v}$'s of the null space to zero, which is what a null space does.

- Now let's write all n of these equations at once. A times the matrix V is A times each of its columns;

$$AV = A\,[\,\vec{v}_1\,|\,\cdots\,|\,\vec{v}_r\,|\,\vec{v}_{r+1}\,|\,\cdots\,|\,\vec{v}_n\,] = [\,\sigma_1\vec{u}_1\,|\,\cdots\,|\,\sigma_r\vec{u}_r\,|\,\vec{0}\,|\,\cdots\,|\,\vec{0}\,]$$

  which has n columns, r stretched $\vec{u}$'s and $n - r$ zeros.

- The right hand side is $U\Sigma$. The jth column of $U\Sigma$ is U times the jth column of Σ, which is $\sigma_j\vec{e}_j$ for $j \leq r$ and $\vec{0}$ after, so it is $\sigma_j\vec{u}_j$ and then zeros. So $AV = U\Sigma$, and multiplying from the right with V^T ($VV^T = I$) we obtain;

$$A = U\Sigma V^T$$

- The columns of U and V after the rth one always meet the zeros of Σ, so we can drop them and lose nothing. This is the **reduced SVD**;

$$A = U_r\Sigma_r V_r^T$$

  where $U_r = [\,\vec{u}_1\,|\,\cdots\,|\,\vec{u}_r\,]$ is $m \times r$, Σ_r is $r \times r$ and $V_r = [\,\vec{v}_1\,|\,\cdots\,|\,\vec{v}_r\,]$ is $n \times r$.

- Gram-Schmidt on the columns of A, the QR decomposition, also gives us an orthonormal basis of $R(A)$. The $\vec{u}_1, \ldots, \vec{u}_r$ of the SVD are a special one, the images of an orthonormal basis of the row space.

Example: Let;

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \\ 2 & 2 \end{bmatrix}$$

which has $m = 3$ and $n = 2$.

a) Find its rank and an orthonormal basis for each of its four subspaces.

- Every row is a multiple of $(1, 1)$ and both columns are $(1, 2, 2)^T$, so $r = 1$ and we can read the bases by eye;

  1) $\vec{v}_1 = (1, 1)^T/\sqrt{2}$ for the row space, the row direction with the length $\sqrt{2}$ divided out,

  2) $\vec{v}_2 = (-1, 1)^T/\sqrt{2}$ for the null space, since $A(-1, 1)^T = (0, 0, 0)^T$,

  3) $\vec{u}_1 = (1, 2, 2)^T/3$ for the column space, since $\|(1, 2, 2)^T\|_2 = (1 + 4 + 4)^{1/2} = 3$,

  4) $\vec{u}_2 = (2, 1, -2)^T/3$ and $\vec{u}_3 = (2, -2, 1)^T/3$ for the left null space.

- For 4) we need $A^T\vec{u} = \vec{0}$, which means $\vec{u}$ is orthogonal to $(1, 2, 2)^T$, since both rows of A^T are $(1, 2, 2)$. Both vectors are, $2 + 2 - 4 = 0$ and $2 - 4 + 2 = 0$, and they are orthogonal to each other, $4 - 2 - 2 = 0$. The dimensions fit, $1 + 1 = 2 = n$ and $1 + 2 = 3 = m$.

b) Find σ_1 with $A\vec{v}_1 = \sigma_1\vec{u}_1$ and write U, Σ and V.

- Let's apply A to $\vec{v}_1$;

$$A\vec{v}_1 = \frac{1}{\sqrt{2}}\begin{bmatrix} 2 \\ 4 \\ 4 \end{bmatrix} = \frac{2}{\sqrt{2}}\begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = \sqrt{2} \times 3\,\vec{u}_1 = 3\sqrt{2}\,\vec{u}_1$$

  so $\sigma_1 = 3\sqrt{2}$, and $A\vec{v}_2 = \vec{0}$ as it should. Then;

$$U = \frac{1}{3}\begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix} \qquad \Sigma = \begin{bmatrix} 3\sqrt{2} & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \qquad V = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

- Notice the shape, Σ is 3×2 like A and only σ_1 is nonzero since $r = 1$.

c) Multiply the reduced SVD back to A.

- With $r = 1$ the reduced SVD keeps one column on each side;

$$A = \vec{u}_1\,\sigma_1\,\vec{v}_1^T = \frac{1}{3}\begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} 3\sqrt{2}\,\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}\begin{bmatrix} 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 2 & 2 \\ 2 & 2 \end{bmatrix}$$

- With rank one we had nothing to choose, every unit vector of the row space goes to a multiple of $\vec{u}_1$. For a larger rank the $\vec{v}$'s must be picked so that their images $A\vec{v}_i$ stay orthogonal, and the recipe of the next pages tells us which ones.

0.1.14.3 A Sum of Rank One Pieces

- We can multiply two matrices as the sum of the columns of the first times the rows of the second. The columns of $U_r\Sigma_r$ are $\sigma_i\vec{u}_i$ and the rows of V_r^T are $\vec{v}_i^T$, so;

$$A = U_r\Sigma_r V_r^T = \sum_{i=1}^{r} \sigma_i\, \vec{u}_i\vec{v}_i^T$$

  where σ_i is a number, so it can stand anywhere in its term, $\vec{u}_i\sigma_i\vec{v}_i^T$ and $\sigma_i\vec{u}_i\vec{v}_i^T$ are the same thing.

- Each $\vec{u}_i\vec{v}_i^T$ is an $m \times n$ matrix whose j^{th} column is the jth entry of $\vec{v}_i$ times $\vec{u}_i$. Every column is a multiple of $\vec{u}_i$, so it is a rank one matrix, and we have A as a sum of r rank one matrices with the largest piece first. Our 3×2 example was a single piece.

- Without proof I present the **Eckart Young theorem**. Keeping the $k < r$ largest terms;

$$A_k = \sum_{i=1}^{k} \sigma_i\, \vec{u}_i\vec{v}_i^T$$

  gives the best rank k approximation of A, no matrix of rank k is closer to A, and what we leave out has the size σ_{k+1} (in the matrix norm of the geometric picture below).

- This is how we compress images. A gray image is a matrix of pixel values, and a 1000×1000 one needs 10^6 numbers. A_{50} needs only $50 \times (1000 + 1000 + 1) = 100050$ of them, about a tenth, and since the σ_i of a picture fall quickly it still looks like the picture.

- We will meet rank one pieces again later in the course. The quasi Newton methods build their matrices by adding rank one matrices $\vec{u}\vec{v}^T$ at every step.

0.1.14.4 How to Compute It

- The big picture tells us what the bases are but not how to find the $\vec{v}$'s when $r > 1$. The trick is A^TA, a square symmetric $n \times n$ matrix. Using $U^TU = I$;

$$A^TA = (U\Sigma V^T)^T(U\Sigma V^T) = V\Sigma^TU^TU\Sigma V^T = V\Sigma^T\Sigma V^T$$

  where $\Sigma^T\Sigma$ is $n \times n$ and diagonal, with $\sigma_1^2, \ldots, \sigma_r^2$ and then zeros.

- Multiplying by V from the right we have $A^TAV = V\Sigma^T\Sigma$, which column by column reads;

$$A^TA\,\vec{v}_i = \sigma_i^2\,\vec{v}_i$$

  So the $\vec{v}_i$ are orthonormal eigenvectors of A^TA and the σ_i^2 are its eigenvalues. The Symmetric Matrices pages give us that such eigenvectors exist, and the eigenvalues are never negative since $\vec{x}^TA^TA\vec{x} = \|A\vec{x}\|_2^2 \geq 0$ makes A^TA psd (the eigenvalue test of the Definite Matrices pages). So their roots are real.

- Then we get the $\vec{u}$'s from the main relation $A\vec{v}_i = \sigma_i\vec{u}_i$;

$$\vec{u}_i = \frac{1}{\sigma_i}\,A\vec{v}_i \qquad \text{for} \;\; i \leq r$$

- These $\vec{u}_i$ are orthonormal, let's check;

$$\vec{u}_i^T \vec{u}_j = \frac{1}{\sigma_i \sigma_j} \vec{v}_i^T A^T A \, \vec{v}_j = \frac{1}{\sigma_i \sigma_j} \vec{v}_i^T \left(\sigma_j^2 \, \vec{v}_j\right) = \frac{\sigma_j}{\sigma_i} \vec{v}_i^T \vec{v}_j = \frac{\sigma_j}{\sigma_i} \delta_{ij} = \delta_{ij}$$

  where δ_{ij} is 1 for $i = j$ and 0 otherwise. There is no hidden assumption in it. The only orthogonality we used is $\vec{v}_i^T \vec{v}_j = \delta_{ij}$, and we have that because we chose the $\vec{v}$'s as orthonormal eigenvectors.

- In the same way we see that the $\vec{u}_i$ are eigenvectors of AA^T with the same eigenvalues;

$$AA^T \vec{u}_i = \frac{1}{\sigma_i} A\big(A^T A \, \vec{v}_i\big) = \frac{1}{\sigma_i} A \, \sigma_i^2 \, \vec{v}_i = \sigma_i^2 \, \vec{u}_i$$

- If $r < m$ we still miss some of the $\vec{u}$'s. We complete them with any orthonormal basis $\vec{u}_{r+1}, \ldots, \vec{u}_m$ of $N(A^T)$, found by Gram-Schmidt or by eye.

- So the recipe we follow is;

  1) find the eigenvalues of $A^T A$ and sort them, σ_i is the root of the ith one,

  2) find orthonormal eigenvectors $\vec{v}_i$ of $A^T A$, the columns of V,

  3) take $\vec{u}_i = A\vec{v}_i/\sigma_i$ for $i \leq r$ and complete them with $N(A^T)$ if $r < m$,

  4) check $A = U\Sigma V^T$.

- For the 3×2 example, with the rows $(1,1)$, $(2,2)$ and $(2,2)$, $A^T A$ has $1 + 4 + 4 = 9$ in every entry, with the eigenvalues 18 and 0 and the eigenvectors $(1,1)^T$ and $(-1,1)^T$. So $\sigma_1 = \sqrt{18} = 3\sqrt{2}$, which is the same σ_1 we found by eye.

Example: Let;

$$A = \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix}$$

a) Find the singular values of A and the matrix V.

- We start with A^TA;

$$A^TA = \begin{bmatrix} 3 & 4 \\ 0 & 5 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix} = \begin{bmatrix} 9+16 & 0+20 \\ 0+20 & 0+25 \end{bmatrix} = \begin{bmatrix} 25 & 20 \\ 20 & 25 \end{bmatrix}$$

- Its characteristic polynomial is;

$$\begin{aligned} \det(A^TA - \lambda I) &= (25-\lambda)^2 - 20^2 \\ &= \lambda^2 - 50\lambda + 625 - 400 \\ &= \lambda^2 - 50\lambda + 225 \\ &= (\lambda - 45)(\lambda - 5) \end{aligned}$$

so the eigenvalues are 45 and 5, and the singular values are $\sigma_1 = \sqrt{45} = 3\sqrt{5}$ and $\sigma_2 = \sqrt{5}$.

- For an eigenvector $(a, b)^T$ of $\lambda = 45$ the rows of $A^TA - 45I$ are $(-20, 20)$ and $(20, -20)$, so $a = b$. For $\lambda = 5$ the rows of $A^TA - 5I$ are $(20, 20)$ twice, so $a = -b$. Normalizing we have;

$$\vec{v}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \qquad \vec{v}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \end{bmatrix} \qquad V = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

b) Find the matrix U and check its columns with AA^T.

- Here $r = 2 = m$, so we get both $\vec{u}$'s from $\vec{u}_i = A\vec{v}_i/\sigma_i$;

$$\begin{aligned} \vec{u}_1 &= \frac{1}{3\sqrt{5}} \frac{1}{\sqrt{2}} \begin{bmatrix} 3 \\ 4+5 \end{bmatrix} = \frac{3}{3\sqrt{10}} \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \frac{1}{\sqrt{10}} \begin{bmatrix} 1 \\ 3 \end{bmatrix} \\ \vec{u}_2 &= \frac{1}{\sqrt{5}} \frac{1}{\sqrt{2}} \begin{bmatrix} -3 \\ -4+5 \end{bmatrix} = \frac{1}{\sqrt{10}} \begin{bmatrix} -3 \\ 1 \end{bmatrix} \end{aligned}$$

- They are orthonormal, $\vec{u}_1^T\vec{u}_2 = (-3+3)/10 = 0$ and $\|\vec{u}_1\|_2^2 = \|\vec{u}_2\|_2^2 = (1+9)/10 = 1$. For the check we need AA^T;

$$AA^T = \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} 3 & 4 \\ 0 & 5 \end{bmatrix} = \begin{bmatrix} 9 & 12 \\ 12 & 41 \end{bmatrix}$$

- Then the $\vec{u}$'s are eigenvectors of AA^T with the same eigenvalues 45 and 5, as they should be;

$$\begin{aligned} AA^T(1,3)^T &= (9+36,\ 12+123)^T = (45, 135)^T = 45\,(1,3)^T \\ AA^T(-3,1)^T &= (-27+12,\ -36+41)^T = (-15, 5)^T = 5\,(-3,1)^T \end{aligned}$$

c) Write the SVD and multiply it back to A.

- Transposing V for the last factor, the SVD is;

$$A = U\Sigma V^T = \frac{1}{\sqrt{10}}\begin{bmatrix}1 & -3\\ 3 & 1\end{bmatrix}\begin{bmatrix}3\sqrt{5} & 0\\ 0 & \sqrt{5}\end{bmatrix}\frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ -1 & 1\end{bmatrix}$$

- Let's multiply it back. The first two factors give $\sqrt{5}/\sqrt{10} = 1/\sqrt{2}$ in front;

$$U\Sigma = \frac{\sqrt{5}}{\sqrt{10}}\begin{bmatrix}3 & -3\\ 9 & 1\end{bmatrix} = \frac{1}{\sqrt{2}}\begin{bmatrix}3 & -3\\ 9 & 1\end{bmatrix}$$
$$U\Sigma V^T = \frac{1}{2}\begin{bmatrix}3 & -3\\ 9 & 1\end{bmatrix}\begin{bmatrix}1 & 1\\ -1 & 1\end{bmatrix} = \frac{1}{2}\begin{bmatrix}3+3 & 3-3\\ 9-1 & 9+1\end{bmatrix} = \frac{1}{2}\begin{bmatrix}6 & 0\\ 8 & 10\end{bmatrix} = \begin{bmatrix}3 & 0\\ 4 & 5\end{bmatrix}$$

  which is A.

- One should not forget the transpose on V. With V itself as the last factor the same product gives the rows $(0,-3)$ and $(5,-4)$, which is not A.

Remark: The eigenvalues of this A, with the rows $(3,0)$ and $(4,5)$, are 3 and 5, since it is triangular, while its singular values are $3\sqrt{5} \approx 6.71$ and $\sqrt{5} \approx 2.24$. They are different things. An eigenvector keeps its direction, $A\vec{x} = \lambda\vec{x}$, while A sends $\vec{v}_i$ to a different direction $\vec{u}_i$.

- For a symmetric matrix the spectral theorem $A = Q\Lambda Q^T$ is already an SVD once we sort by $|\lambda_i|$ and move the minus signs of the negative eigenvalues into U, so there $\sigma_i = |\lambda_i|$. Our A is not symmetric.

- What survives in general is the product. From $U^TU = I$ we have $(\det U)^2 = 1$, so $\det U = \pm 1$ and the same for V. Then;

$$|\det A| = |\det U|\,|\det \Sigma|\,|\det V^T| = \sigma_1\sigma_2$$

  and here $\sigma_1\sigma_2 = 3\sqrt{5} \times \sqrt{5} = 15 = |3 \times 5| = |\det A|$.

0.1.14.5 The Geometric Picture

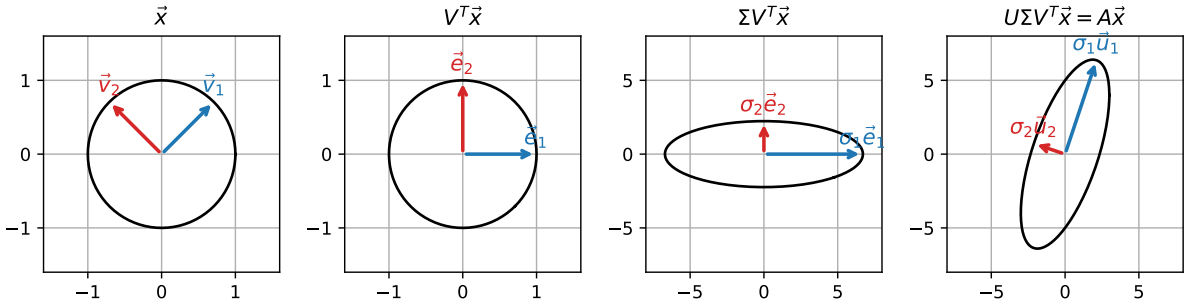
- Reading $A\vec{x} = U\big(\Sigma\,(V^T\vec{x})\big)$ from the inside out, we see A act in three steps;

  1) V^T rotates (or reflects) $\vec{x}$ in $\mathbb{R}^n$. Its rows are the $\vec{v}_i^T$, so $V^T\vec{v}_i = \vec{e}_i$ and each $\vec{v}_i$ is turned onto an axis.

  2) Σ stretches along the axes, $\vec{e}_i$ to $\sigma_i\vec{e}_i$. When $m > n$ it also adds zero components, and when $m < n$ it drops components.

  3) U rotates (or reflects) in $\mathbb{R}^m$, taking each $\vec{e}_i$ to $\vec{u}_i$.

- So; A takes the unit circle to an ellipse whose semi axes are $\sigma_1\vec{u}_1$ and $\sigma_2\vec{u}_2$. For the matrix of the example, A with the rows $(3,0)$ and $(4,5)$, $\vec{v}_1 = (1,1)^T/\sqrt{2}$, $\vec{u}_1 = (1,3)^T/\sqrt{10}$, $\sigma_1 = 3\sqrt{5}$ and $\sigma_2 = \sqrt{5}$, V^T rotates by $-45°$ and U rotates by about $71.6°$, and the circle with its three steps is shown in the following graph as;



- Since U does not change lengths and $V^T\vec{x}$ is again a unit vector, the most A can stretch a unit vector is the longest semi axis;

$$\max_{\|\vec{x}\|_2=1} \|A\vec{x}\|_2 = \sigma_1 = \|A\|_2$$

  which is reached at $\vec{x} = \vec{v}_1$, and we call it the 2-norm of the matrix. For the example $\|A\|_2 = 3\sqrt{5} \approx 6.71$.

- The ratio of the longest semi axis to the shortest is the **condition number**;

$$\kappa(A) = \frac{\sigma_1}{\sigma_r}$$

  Meaning how much more A stretches its best direction than its worst one. For the example $\kappa = 3\sqrt{5}/\sqrt{5} = 3$. A large κ means a long and thin ellipse, and we will meet it in every method later, where it slows the iterations down and blows rounding errors up.

0.1.14.6 The Pseudo Inverse

- Now let's use the SVD to solve $A\vec{x} = \vec{b}$ for any A, square or not, invertible or not. Multiplying $U\Sigma V^T\vec{x} = \vec{b}$ by U^T from the left we have;

$$\Sigma V^T\vec{x} = U^T\vec{b}$$

- Let's write $\vec{y} = V^T\vec{x}$ and $\vec{c} = U^T\vec{b}$, the coordinates of $\vec{x}$ and $\vec{b}$ in the two bases. Then $\Sigma\vec{y} = \vec{c}$ is a diagonal system;

$$\sigma_i\, y_i = c_i \quad \text{for} \quad i \le r \qquad\qquad 0 = c_i \quad \text{for} \quad i > r$$

- The first r rows we solve, $y_i = c_i/\sigma_i$. The rows after r we can not do anything about, they are the part of $\vec{b}$ in $N(A^T)$ that no $\vec{x}$ reaches. And the y_i with $i > r$ do not appear at all, since they multiply the zero columns of Σ. They are the null space directions and we set them to zero.

- So we invert what can be inverted, $\vec{y} = \Sigma^+\vec{c}$, where Σ^+ is the $n \times m$ matrix with $1/\sigma_1, \ldots, 1/\sigma_r$ on its diagonal and zeros everywhere else. Going back with $\vec{x} = V\vec{y}$ we have;

$$\vec{x}^* = V\Sigma^+U^T\vec{b} = A^+\vec{b} \qquad \text{with} \qquad A^+ = V\Sigma^+U^T$$

  where A^+ is the **pseudo inverse** of A, also called the Moore Penrose inverse. Books often write it as $A^\dagger$, but in these notes the dagger is the conjugate transpose, so I write A^+.

- Since U and V preserve lengths, $\|A\vec{x} - \vec{b}\|_2 = \|\Sigma\vec{y} - \vec{c}\|_2$ and $\|\vec{x}\|_2 = \|\vec{y}\|_2$. Then;

  1) $\vec{x}^*$ makes the rows $i \le r$ of $\Sigma\vec{y} - \vec{c}$ zero, and the rows $i > r$ do not depend on $\vec{y}$. So $\|A\vec{x} - \vec{b}\|_2$ can not be made smaller, $\vec{x}^*$ is a least squares solution.

  2) Every other least squares solution differs from $\vec{x}^*$ only in the free y_i, a part in $N(A)$, which does not change $A\vec{x}$ and only adds length. So among all least squares solutions $\vec{x}^*$ has the smallest norm.

- When A has independent columns, $r = n$, the least squares solution is unique and A^+ is the matrix of the normal equations of the Least Squares Data Fitting topic. With the reduced SVD $A = U_n\Sigma_nV^T$, where Σ_n and V are $n \times n$ and invertible;

$$\begin{aligned}(A^TA)^{-1}A^T &= (V\Sigma_nU_n^TU_n\Sigma_nV^T)^{-1}\,V\Sigma_nU_n^T \\ &= V\Sigma_n^{-2}V^T\,V\Sigma_nU_n^T \\ &= V\Sigma_n^{-1}U_n^T = A^+\end{aligned}$$

  where in the second step I have used $U_n^TU_n = I$ and $V^{-1} = V^T$. And when A is square and invertible, $\Sigma^+ = \Sigma^{-1}$ and $A^+ = A^{-1}$.

Example: Let A be the 3×2 matrix of rank one of the big picture, with $\vec{u}_1 = (1,2,2)^T/3$, $\vec{u}_2 = (2,1,-2)^T/3$, $\vec{u}_3 = (2,-2,1)^T/3$, $\vec{v}_1 = (1,1)^T/\sqrt{2}$ and $\sigma_1 = 3\sqrt{2}$, and let $\vec{b} = (0,3,6)^T$.

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \\ 2 & 2 \end{bmatrix}$$

a) Find $\vec{x}^* = A^+\vec{b}$.

- First we find the coordinates of $\vec{b}$, $\vec{c} = U^T\vec{b}$, whose entries are $c_i = \vec{u}_i^T\vec{b}$;

$$\vec{c} = \frac{1}{3}\begin{bmatrix} 0+6+12 \\ 0+3-12 \\ 0-6+6 \end{bmatrix} = \begin{bmatrix} 6 \\ -3 \\ 0 \end{bmatrix}$$

- Only σ_1 is nonzero, so Σ^+ is 2×3 with $1/\sigma_1$ in its corner. Then $y_1 = c_1/\sigma_1$ and $y_2 = 0$;

$$y_1 = \frac{6}{3\sqrt{2}} = \sqrt{2} \qquad \Rightarrow \qquad \vec{x}^* = V\vec{y} = \sqrt{2}\,\vec{v}_1 = \sqrt{2}\,\frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

- Notice that $c_2 = -3$ is the part of $\vec{b}$ that no $\vec{x}$ reaches. The residual is $\vec{b} - A\vec{x}^* = (0,3,6)^T - (2,4,4)^T = (-2,-1,2)^T = -3\,\vec{u}_2$, which lies in the left null space.

b) Show that every $\vec{x}$ with $x_1 + x_2 = 2$ is a least squares solution and that $\vec{x}^*$ is the shortest one of them.

- Every $A\vec{x}$ is $(x_1 + x_2)(1,2,2)^T$, so with $t = x_1 + x_2$ we have the squared residual;

$$\begin{aligned} \|A\vec{x} - \vec{b}\|_2^2 &= t^2 + (2t-3)^2 + (2t-6)^2 \\ &= t^2 + 4t^2 - 12t + 9 + 4t^2 - 24t + 36 \\ &= 9t^2 - 36t + 45 \end{aligned}$$

- Setting the derivative to zero we get $18t - 36 = 0$ and $t = 2$. So every $\vec{x}$ with $x_1 + x_2 = 2$ is a least squares solution, with the squared residual $9 \times 4 - 72 + 45 = 9$, so $\|A\vec{x} - \vec{b}\|_2 = 3$ for all of them.

- Their lengths are $\|\vec{x}\|_2^2 = x_1^2 + (2 - x_1)^2 = 2x_1^2 - 4x_1 + 4$, which is smallest at $4x_1 - 4 = 0$, $x_1 = 1$. That is $\vec{x} = (1,1)^T$ with $\|\vec{x}\|_2 = \sqrt{2}$, the $\vec{x}^*$ of part a. For comparison $(2,0)^T$ has the length 2 and $(3,-1)^T$ has $\sqrt{10}$.

- Here A^TA has 9 in every entry, so $\det(A^TA) = 0$ and the normal equations $9x_1 + 9x_2 = 18$ only give the line $x_1 + x_2 = 2$. The pseudo inverse picks the shortest point of that line.

Worked Examples

Example: Let;

$$A = \begin{bmatrix} 1 & 0 \\ 2 & 2 \\ 0 & 1 \end{bmatrix}$$

which has $m = 3$ and $n = 2$.

a) Find the singular values of A and the matrix V.

- A^TA is 2×2, since $n = 2$;

$$A^TA = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1+4+0 & 0+4+0 \\ 0+4+0 & 0+4+1 \end{bmatrix} = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}$$

- Then its eigenvalues are;

$$\det(A^TA - \lambda I) = (5-\lambda)^2 - 16 = 0 \quad \Rightarrow \quad 5 - \lambda = \pm 4 \quad \Rightarrow \quad \lambda_1 = 9, \;\; \lambda_2 = 1$$

so $\sigma_1 = 3$ and $\sigma_2 = 1$. Both are nonzero, so the rank is $r = 2$.

- For $\lambda = 9$ the rows of $A^TA - 9I$ are $(-4, 4)$ and $(4, -4)$, so $a = b$. For $\lambda = 1$ the rows of $A^TA - I$ are $(4, 4)$ twice, so $a = -b$. Normalizing we have;

$$\vec{v}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \qquad \vec{v}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \end{bmatrix} \qquad V = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

b) Find $\vec{u}_1$ and $\vec{u}_2$ from $A\vec{v}_i = \sigma_i \vec{u}_i$, and complete U with the left null space.

- For $i \leq r$ the $\vec{u}$'s are $\vec{u}_i = A\vec{v}_i/\sigma_i$;

$$\vec{u}_1 = \frac{1}{3} \frac{1}{\sqrt{2}} \begin{bmatrix} 1+0 \\ 2+2 \\ 0+1 \end{bmatrix} = \frac{1}{3\sqrt{2}} \begin{bmatrix} 1 \\ 4 \\ 1 \end{bmatrix}$$

$$\vec{u}_2 = \frac{1}{1} \frac{1}{\sqrt{2}} \begin{bmatrix} -1+0 \\ -2+2 \\ 0+1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

- Let's check them, $\vec{u}_1^T \vec{u}_2 = (-1+0+1)/(3 \times 2) = 0$, $\|\vec{u}_1\|_2^2 = (1+16+1)/18 = 1$ and $\|\vec{u}_2\|_2^2 = (1+0+1)/2 = 1$.

- Now $r = 2 < m = 3$ and one $\vec{u}$ is missing. By the big picture it is an orthonormal basis of the left null space $N(A^T)$, whose dimension is $m - r = 3 - 2 = 1$. So we solve $A^T\vec{z} = \vec{0}$;

$$z_1 + 2z_2 = 0$$
$$2z_2 + z_3 = 0$$

- The two rows give $z_1 = -2z_2$ and $z_3 = -2z_2$, and z_2 is free. I choose $z_2 = -1$ so that the first entry comes out positive, and we have $\vec{z} = (2, -1, 2)^T$, whose length is $(4 + 1 + 4)^{1/2} = 3$, so;

$$\vec{u}_3 = \frac{1}{3}\begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}$$

- It is orthogonal to the other two, $\vec{u}_3^T \vec{u}_1 = (2 - 4 + 2)/(9\sqrt{2}) = 0$ and $\vec{u}_3^T \vec{u}_2 = (-2 + 0 + 2)/(3\sqrt{2}) = 0$, as the Four Subspaces pages promise. The left null space is orthogonal to the column space, which $\vec{u}_1$ and $\vec{u}_2$ span.

c) Write the full SVD, and multiply the reduced SVD back to A as a sum of rank one pieces.

- Σ has the shape of A, 3×2, with the row of zeros at the bottom. So the full SVD is $A = U\Sigma V^T$ with;

$$U = \begin{bmatrix} 1/(3\sqrt{2}) & -1/\sqrt{2} & 2/3 \\ 4/(3\sqrt{2}) & 0 & -1/3 \\ 1/(3\sqrt{2}) & 1/\sqrt{2} & 2/3 \end{bmatrix} \qquad \Sigma = \begin{bmatrix} 3 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \qquad V = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

- The column $\vec{u}_3$ only meets the zero row of Σ, so the reduced SVD drops it, $A = \sigma_1 \vec{u}_1 \vec{v}_1^T + \sigma_2 \vec{u}_2 \vec{v}_2^T$. The two pieces are;

$$\sigma_1 \, \vec{u}_1 \vec{v}_1^T = 3 \times \frac{1}{3\sqrt{2}} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 4 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 4 & 4 \\ 1 & 1 \end{bmatrix}$$

$$\sigma_2 \, \vec{u}_2 \vec{v}_2^T = 1 \times \frac{1}{\sqrt{2}} \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \begin{bmatrix} -1 & 1 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 1 & -1 \\ 0 & 0 \\ -1 & 1 \end{bmatrix}$$

- Adding the two we have;

$$\frac{1}{2}\begin{bmatrix} 1+1 & 1-1 \\ 4+0 & 4+0 \\ 1-1 & 1+1 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 2 & 0 \\ 4 & 4 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2 & 2 \\ 0 & 1 \end{bmatrix} = A$$

- Notice that we never needed $\vec{u}_3$ to build A back. It is there so that U is a square orthogonal matrix, and in a least squares problem it measures the part of a $\vec{b}$ that no $\vec{x}$ can reach.

Your Turn

Let;

$$A = \begin{bmatrix} 4 & 1 \\ -1 & -4 \end{bmatrix}$$

a) Find the singular values of A and the matrix V.

b) Find the matrix U.

c) Multiply $U\Sigma V^T$ back to A.

d) Check that $\sigma_1\sigma_2 = |\det A|$.

Your Turn

Let;

$$A = \begin{bmatrix} 2 & 4 \\ 1 & 2 \\ 2 & 4 \end{bmatrix} \qquad \vec{b} = \begin{bmatrix} 13 \\ 11 \\ 4 \end{bmatrix}$$

a) Find $A^T A$, its eigenvalues and the rank of A, and write the reduced SVD $A = \sigma_1 \vec{u}_1 \vec{v}_1^T$.

b) Find $\vec{x}^* = A^+ \vec{b}$.

c) Find the residual $\vec{b} - A\vec{x}^*$, show that it lies in $N(A^T)$ and explain why this makes $\vec{x}^*$ a least squares solution.

d) Show that every $\vec{x}$ with $x_1 + 2x_2 = 5$ is a least squares solution, and that $\vec{x}^*$ is the shortest of them.

0.1.15 Summary

Supremum and Infimum

$\sup A$ is the smallest upper bound of A and $\inf A$ is the largest lower bound. They do not need to be in A. If they are, they are the maximum and the minimum. A minimizer exists exactly when the infimum of the values is reached.

Vector Space, Subspace and Affine Space

$$\vec{x}+\vec{y}\in V \qquad \alpha\,\vec{x}\in V \qquad \vec{x}_0+V=\{\vec{x}_0+\vec{v}:\vec{v}\in V\}$$

A subset of a vector space is a subspace if it is closed under both operations, and then it has $\vec{0}$ in it. An affine space is a subspace shifted by $\vec{x}_0$. The solutions of a linear equation form one.

Span and Linear Independence

$$\operatorname{span}\{\vec{x}_1,\dots,\vec{x}_k\}=\Big\{\sum_i \alpha_i\,\vec{x}_i\Big\} \qquad \sum_i \alpha_i\,\vec{x}_i=\vec{0} \;\Rightarrow\; \alpha_i=0 \text{ for all } i$$

Dependent vectors have one that is a linear combination of the others.

Basis, Coordinates and Dimension

$$\vec{x}=\sum_i \alpha_i\,\vec{x}_i \qquad [\vec{x}]_{\mathcal{B}}=(\alpha_1,\dots,\alpha_k)^T=A^{-1}\,\vec{x}$$

A basis is linearly independent and spans V, and the columns of A are its vectors. The coordinates are unique. Every basis has the same number of vectors, the dimension, and $\dim\mathbb{R}^n=n$. More than $\dim V$ vectors are always dependent and fewer never span V.

Metric and Norm

$$d(x,y)\le d(x,z)+d(z,y) \qquad \|\alpha\,\vec{x}\|=|\alpha|\,\|\vec{x}\| \qquad \|\vec{x}+\vec{y}\|\le\|\vec{x}\|+\|\vec{y}\| \qquad d(\vec{x},\vec{y})=\|\vec{x}-\vec{y}\|$$

A metric is the distance between two points of any set. It is symmetric and it is zero only for equal points. A norm is the length of one vector, zero only for $\vec{0}$, and it needs a vector space. Every norm gives a metric.

L_p Norms and Unit Balls

$$\|\vec{x}\|_p=\Big(\sum_i |x_i|^p\Big)^{1/p} \qquad \|\vec{x}\|_1=\sum_i |x_i| \qquad \|\vec{x}\|_2=\Big(\sum_i x_i^2\Big)^{1/2} \qquad \|\vec{x}\|_\infty=\max_i |x_i|$$

They are norms for $p\ge 1$. The unit ball $\{\vec{x}:\|\vec{x}\|\le 1\}$ in $\mathbb{R}^2$ is a standing square for $p=1$, a disc for $p=2$ and a full square for $p=\infty$. Any two norms on $\mathbb{R}^n$ are equivalent;

$$\alpha\,\|\vec{x}\|_a\le\|\vec{x}\|_b\le\beta\,\|\vec{x}\|_a \qquad \|\vec{x}\|_\infty\le\|\vec{x}\|_2\le\|\vec{x}\|_1\le n\,\|\vec{x}\|_\infty$$

so a set that fits into a ball of one norm is bounded in all of them.

Inner Product

$$\langle\vec{x},\vec{y}\rangle=\vec{x}^T\vec{y}=\sum_i x_i\,y_i \qquad \cos\theta=\frac{\langle\vec{x},\vec{y}\rangle}{\|\vec{x}\|_2\,\|\vec{y}\|_2} \qquad \langle\vec{x},\vec{x}\rangle=\|\vec{x}\|_2^2$$

Orthogonal vectors have $\langle\vec{x},\vec{y}\rangle=0$. An inner product space is a normed space and a normed space is a metric space.

Inner Product Space

An inner product is positive ($\langle \vec{x}, \vec{x} \rangle = 0$ only for $\vec{x} = \vec{0}$), symmetric and linear in the first slot. The dot product is the one of $\mathbb{R}^n$. In $\mathbb{C}^n$ it is $\langle \vec{x}, \vec{y} \rangle = \vec{y}^{\dagger} \vec{x}$, conjugate symmetric and conjugate linear in the second slot.

Cauchy Schwarz and Pythagoras

$$|\langle \vec{x}, \vec{y} \rangle| \leq \|\vec{x}\| \, \|\vec{y}\| \qquad \vec{x} \perp \vec{y} \;\; \Rightarrow \;\; \|\vec{x} + \vec{y}\|^2 = \|\vec{x}\|^2 + \|\vec{y}\|^2$$

The two sides of Cauchy Schwarz are equal exactly for parallel vectors.

Orthogonal Complement

$$S^{\perp} = \{\vec{x} \in V : \langle \vec{x}, \vec{y} \rangle = 0 \;\text{ for all }\; \vec{y} \in S\}$$

It is a subspace. Two subspaces are orthogonal if every vector of one is orthogonal to every vector of the other.

Orthogonal Bases

$$\vec{x} = \sum_i \alpha_i \, \vec{x}_i \qquad \alpha_i = \frac{\langle \vec{x}, \vec{x}_i \rangle}{\|\vec{x}_i\|^2} \qquad \vec{x} = \sum_i \langle \vec{x}, \vec{q}_i \rangle \, \vec{q}_i \qquad \langle \vec{q}_i, \vec{q}_j \rangle = \delta_{ij}$$

Mutually orthogonal nonzero vectors are linearly independent, so n of them are a basis of $\mathbb{R}^n$. The coordinates are one inner product each, the Fourier coefficients, with no system to solve.

Projection

$$\text{proj}_{\vec{u}} \, \vec{x} = \frac{\langle \vec{x}, \vec{u} \rangle}{\|\vec{u}\|^2} \, \vec{u} \qquad \frac{\langle \vec{x}, \vec{u} \rangle}{\|\vec{u}\|} = \|\vec{x}\| \cos\theta \qquad (\vec{x} - \text{proj}_{\vec{u}} \, \vec{x}) \perp \vec{u}$$

The scalar projection is negative when the angle is above $90°$.

Projection Matrix

$$P = \vec{q} \, \vec{q}^{\,T} \qquad P\vec{x} = (\vec{q}^{\,T} \vec{x}) \, \vec{q} \qquad P^2 = P$$

For a unit vector $\vec{q}$ the outer product $\vec{q} \, \vec{q}^{\,T}$ is a symmetric $n \times n$ matrix. $I - P$ projects onto the vectors orthogonal to $\vec{q}$.

Gram-Schmidt

$$\vec{z}_1 = \vec{x}_1 \qquad \vec{z}_k = \vec{x}_k - \sum_{j<k} \frac{\langle \vec{x}_k, \vec{z}_j \rangle}{\|\vec{z}_j\|^2} \, \vec{z}_j \qquad \vec{y}_k = \frac{\vec{z}_k}{\|\vec{z}_k\|}$$

From each vector we take away its shadows on the ones already done. The first k new vectors span the same space as the first k old ones, and $\vec{z}_k = \vec{0}$ means $\vec{x}_k$ depends on the vectors before it.

QR Decomposition

$$A = QR \qquad r_{jj} = \|\vec{z}_j\| \qquad r_{ij} = \langle \vec{a}_j, \vec{q}_i \rangle \;\; (i < j) \qquad R = Q^T A$$

Q holds the orthonormal vectors of Gram-Schmidt on the columns of A and R is upper triangular. $A\vec{x} = \vec{b}$ becomes $R\vec{x} = Q^T \vec{b}$, solved from the bottom up, and the least squares problem becomes $R\vec{\alpha} = Q^T \vec{y}$ without forming $X^T X$.

Orthogonal Matrices

$$Q^TQ = QQ^T = I \qquad Q^{-1} = Q^T \qquad \sum_i \vec{q}_i\,\vec{q}_i^{\,T} = I \qquad \|Q\vec{x}\| = \|\vec{x}\|$$

A square matrix with orthonormal columns has orthonormal rows too. It only rotates or reflects. In $\mathbb{C}^n$ it is a unitary matrix, $Q^\dagger Q = QQ^\dagger = I$.

The Four Subspaces

$$R(A) = \{A\vec{x}\} \subseteq \mathbb{R}^m \qquad N(A) = \left\{\vec{x} : A\vec{x} = \vec{0}\right\} \subseteq \mathbb{R}^n \qquad R(A^T) \subseteq \mathbb{R}^n \qquad N(A^T) \subseteq \mathbb{R}^m$$

The column space is the span of the columns, the row space the span of the rows. $A\vec{x} = \vec{b}$ can be solved exactly when $\vec{b} \in R(A)$, and then all of its solutions are $\vec{x}_0 + N(A)$.

Rank and Rank Nullity

$$r = \dim R(A) = \dim R(A^T) \le \min\{m, n\} \qquad \dim N(A) = n - r \qquad \dim N(A^T) = m - r$$

The column rank and the row rank are always equal.

Orthogonal Decomposition

$$N(A) = R(A^T)^\perp \qquad N(A^T) = R(A)^\perp \qquad \mathbb{R}^n = R(A^T) \oplus N(A) \qquad \mathbb{R}^m = R(A) \oplus N(A^T)$$

Every $\vec{b} \in \mathbb{R}^m$ splits in exactly one way into $\vec{b} = \vec{q} + \vec{r}$ with $\vec{q} \in R(A)$, $\vec{r} \in N(A^T)$ and $\vec{q} \perp \vec{r}$. A sends the row space onto the column space and the null space to $\vec{0}$.

Eigenvalues and Eigenvectors

$$A\vec{x} = \lambda\,\vec{x}, \quad \vec{x} \neq \vec{0} \qquad p(\lambda) = \det(A - \lambda I) = 0 \qquad \mathcal{E}_\lambda = N(A - \lambda I)$$

An eigenvector is a direction that A only stretches. A real matrix may have complex eigenvalues.

Diagonalization

$$AX = X\Lambda \qquad A = X\Lambda X^{-1} \qquad A^k = X\Lambda^k X^{-1}$$

The columns of X are the eigenvectors and $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$. It works exactly when the geometric multiplicity $\dim \mathcal{E}_{\lambda_i}$ equals the algebraic multiplicity of every eigenvalue, always so for distinct eigenvalues. Otherwise there is only the Jordan form.

Determinant and Trace

$$\det A = \prod_i \lambda_i \qquad \operatorname{tr} A = \sum_i \lambda_i \qquad p(\lambda) = \lambda^2 - (\operatorname{tr} A)\,\lambda + \det A \quad \text{for} \quad 2 \times 2$$

A is invertible exactly when 0 is not an eigenvalue.

Spectral Theorem

$$A^T = A \quad \Rightarrow \quad Q^TAQ = \Lambda \qquad A = Q\Lambda Q^T = \sum_i \lambda_i\,\vec{q}_i\vec{q}_i^{\,T}$$

A symmetric matrix has n real eigenvalues and orthonormal eigenvectors, the columns of the orthogonal Q. Eigenvectors of different eigenvalues are orthogonal by force. For a Hermitian $A^\dagger = A$ the same holds with a unitary Q.

Sum of Projections

$$A\vec{x} = \sum_i \lambda_i \left(\vec{q}_i^{\,T}\vec{x}\right)\vec{q}_i = \sum_i \alpha_i\,\vec{q}_i \qquad\qquad \alpha_i = \lambda_i\,\vec{q}_i^{\,T}\vec{x} = \langle A\vec{x}, \vec{q}_i\rangle$$

A projects $\vec{x}$ onto each eigen direction and stretches the piece by its eigenvalue, and the α_i are the Fourier coefficients of $A\vec{x}$. In $\mathbb{R}^2$ it sends the unit circle to an ellipse with the eigenvectors as its axes and the half axes $|\lambda_i|$.

Quadratic Form

$$\vec{x}^{\,T}A\vec{x} = \sum_{i,j} a_{ij}\,x_i x_j = \vec{x}^{\,T}H\vec{x} \qquad\qquad H = \frac{1}{2}\left(A + A^T\right)$$

The form only sees the symmetric version H, so without loss of generality the matrix of a quadratic form is symmetric.

Definite Matrices

A symmetric A is pd if $\vec{x}^{\,T}A\vec{x} > 0$ for all $\vec{x} \neq \vec{0}$, psd if $\vec{x}^{\,T}A\vec{x} \geq 0$, nd if $\vec{x}^{\,T}A\vec{x} < 0$ for all $\vec{x} \neq \vec{0}$, nsd if $\vec{x}^{\,T}A\vec{x} \leq 0$, and indefinite if the form takes both signs. A is nd exactly when $-A$ is pd.

Eigenvalue Test

$$\vec{x}^{\,T}A\vec{x} = \sum_i \lambda_i\,y_i^2 \qquad \vec{y} = Q^T\vec{x}$$

pd if all $\lambda_i > 0$, psd if all $\lambda_i \geq 0$, nd if all $\lambda_i < 0$, nsd if all $\lambda_i \leq 0$, indefinite if some $\lambda_i > 0$ and some $\lambda_j < 0$. A pd matrix has $\det A > 0$ and is invertible.

Sylvester's Criterion

With the leading principal minors P_k, the determinants of the upper left $k \times k$ blocks, a symmetric A is pd if and only if all $P_k > 0$, and nd if and only if $(-1)^k P_k > 0$. With $\det A \neq 0$ and neither pattern it is indefinite. Nonnegative minors prove nothing about semidefiniteness. For a 2×2 matrix pd means $\operatorname{tr} A > 0$ and $\det A > 0$.

Cholesky Decomposition

$$A = LL^T = \tilde{L}D\tilde{L}^T \qquad L = \tilde{L}D^{1/2} \qquad L\vec{y} = \vec{b},\;\; L^T\vec{x} = \vec{y}$$

A symmetric A is pd if and only if it has a lower triangular L with a positive diagonal. A root of a number ≤ 0 on the way means A is not pd. $A\vec{x} = \vec{b}$ is solved from the top down, then from the bottom up, with no inverse.

Singular Value Decomposition

$$A = U\Sigma V^T = \sum_{i=1}^{r} \sigma_i\,\vec{u}_i\vec{v}_i^{\,T} \qquad A\vec{v}_i = \sigma_i\,\vec{u}_i \qquad \sigma_1 \geq \cdots \geq \sigma_r > 0$$

Works for every $m \times n$ matrix. U and V are orthogonal, Σ has the shape of A with the σ_i on its diagonal. $\vec{v}_1, \ldots, \vec{v}_r$ span the row space, $\vec{v}_{r+1}, \ldots, \vec{v}_n$ the null space, $\vec{u}_1, \ldots, \vec{u}_r$ the column space and $\vec{u}_{r+1}, \ldots, \vec{u}_m$ the left null space.

Computing the SVD

$$A^T A\,\vec{v}_i = \sigma_i^2\,\vec{v}_i \qquad \vec{u}_i = \frac{1}{\sigma_i}\,A\vec{v}_i \qquad AA^T\vec{u}_i = \sigma_i^2\,\vec{u}_i$$

Singular values are not eigenvalues, except $\sigma_i = |\lambda_i|$ for a symmetric matrix. For a square matrix $|\det A| = \prod_i \sigma_i$.

Best Rank k Approximation

$A_k = \sum_{i=1}^{k} \sigma_i \, \vec{u}_i \vec{v}_i^T$ is the closest matrix of rank k to A, with the error σ_{k+1}.

Matrix Norm and Condition Number

$$\|A\|_2 = \max_{\|\vec{x}\|_2=1} \|A\vec{x}\|_2 = \sigma_1 \qquad \kappa(A) = \frac{\sigma_1}{\sigma_r}$$

A takes the unit circle to an ellipse with the semi axes $\sigma_i \vec{u}_i$.

Pseudo Inverse

$$A^+ = V\Sigma^+ U^T \qquad \vec{x}^* = A^+ \vec{b} \qquad A^+ = (A^T A)^{-1} A^T \text{ for independent columns}$$

Σ^+ is $n \times m$ with $1/\sigma_i$ on its diagonal. $\vec{x}^*$ is the least squares solution with the smallest norm, and $A^+ = A^{-1}$ for an invertible A.